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Un modèle mathématique d'impression et d'imagerie pour l'authentification des codes de détection de copie

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UNIVERSITÉ DE GENÈVE
Département d'Informatique

FACULTÉ DES SCIENCES
Professeur Slava Voloshynovskiy

un modèle mathématique d'impression et d'imagerie pour l'authentification des codes de détection de copie

THÈSE

présentée à la Faculté des Sciences de l'Université de Genève
pour obtenir le grade de Docteur ès Sciences, mention Informatique

par

Joakim Tutt
de
Sirnach (TG)

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Thèse de Monsieur Joakim TUTT

intitulée :

«un Modèle Mathématique d'Impression et d'Imagerie pour l'Authentification des Codes de Détection de Copie»

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Genève, le 8 janvier 2026

Thèse - 5965 -

La Doyenne

N.B. - La thèse doit porter la déclaration précédente et remplir les conditions énumérées dans les "Informations relatives aux thèses de doctorat à l'Université de Genève".

À ma femme, Thi Xa, ton incroyable détermination et ton courage sont une source infinie d'énergie et d'inspiration qui me portent en avant.

Préface

Me revoilà face à moi-même, six années ont passé. Je contemple cette thèse qui représente la quintessence de toutes ces années de recherche, le résumé écrit d'une sacré aventure et je réalise qu'une page est en train d'être tournée. Aujourd'hui, je ne saurais dire si le temps passé fut long ou court. Il fut riche, intense, plein de vie et d'émotions. J'ai attendu le dernier moment pour écrire ces mots. Il fallait que tout soit écrit, que l'expérience soit complète afin d'en mesurer toute l'ampleur. La thèse a été pour moi un exercice transformatif. Je le dis sans honte, je n'avais pas compris. Au sortir d'études brillantes en mathématiques, je n'avais aucune idée de ce que la recherche représentait et de ce qu'elle demandait. Je faisais de la science comme on visite un musée, attiré par la beauté de ses théories, ému par tel théorème brillant élaborant des connexions profondes entre des sujets en apparence distincts. Un touriste scientifique en quelque sorte.

Il a fallu un fin pédagogue pour que j'accepte de changer mon point de vue. Porter mon regard non pas seulement sur les belles théories construites par d'autres, mais sur les terrains vagues et imparfaits que sont les questions ouvertes et les champs de recherche vierges. Je ne remercierai jamais assez Slava d'avoir su m'amener patiemment à apprivoiser ce monde étrange pour moi, fait de doutes et d'intuitions, d'essais et d'erreurs, où l'on observe les magnifiques édifices construits par d'autre et l'on travaille dur pour construire son propre château de sable. J'ai alors découvert une sensation que je ne connaissais pas : J'ai goûté à la satisfaction immense d'avoir construit quelque chose de mes mains. Est-ce donc ça le goût de la recherche ?

Arrivé au terme de ces six années de recherche, je constate ce profond changement de perspective qui s'est produit en moi tout au long de ce parcours. L'impression d'avoir laissé derrière moi mon éternel statut d'étudiant pour laisser place au modeste chercheur que je suis aujourd'hui. Je ressors enrichi par le long travail accompli, par tous mes doutes, toutes mes errances et recherches infructueuses, mais également fort de toutes mes réussites et mes compréhensions. En guise de conclusion, me revient ce magnifique poème de Jean de La Fontaine et qui dit mieux que moi tout mon sentiment pour ces nombreuses années de recherche.

Le Laboureur et ses enfants

Travaillez, prenez de la peine :
C'est le fonds qui manque le moins.
Un riche Laboureur, sentant sa mort prochaine,
Fit venir ses enfants, leur parla sans témoins.
Gardez-vous, leur dit-il, de vendre l'héritage
Que nous ont laissé nos parents.
Un trésor est caché dedans.
Je ne sais pas l'endroit ; mais un peu de courage
Vous le fera trouver, vous en viendrez à bout.
Remuez votre champ dès qu'on aura fait l'Oût.
Creusez, fouillez, bêchez ; ne laissez nulle place
Où la main ne passe et repasse.
Le père mort, les fils vous retournent le champ
Deçà, delà, partout ; si bien qu'au bout de l'an
Il en rapporta davantage.
D'argent, point de caché. Mais le père fut sage
De leur montrer avant sa mort
Que le travail est un trésor.

Jean de La Fontaine, Fables de la Fontaine, 1668

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Une thèse est un travail collectif. Tant de personnes ont contribué à ce que ce travail existe aujourd'hui. Ces quelques lignes sont là pour vous remercier. Vous remercier de votre soutien et de votre aide dans ce long parcours que j'ai mené au long des six dernières années. Vous en avez été une part importante, je vous l'assure et je ne saurais trop insister sur l'importance d'être bien accompagné lorsque l'on se lance dans une telle aventure. Par avance je tiens à m'excuser auprès de toutes les personnes que j'aurai oubliées, ce n'est certainement pas par choix mais simplement ma mémoire qui m'aura joué des tours lors de l'écriture de ces lignes.

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Finalement, un immense merci à Slava, mon directeur de thèse. Je ne saurais exprimer ici le rôle central que tu as joué dans la réussite de mon parcours de thèse. Ta confiance et ton optimisme constant lors de nos réunions de recherche m'ont beaucoup aidé à progresser sereinement même lorsque j'avais l'impression de n'aller nulle part. Ta disponibilité quasi permanente en cas de besoin m'a permis de me débloquer à de nombreuses reprises. Enfin, ta compréhension, ton soutien moral et ta gentillesse naturelle m'ont permis de traverser les obstacles qui se sont dressés sur ma route et d'arriver aujourd'hui au terme de ce travail.

Résumé

Cette thèse étudie le problème de la lutte contre la contrefaçon des objets et des méthodes numériques permettant d'en assurer la protection. Elle a pour but le développement de stratégies d'authentification sécurisées et fiables et fait partie d'un plus large projet d'analyse théorique des systèmes d'authentification basés sur les algorithmes d'apprentissage profond du point de vue de la théorie de l'information. La contrefaçon des biens représente un défi majeur pour le développement de nos sociétés modernes. En effet, ces attaques représentent une menace croissante pour les industries tant pour leur image que pour le bien-être de leur clientèle, en particulier lorsque les attaques ciblent des produits sensibles comme les médicaments ou les systèmes de sécurité. Ce problème est d'autant plus important aujourd'hui, avec une économie mondialisée et des biens qui circulent entre plusieurs pays, multipliant par là-même les possibilités d'attaques et les vulnérabilités, tout en compliquant d'avantage le suivi et l'identification des biens.

Parmi la multitude de technologies développées dans le but de sécuriser les chaînes de production, une invention récente apparue en 2004 a démontré un intérêt certain, se positionant comme une solution simple à implémenter dans une chaine de production tout en garantissant un haut niveau de suivi et de fiabilité. Cette technologie se nomme le Code de Détection de Copie (CDP en anglais). Néanmoins, sa relative jeunesse comparée à d'autres systèmes de protection existants font que la littérature concernant les garanties de sécurité de cette approche sont majoritairement inexplorés. En particulier, l'étude des forces et faiblesses et la stabilité de cette technologie n'ont jusqu'à ce jour été testé que de manière empirique sur des jeux de données privés et dans le secret des laboratoires de recherche industriels. De plus, les récents progrès concernant la génération d'images synthétiques par des techniques d'apprentissage profond a ouvert la voie à de nouvelles attaques qui représentent une menace sérieuse pour les systèmes de protection basés sur la technologie CDP. Ainsi, il y a aujourd'hui un grand besoin de développer des recherches ouvertes et indépendantes concernant les garanties de sécurité et la performance des systèmes d'authentification basés sur les CDP. Ces recherches doivent permettre une analyse neutre de cette technologie, basée sur des jeux de données en libre accès et avec des recherches reproductibles menées par des chercheurs indépendants.

Le travail que nous présentons dans cette thèse est une contribution à cet effort global. L'objectif principal est de mener une recherche approfondie sur le sujet du canal d'impression-

imagerie de l'authentification par CDP et de proposer de nouveaux outils et méthodes de mesures de ses caractéristiques et de leurs statistiques. Ces nouvelles approches sont étudiées tant du point de vue théorique, basées sur le développement de modèles permettant l'analyse et la discussion des garanties de sécurités de ce système que du point de vue expérimental, en testant ces modèles sur des jeux de données réelles imprimées et capturées dans des conditions réelles. La thèse propose également la mise en place de nouvelles stratégies pour l'authentification des CDP basées sur les modèles théoriques étudiés, et démontre que ces nouvelles stratégies permettent de lutter de manière efficace contre les plus puissantes attaques basées sur l'apprentissage profond.

Dans une première partie, la thèse se concentre sur la modélisation du canal d'impression-imagerie des CDP par une paire d'encodeur-décodeur basés sur un modèle d'apprentissage profond. La motivation derrière cette approche repose sur la complexité de la modélisation des phénomènes physiques et l'impressionnante capacité qu'ont les architectures basées sur l'apprentissage profond à générer des photos de haute qualité quasiment indiscernables de photos originales. Ce modèle nous donne ainsi accès à un clone numérique du processus d'impression-imagerie, que nous analysons en détail et comparons avec des jeux de données réelles. Nous proposons ensuite diverses approches permettant d'utiliser cette architecture pour renforcer les systèmes d'authentification basés sur les CDP.

Dans une seconde partie, la thèse introduit le modèle BPC, un modèle mathématique qui décrit le canal d'impression-imagerie des CDP comme un modèle sur réseau où chaque symbole binaire du motif original se transforme avec une certaine probabilité. Ce modèle est largement inspirés des modèles sur réseaux en théorie des probabilités et par la théorie de la communication développée par Shannon. Ce modèle nous permet de prédire le comportement d'un schéma d'authentification de CDP et donne de nouvelles clés de compréhension expliquant les raisons pour lesquelles les stratégies classiques d'authentification peinent à repérer les attaques basées sur l'apprentissage profond. Finalement, cette seconde partie propose de nouveaux systèmes d'authentification et démontre expérimentalement la supériorité de ces nouvelles approches.

Dans la troisième partie, la thèse étudie la question des garanties de performances des systèmes d'authentification par CDP. En faisant appel à la théorie des tests d'hypothèses en statistiques et en se basant sur le modèle BPC introduit précédemment, cette partie donne des critères permettant d'optimiser les stratégies d'authentification basés sur les CDP et des mesures théoriques permettant de garantir la détection correcte de contrefaçons.

Abstract

This thesis takes place in the field of Physical Object Security in view of the development of secure and reliable authentication schemes and is a part of the broader project of Information-theoretic analysis of deep identification systems. The problem of object counterfeiting represents a major threat for the development of our modern societies. These attacks constitute a menace for the industrie's development and reputation and constitute a risk for the consumer especially when counterfeiters target vital industries such as drugs or security devices. The problem of object counterfeiting is further exacerbated today as we live in a decentralized world where goods and products are being shipped all across the globe, multiplying the possibilities of attack against production chains and complicating the task of object tracking and identification.

Among the great variety of technologies available for the fight against object counterfeiting, a recent anticopy technology emerged in 2004 which showed great promise as an interesting trade-off offering high level of security against product counterfeiting while still maintaining a relative simplicity in implementation for mass-manufacturing industries. This technology is called the Copy Detection Pattern (CDP). However, due to its relative youth when compared to other well-known technology, the approach based on CDP still lacks a deeper study of its strengths and weaknesses and its stability has only been tested empirically on small proprietary datasets, kept in the secrecy of private industrial research laboratories. Moreover, the recent breakthrough in synthetic image generation based on deep learning architectures were shown to be a major threat to the classical methods used for the authentication of CDP. As such, there is a pressing need in developing open researches on the security guarantees provided by the CDP authentication scheme that would allow for reproducible experiments and fair comparison of its capabilities by independent researchers.

The work we present in this thesis is a contribution to this general effort. Its main objective is to provide an in-depth study of the printing-imaging channel associated to the CDP authentication framework both theoretically and experimentally so as to develop a better understanding of this technology's main challenges and ultimately help fighting against the new threats posed by modern attacks. In this regard, this thesis proposes novel directions for the description and analysis of the printing-imaging channel of CDP, providing new tools and measures of the channel's features and statistics. These new approaches are studied both from a theoretical point of view, developing sound foundations to discuss the problem of

CDP-based framework security guarantees, and experimentally, by testing them on publicly available datasets of CDP. The thesis also adresses the question of applying these new tools to enhance the classical authentication scheme and demonstrates that the proposed novel approaches are actually able to detect even the strongest attacks on CDP known as of today.

In a first part, the thesis concentrates on modelling the printing-imaging channel of CDP using a deep-learning encoder-decoder architecture. This approach is motivated by the multiple breakthrough in image generation and provides a fully digital clone of the printing-imaging procedure typically encountered in CDP authentication. We then provide a comparison study of the digital clone approach with the real setup of CDP and explore multiple directions for the use of this architecture to propose enhanced authentication schemes for CDP.

In a second part, the thesis introduces the BPC model, a probabilistic model that describes the printing-imaging channel of CDP, inspired by lattice models and information theory. This model allows us to predict the behaviour of a given CDP scheme and gives new insight on why classical authentication schemes fail against high quality fakes and points at possible directions of enhancement. New authentication schemes are then proposed and experiments demonstrate the superiority of these new approaches.

Finally, we tackle the question of theoretical performance guarantees for the authentication of CDP. Relying on statistical binary hypothesis testing for the BPC model, we derive optimal bounds and explore questions regarding the stability and reliability of CDP frameworks. This study provides criteria for the optimisation of CDP-based framework performances.

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Chapter 1

Introduction

In this Chapter, we present the field of Digital Security for Physical Objects, in which this thesis takes place. After briefly introducing the different challenges and evolutions of the field in Section 1.1, we detail the main anti-counterfeiting technologies that are being used nowadays in Section 1.2. We then focus our attention on one particularly promising technology, the Copy Detection Patterns (CDP), describing the general scheme behind its use, detailing the various studies that have been conducted on this technology and pointing out its advantages and limitations in Section 1.3. The scope of the present work is described in Section 1.4, where we take the limitations and unsolved challenges that CDP face as a starting point for our research. Finally, Sections 1.5 and 1.6 present an outline of the thesis and a list of its contributions.

1.1 Physical Object Security

This Section presents the field of physical object security, in view of the digitalization of modern society. While this evolution has had a positive impact on the enhancement of overall accessibility and connectivity across the globe for goods and services, it has, at the same time, introduced many new challenges regarding the security and the fight against counterfeiting and forgery. These new challenges call for new security solutions that can face these threats in both digital and physical world. We give a brief overview of the multiple aspects of this complex problem organized around key topics.

Digital Society

In recent years, our societies have seen tremendous changes in the way they interact with the world by almost systematically relying on the use of digital technology for most interactions and production systems. This evolution has been studied and documented in the scope of Society 4.0 [1, 2, 3] and led to the development of industries with cheaper production costs,

more environmentally efficient production pipelines and closer end-to-end connection from customer to producer.

Distributed Society

At the same time, the global economy has become highly distributed worldwide, driven by improvements in communication and production technologies [4]. It is today economically viable to produce goods in one region of the world, exploiting available resources, and ship them for consumption in another. This has conducted to a geographically dispersed production-consumption model, as seen, for instance, in China's manufacturing capabilities producing goods for consumers spread across Europe and the United States. Consequently, our society is interconnected, and production chains are spread all over the globe, heavily relying on advanced technologies to ensure virtual proximity and seamless integration.

Recent Progresses in Digital Technologies

The revolution in the way industry produces goods and distributes them was made possible by a serie of progresses in digital technologies. In [5], the authors highlight six design principles that characterize this change namely: interoperability, virtualization, decentralization, real-time capability, service orientation and modularity. These characteristics rely on digital innovations such as:

- *Cyber-Physical Systems (CPS):* Systems in which physical processes and objects are linked to a digital identity and are able to inform and update in real-time about their current status and needs. Typical technologies are RFID, NFC and digital sensors.
- *Internet of Things (IOT):* This innovation allows CPS to connect over the internet and operate in coordination by directly adressing current status, needs and demands. Barcodes, QR codes and embedded chips are widely used nowadays and allow for digital identification and authentication.
- *Internet of Services (IOS):* Connects producers and consumers over the internet and allows for personalized and customizable production as well as end-to-end traceability for the customer. Industries and societies nowadays typically use websites and smartphone apps as digital storefronts for their services.
- *Smart Factories:* These are factories which encompass all previous innovations and organize in an automated way, relying on humans only for high-level decisions.

The recent breakthrough in the development of AI technologies further accelerates this trend by enabling the automatization of increasingly complex decision tasks in the production pipeline. Some authors even refer to the appelation Industry 5.0 as more and more advanced automated systems based on AI are being used [6].



Figure 1.1. A diagram illustrating the various design principles of industry 4.0 as shown in [1].

Challenges, Threats and Limits

All these advances have brought many benefits to the society, but these advantages come with new challenges and threats. The same technologies enabling faster, cheaper, and more precise production also provide new opportunities for counterfeiters [7, 8, 9]. Typical counterfeitings of goods involve reverse engineering genuine products, industrial espionage and competitive intelligence from former employees. Fig. 1.2 shows the different type of attacks typically encountered by industries in the past ten years. Classical countermeasures against such attacks, rely on digital product identification (holograms, RFID, data matrix codes, ...) and track-and-trace systems based on security markers.

With the global automatisation of production pipelines nowadays, the authentication and tracing needs to be done digitally and is being challenged by novel attack schemes. Traditional anti-counterfeiting technologies such as RFIDs, holograms, and printed QR codes, originally developped for identification purposes, now reveal several limitations. These older technologies face vulnerabilities to cloning especially in the face of the fast-developping AI capabilities. Their susceptibility to being copied and manipulated further demonstrates the issue in an environment where counterfeiters are equipped with advanced digital tools.

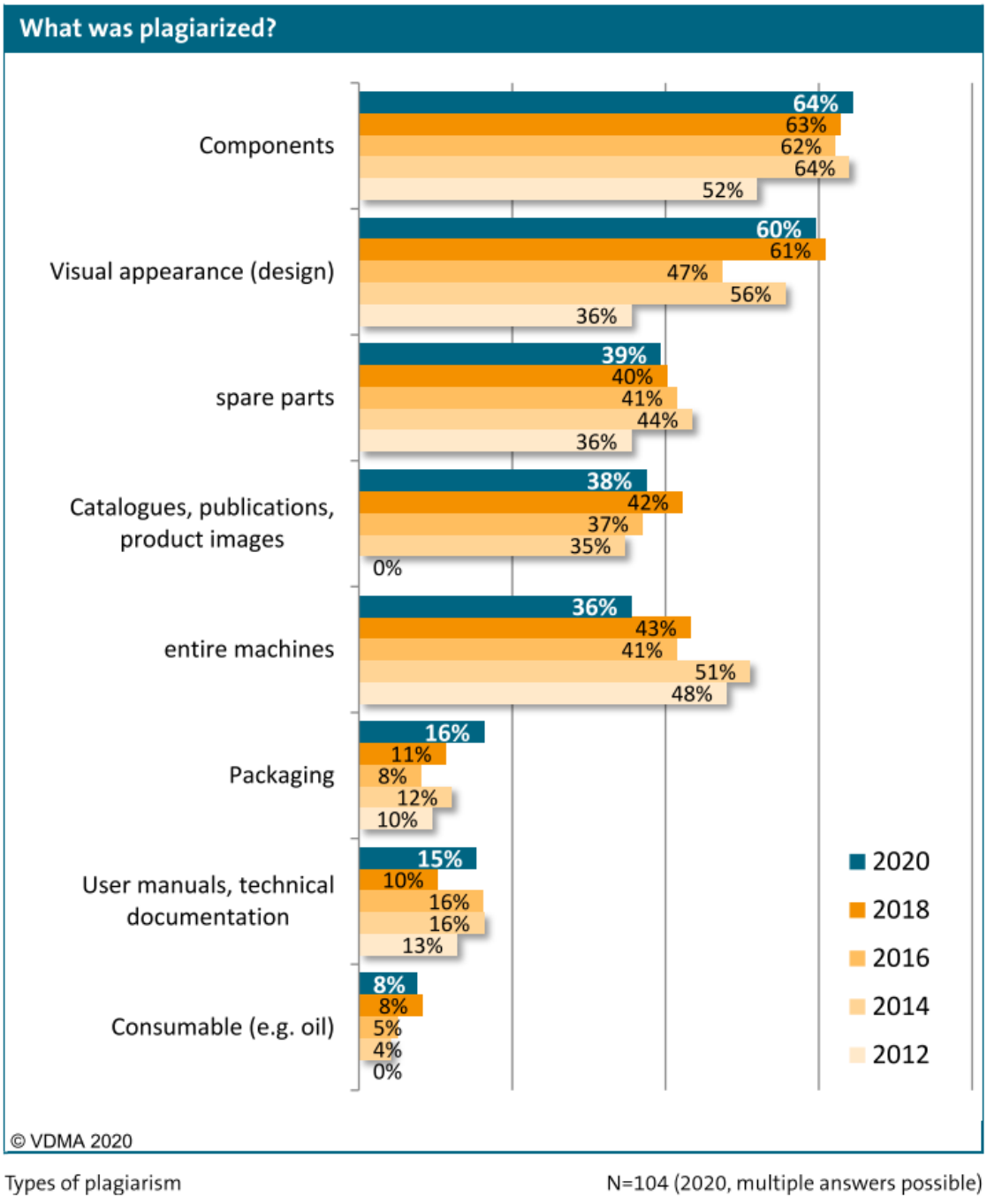


Figure 1.2. A bar graph showing the main targets of attacks when producing counterfeited products and its evolution across the years. The percentages represent the proportion of industries affected by each type of attack. Illustration taken from [8].

1.2 State-of-the-Art in Physical Object Protection

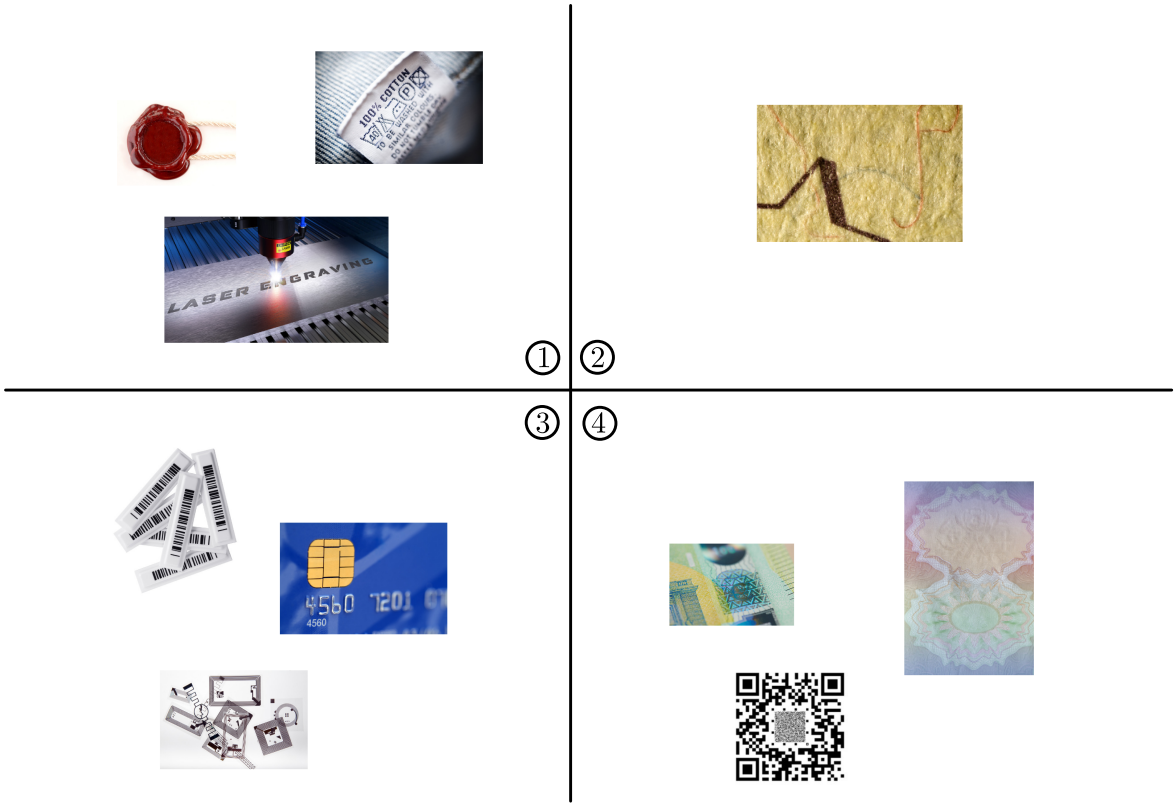


Figure 1.3. The four main categories of technologies for the Physical Object Protection. The illustrating images were taken from [10].

In this Section, we detail the different technologies available nowadays to protect physical objects against counterfeiting and review their actual advantages and drawbacks. The technologies involved are very diverse and, based on the description done in [10, 11], can be broken down into four main categories as illustrated in Fig. 1.3:

1. *Mechanical Technologies:* examples are Labels, Tags, Laser Engraving, Anti-Alteration Devices, Seals, Security Threads and Security Film.
2. *Chemical and Physical Technologies:* such as DNA Coding, Chemical Encoding, Glue Coding and Surface Fingerprinting.
3. *Electronic Technologies:* which comprise all Chip-Based Radio Frequency Identifiers (RFID) Technologies, Battery-Assisted Passive (BAP) Tags, Near Field Communication (NFC), Electronic Seals, Magnetic Stripes and Contact Chips.
4. *Marking Technologies:* which take the form of Optical Memory Stripe, Machine-Readable Codes, One and Two Dimensional Barcodes, Security Holograms, Special Inks such as

Ultraviolet/Infrared/Magnetic Inks, Encrypted Images and Watermarks, Microtexts, Guilloche, Unique Identifier Marks and Copy-Detection Patterns.

We now give an overview of each of these fields individually and discuss their potential advantages and drawbacks in fighting against counterfeited goods and highlight the position of Copy Detection Patterns (CDP) as a balanced solution for practical applications. Each technology is being evaluated according to the following criteria:

1. *Security Level:* Effectiveness in preventing counterfeit replication.
2. *Appearance Impact:* Whether the feature alters the object's appearance.
3. *Cost Efficiency:* Cost of implementation and maintenance.
4. *Equipment and Expertise:* Need for specialized equipment or expert skills.
5. *Eco-Friendliness:* Environmental impact and recyclability.
6. *Scalability:* Feasibility for large-scale production and mass manufacturing.

Mechanical Technologies

Mechanical technologies rely on extra elements added to the product whose presence and integrity serve as tamper-proofs for the product [12]. Typical technologies being used are seals, security threads and films or anti-alteration devices. Labels, tags and engravings, on the other hand, are used to ascertain the quality and origin of the product. They are mainly used as authenticity certificates for the product.

1. *Security Level:* Generally moderate. Physical labels and tags can be tampered with.
2. *Appearance Impact:* Significant in cases of labels, tags, and laser engraving, potentially altering the product's appearance.
3. *Cost Efficiency:* Varies, labels and tags are affordable, while engraving and advanced security threads can be costly.
4. *Equipment and Expertise:* Readable by human or with mobile phone cameras.
5. *Eco-Friendliness:* Mixed, engravings and tags can be reused, but security threads, labels and films require extra production and recycling pipelines.
6. *Scalability:* Feasible but may depend on product type and appearance limitations.

Chemical and Physical Technologies

These technologies ensure the security of the products by relying on unique markers based on physical or chemical reactions [13]. DNA Coding Functions are high technology molecular tags, which are inherently immutable and act as highly reliable certificates of authenticity. Genotyping and Isotopes use the molecules within the product as a fingerprint for identification and traceability. The idea is similar with Surface Fingerprinting which uses the imperfections of product's surface as a Physical Unclonable Function (PUF).

1. *Security Level:* Very high, especially for DNA and chemical encodings. Surface fingerprinting guarantees uniqueness in identification but may be limited in application.
2. *Appearance Impact:* Often minimal for DNA and chemical encodings; surface fingerprinting is non-invasive.
3. *Cost Efficiency:* Generally expensive due to specialized material and technology costs.
4. *Equipment and Expertise:* Exclusively machine-readable, requiring specialized reading devices and sometimes expert handling.
5. *Eco-Friendliness:* Mixed, DNA and chemical encodings may not be eco-friendly due to the type of material being used, while surface fingerprinting has no impact.
6. *Scalability:* Challenging for mass production due to the high cost in infrastructure needs.

Electronic Technologies

Electronic technologies are based on electronic signals for identification and authentication. They rely on passive or active chips such as RFID chips [14], BAP tags, NFC [15], and Electronic Seals. By being able to communicate directly through digital means, they are the most proeminent actors in the Industry 4.0 development as they allow for remote digital authentication, live status update and interaction through the Internet of Things.

1. *Security Level:* Moderate. While unique identifiers add security, these tags are potentially vulnerable to cloning or hacking for read-write systems unless properly protected with advanced security measures.
2. *Appearance Impact:* Minimal, as they are typically embedded into the product such as in credit cards and ID documents.
3. *Cost Efficiency:* Varying, but still requires chip manufacturing, deployment and management.
4. *Equipment and Expertise:* Specialized equipment for reading; sometimes accessible via mobile devices.
5. *Eco-Friendliness:* Generally poor, the objects are difficult to recycle due to the presence of electronic components. Implementing a full recycling loop adds additional costs and challenges.
6. *Scalability:* Limited by cost and recycling constraints; requires investment in infrastructure for large-scale applications.

Marking Technologies

Marking technologies base the security of products by adding a special mark directly on the product or on its package to act as a certification of its origin, or contain relevant information

about the product, similarly to Tags. Typical examples of such technologies are 1D and 2D Barcodes or any machine-readable code. In order to be resistant to counterfeiting, these marks need to include patterns that are hard to imitate or based on secret-key features. This extra security may be implemented in various ways such as using:

- *Special material:* Special Inks, Holograms, Special Printings such as Guilloche;
- *Secret key features:* Copy-Detection Patterns, Encrypted Images or Watermarks.

These options vary widely in cost, visibility, and ease of reproduction.

1. *Security Level:* Varies. QR codes and similar marks used for product identification are easily cloned, while holograms and specific inks offer moderate security but require higher knowledge and equipment.
2. *Appearance Impact:* Printed barcodes are generally well accepted on products and do not require extra manufacturing step, while holograms and special inks may alter the product's appearance.
3. *Cost Efficiency:* QR codes and watermarks are inexpensive; holograms and specialty inks can be costly due to proprietary restrictions and need of specialized equipment and material.
4. *Equipment and Expertise:* QR codes are easily machine-readable; holograms and inks often require human inspection or specialized reading devices.
5. *Eco-Friendliness:* Printed barcodes have no extra impact on ecology as they integrate seamlessly in the production pipeline. Holograms and special inks have more impact due to manufacturing requirements.
6. *Scalability:* Highly feasible, for QR codes and CDP, which integrate seamlessly into production processes.

Justification for Copy Detection Patterns as an Interesting Tradeoff

Copy Detection Patterns [16], as part of the Marking Technologies presented above appears as a very interesting tradeoff when trying to develop anticounterfeiting technologies. Indeed, while having a very low production cost, this technology can effectively offer high security for product authentication. We give here a detailed analysis of this technology:

1. *Security Level:* High security due to the difficulty of duplicating random patterns. Cloning is challenging without having an access to the original templates.
2. *Appearance Impact:* Minimal, especially when integrated within QR codes, preserving product aesthetics.
3. *Cost Efficiency:* Low-cost implementation, compatible with existing printing technologies.

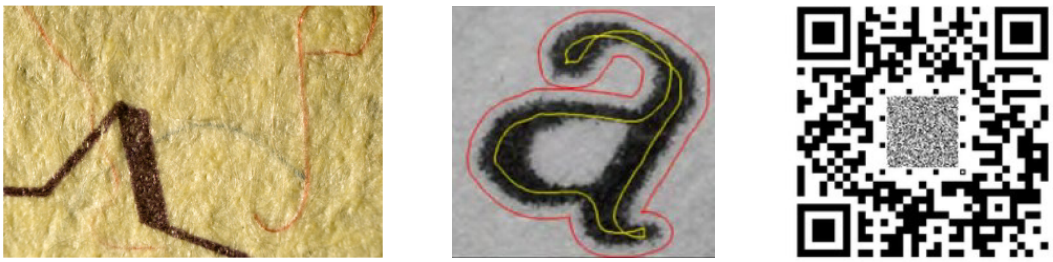
4. *Equipment and Expertise:* Codes are machine-readable and require no specialized training or equipment for verification.
5. *Eco-Friendliness:* High; the patterns are printed and thus require no extra material added to the product.
6. *Scalability:* Ideal for large-scale production; easy integration into existing workflows.

This analysis reveals that while existing technologies each have strengths, they also face limitations in one or more areas and typically suffer from the tradeoff between security, eco-friendliness and scalability. In contrast, CDP offer a simple and ecological solution that aligns with modern requirements for security, cost-effectiveness, sustainability, and scalability, presenting a viable choice for anti-counterfeiting efforts across various industries. We summarize all these results in Table 1.1 where a grade is given for each criterion.

Table 1.1. Comparative analysis of physical object protection technologies. Each criterion is associated with a grade: 1 (bad), 2 (average) and 3 (good).

	Mechanical	Chemical/Physical	Electronic	Marking	CDP
Security	2	3	2	2	2
Appearance	1	3	3	3	3
Cost	2	1	2	2	3
Equipment	3	1	2	2	3
Ecology	2	2	1	2	3
Scalability	2	1	2	3	3

1.3 Copy-Detection Patterns Technology



(a) Substrate (b) Forensic (c) Anticopy

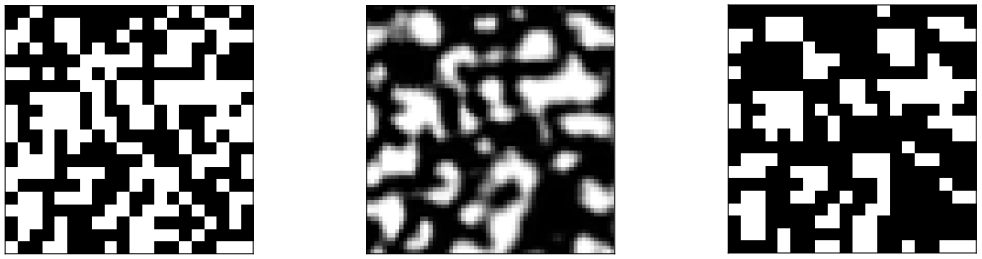
Figure 1.4. The three different approaches to physical object protection based on printed features.

We focus in this Section on the Copy-Detection Patterns technology and its development as an anti-counterfeiting measure for Physical Object Security. This technology has emerged in the process of developping security features based on the natural randomness induced by the printing process. We present here an overview of this field of research following literature reviews such as [17].

Physical Unclonable Functions and Copy Sensitive Graphical Code

The field of anticounterfeiting measures based on printed security elements has seen a great development through the last fifty years with the advent of digital authentication techniques. These techniques can be divided in different approaches but they share the principle of basing their security on the randomness and natural variability of the printing process captured using imaging devices such as scanners or mobile phones. A celebrated notion is the idea of Physical Unclonable Functions (PUFs), that is a physical process (such as printing, engraving, etc.) that produces an outcome whose features are random and makes it so that any repetition of the same process would lead to a different set of random features. In [17], the authors describe three types of approaches for authentication and identification based on this principle (see Fig. 1.4 for an illustration):

- *Substrate Based Approach:* an authentication scheme based on the registration of the inherent randomness of the surface and its interaction with the ink.
- *Forensic Approach:* where the identification is based on a statistical study of the particular randomness of each printing-imaging device and the verification of the presence of these characteristics during product identification.
- *Anticopy Approach:* where the authentication is based on the comparison between a digital reference figure and its printed counterpart, measuring the average amount of deviation with respect to the digital reference.



(a) Digital template (b) Printed CDP (c) Estimated template

Figure 1.5. An illustration of the anticopy authentication scheme based on CDP.

Historically, this kind of processes has been studied by trying to understand the unique interaction between ink and substrate both experimentally [18, 19, 20, 21, 22, 23] and theoretically [24, 25]. Substrate based and Forensic authentication schemes, inspired by the development of digital identification based on fingerprints and other biometric technologies have been and still are the topic of many researches. These technologies are based on the digital registration at microscopic scale of the paper randomness [26, 27] or ink deviation [28, 29, 30]. Modern smartphone cameras today allow this identification to be done without special equipment [31, 32]. The advantage of this approach is the total unclonability of the registered randomness. At the same time, this scheme requires the individual registration of each particular security element during the production and thus ends up being very restrictive in the production pipeline for large scale applications. In addition, the registered references need to be accessible on a server during the verification stage in order to compare the actual probe under examination and its reference.

An approach which tries to alleviate these restrictions is the anticopy authentication scheme as illustrated in Fig. 1.5. These schemes rely on secret key elements such as fixed random patterns [16, 33, 34, 35], high-resolution printed elements [36] or print-resistant watermarking elements [37] that are hidden within a printed graphical element. The idea behind the authentication scheme is to measure the amount of information that is retained by the printing-imaging process from the digital secret key. While some information is destroyed by the random nature of the printing process, a part of the digital information can still be extracted. The security of this scheme is based on the fact that retrieving the secret key based on the printed counterpart is statistically impossible and thus, the attacker's steps of estimating the secret key and reprinting the element on a counterfeit product will completely destroy the secret key information. This approach is thus a tradeoff between information preservation and information loss in order to maximize the probability of detection of counterfeit products [25]. The advantage of such a scheme is the possibility to rely only on digital secret keys for authentication and thus does not require a massive enrollment procedure as for the forensic approach. With the advent of machine learning algorithms, it has however been shown that powerful estimation attacks can be performed which are able to fool the actual protection schemes.

CDP Lifecycle

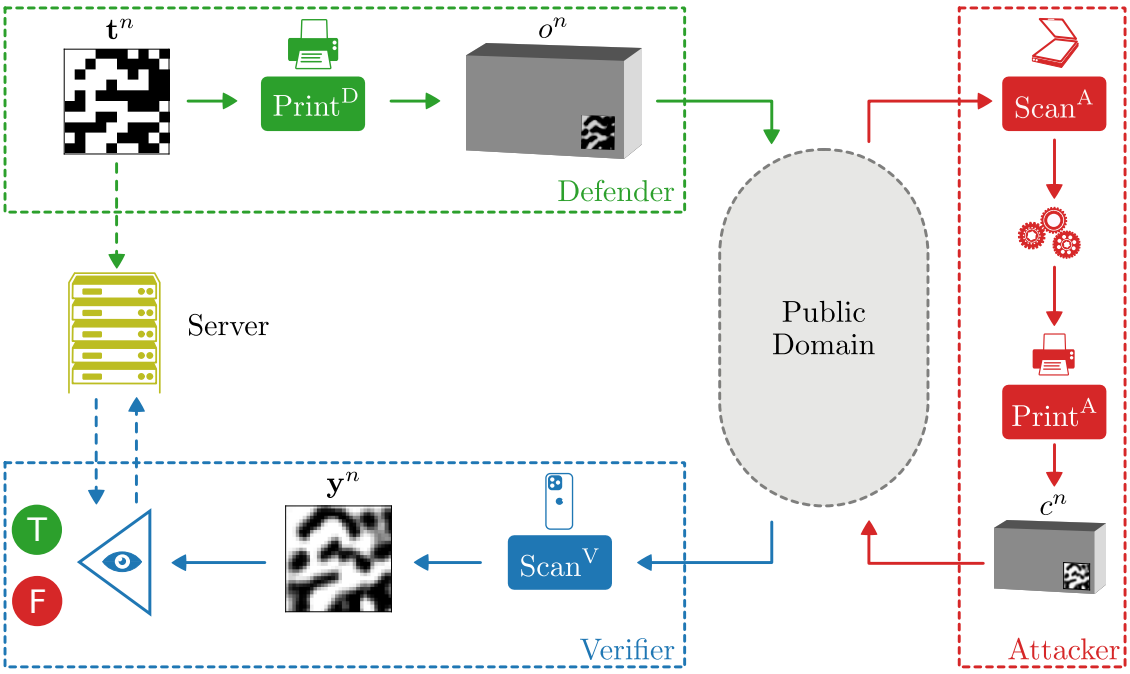


Figure 1.6. CDP lifecycle seen as a two-sided three-player game. The Defender (in green) and the Verifier (in blue) play together against the Attacker (in red).

The Copy-detection patterns technology was first introduced in 2004 by Justin Picard in [16] as an anticopy technology based on maximal entropy data matrix with binary entries. The lifecycle of a CDP-based authentication scheme is illustrated in Fig. 1.6 and can be described as a 2-sided 3-player game with a Defender, an Attacker and a Verifier where both the Defender and the Verifier play together against the Attacker. The roles are the following:

- *Defender (green):* Generates reference templates, which are random binary matrices with maximal entropy based on a secret key, stores them in a protected server for later authentication and prints the CDP together with an identification code n referencing the key on the product using the highest resolution available on the Defender's printer (denoted as $Print^D$). The resolution is chosen to complicate the reproduction of the CDP by the Attacker. At the same time, the printing process at this resolution is information lossy. The couple (CDP, identification code) is generally rendered in the form of an embedded security CDP in the middle of a QR code as proposed in [16]. Finally, the product is sent to the public domain.
- *Attacker (red):* Collects products directly from the public domain and tries to replicate the security CDP element on a new counterfeited product. The reproduction of the CDP element might be done into two separate steps where:

1. The reference template is estimated from the printed CDP using, for instance, a high-resolution scanner $Scan^A$ and special processing;
2. The estimated template is used to generate a counterfeit CDP element which is then printed on the counterfeited product using a printer $Print^A$ that matches the capabilities of the Defender's equipment $Print^D$.

Modern attacking strategies represent a challenging threats nowadays and are presented in detail in the next subsection.

- *Verifier (blue):* Receives an unauthentified product which may either be genuine or counterfeited. An estimation of the key is then performed based on the security CDP using the Verifier's scanner or mobile phone (denoted as $Scan^V$). The scanned image is then processed in a detection module and sent to the server for comparison with the original digital template. If the distorsion is below a certain threshold, the product is authentified as genuine. Otherwise, it is authentified as a fake.

The challenge for the Defender and the Verifier is to produce CDP that are hard to copy while still maintaining high information on the template used to generate them. Until recently, open scientific researches on this particular topic have been sparse [38, 39, 40], especially because of the lack of publicly available CDP datasets. The situation has changed since then with the production of various datasets of CDP based on scanner acquisition [41, 42, 43, 44] and mobile phone acquisition [45, 46], thanks notably to the research fund [47] allocated for the project that financed this work by the Swiss National Foundation (SNF).

Vulnerabilities and Challenges

This section focuses on potential attack mechanisms employed by an adversary to replicate the CDP and attack the cycle described in the previous section. To achieve successful CDP replication, we describe three primary strategies that an attacker may employ based on their expertise, resources, and domain knowledge as illustrated in Fig. 1.7. These strategies, detailed below, vary in complexity, effectiveness, and in the extent of post-processing applied:

(S1) *Copy Machine Strategy:* The simplest form of attack, where the attacker places the original object containing the CDP on a high-resolution photocopy machine (typically 600–2400 dpi). The machine scans and prints the object, potentially using built-in contrast enhancement, morphological adjustments, or other processing techniques to mitigate the dot gain introduced during reproduction. This method relies heavily on the copier's built-in capabilities, producing a replica with moderate fidelity. Such attacks have been studied and documented in [16, 46].

(S2) *Enhanced Scanning and Post-Processing:* A more advanced approach, where the attacker utilizes a high-resolution scanner (e.g., 2400 dpi or higher) and performs custom post-processing using specific image-processing toolboxes. The goal of this strategy is to

enhance the quality of the scanned image through manual adjustments, enabling a higher fidelity replication compared to the copy machine strategy and yielding stronger attacks once reprinted. Such attacks are studied in [16, 48, 49]. Similar studies were conducted on 2-Level QR code which share the same principle [39, 50] which we add here for completeness.

(S3) *Machine Learning-Based Pre-Processing:* The most sophisticated attack strategy known today, leveraging machine learning techniques to approximate the original CDP. Here, the attacker assumes access to a training dataset of paired samples $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{N_{train}}$, where each $\mathbf{x}^n$ represents a scanned image of a synthetic CDP, and $\mathbf{t}^n$ is the corresponding digital template. These pairs may be artificially generated by printing random patterns on equipment similar to the defender's and scanning the outputs. With sufficient data (e.g., 1000–5000 samples), the attacker can train a neural network to predict the original digital CDP from scanned images. Once trained, the model is applied to real-world scans, generating digital templates that approximate the authentic CDP. These very powerful attacks can achieve a Hamming distance as low as 5% between the authentic template $\mathbf{t}^n$ and its synthetic estimated counterpart $\hat{\mathbf{t}}^n$, a significantly smaller error than the typical $20 - 30\%$ Hamming distances produced by simpler methods. Such minimal discrepancy poses a formidable challenge for defenders, as it complicates the task of distinguishing genuine CDP from machine-learned replicas. Such attacks have been heavily studied in the past five years by various authors [41, 42, 43, 44, 49, 51, 52, 53].

The Verifier's objective, thus, is to assess the scanned CDP input probe $\mathbf{y}$ obtained with $Scan^D$ and determine its authenticity, specifically identifying if it originates from the original printed CDP or from any of the attack strategies outlined (strategies 1, 2, or 3) without prior knowledge of the exact replication method employed. The lower the Hamming distance of the estimated template by the attacker is, the more similar the reproduced CDP will be to the original, and consequently, the more difficult it will be to identify discrepancies. In contrast, higher Hamming distances, as observed in less advanced attacks, facilitate the detection due to the more substantial deviations from the original.

The pressing task for the Verifier and the Defender is to establish robust detection mechanisms capable of recognizing authentic CDP amidst these potential imitations, especially those produced by machine learning techniques that achieve high fidelity to the original.

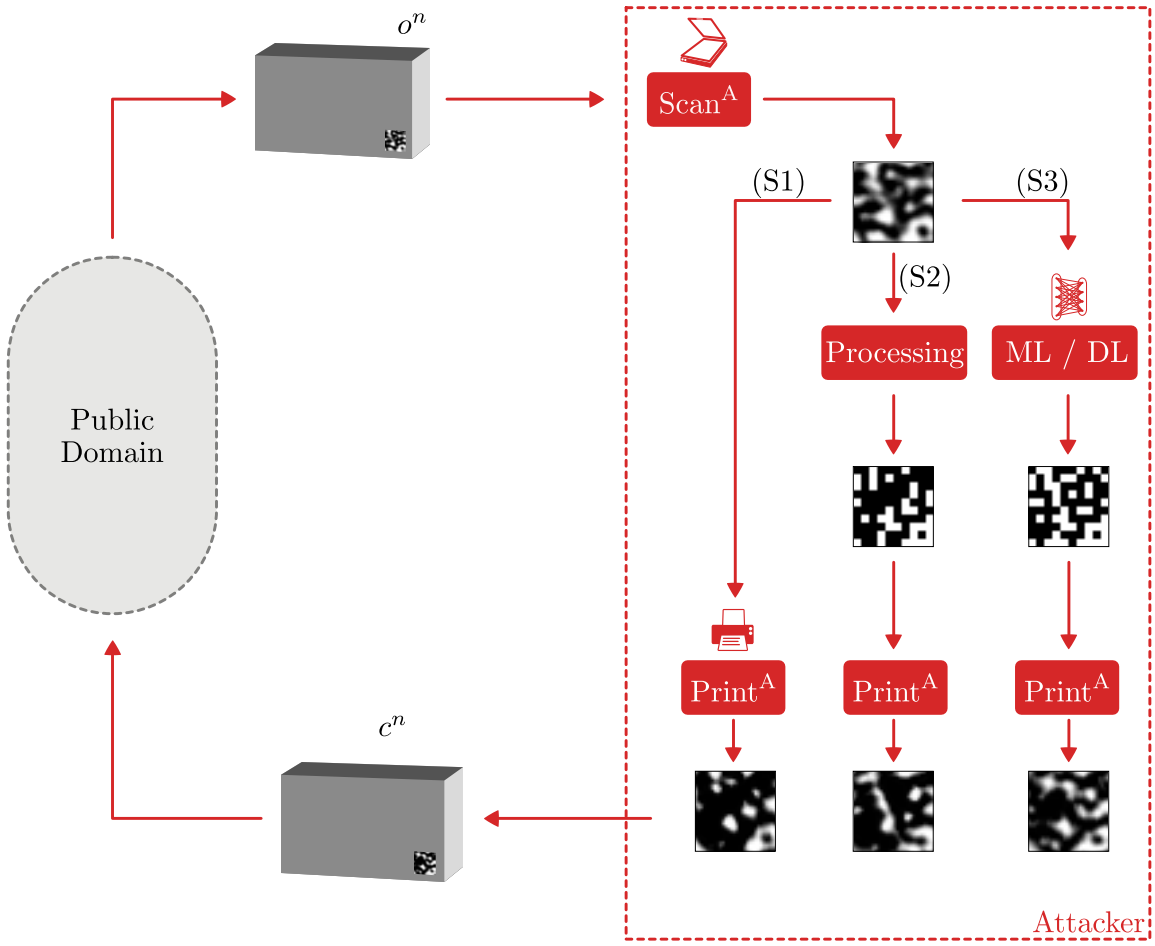


Figure 1.7. An illustration of the three different attacking strategies against a CDP production pipeline. Strategy (S1) is a simple scan and reprint attack, Strategy (S2) includes a standard image processing step and Strategy (S3) relies on machine-learning and deep-learning processing.

1.4 Scope of the Thesis

In this thesis, we focus on the authentication technology based on CDP, which is a novel method of securing physical products. CDP are printable patterns that can be easily authentified using smartphones or high resolution scanners. Originally introduced in [16], this technology is typically confronted nowadays to powerful deep-learning based attacks as described in section 1.3 which demonstrates the vulnerability of such scheme.

The aim of this work is to give a detailed study of strengths and weaknesses of CDP authentication, and propose new approaches and schemes in order to enhance the robustness of this technology in front of the most advanced attacks. The scope of this thesis thus revolves around the comprehension of the printing-imaging pipeline and its theoretical description based on deep learning apparatus and mathematical modelling. The predictions of these models are then compared to actual datasets of industrially produced CDP acquired using modern acquisition devices.

In doing so, this work also demonstrates the limitations of classical authentication schemes in the context of modern counterfeiting attacks. By comparing the older approaches and the new ones, this thesis provides a comprehensive description of the challenges faced by this technology as of today and proposes new tools and approaches that help face and understand these problems.

1.5 Thesis Outline

The CDP technology represents a very interesting tradeoff in the field of Physical Object Security. By being a very cheap and easy-to-integrate solution on the packaging of many products, it proposes a very simple yet strong solution to the problem of product counterfeiting. Recent researches [43] however, have shown that the standard approaches to CDP security [16] can be fooled using advanced deep-learning binarization attacks, and creates a potential breach in the security of such frameworks. This weakness calls for a better understanding of the printing-imaging process being used in the CDP technology in order to strengthen the ability of authentication systems to detect such high quality fakes. This thesis is organized around the problem of creating such a theoretical framework for the description of the printing-imaging channel pipeline of CDP. It approaches this problem in two different ways:

- Chapter 2 provides all necessary foundations for this thesis.
- Chapter 3 is based on a deep learning framework, benefitting from all recent advancements in synthetic image generations, to achieve reliable simulations of the printing-imaging process.
- Chapters 4, 5 and 6 describe the problem using a probabilistic model inspired by information theory.

The approaches presented in the thesis are also studied with the goal of introducing systematic frameworks for the development of new authentication algorithms, enhancing the ability to discriminate original and fake CDP for the most powerful attacks known as of today.

In the last 10 years, the whole field of computer vision has seen tremendous changes with the advent of deep learning for the generation of artificial images [54, 55, 56, 57, 58, 59]. Although these deep models act as black-boxes and are thus theoretically unpredictable, their staggering capacity at rendering realistic images largely balances this drawback and inspired us to explore their capabilities when used in a verification pipeline. In Chapter 3, the printing-imaging channel is modelled using a deep-learning based approach, mimicking the process with an encoder-decoder architecture based on Unet. This model is used to explore the TURBO framework [60], and computes a collection of metrics based on the latent representations of these deep networks. This chapter also studies the usage of autoencoders to generate digital augmentations and measures the quality of the images generated by comparing them to the originals. Finally, these digital augmentations are being used to train deep net two-class classifiers and studies the ability of such networks to correctly discriminate originals and fakes.

Having studied the limitations of Deep Learning based architectures to distribution shift [61] and its intrinsic impossibility to theoretically ensure reliability, we turned to a more traditional statistical approach for the description of the printing-imaging channel of CDP. Based on statistical studies of bit-flipping and exploiting well-defined redundancies in CDP, Chapter 4 describes the Binary Pattern-based Channel (BPC) model, a probabilistic model inspired by the Binary Symmetric Channel model from Information Theory. Extending the model by introducing the notion of pattern-based reliability, this Chapter introduces a formal mathematical description of the printing-imaging channel. The BPC model provides a new paradigm for the analysis of CDP authentication strengths and weaknesses, introducing new measures and statistics such as the reliability codebook and the Posterior Log-Likelihood (PLL) score. The Chapter also demonstrates the soundness of this approach by exploring the stability of the pattern-reliability across various datasets of original and fake CDP.

In Chapter 5, the mathematical model introduced in Chapter 4 is used to develop two novel interpretable authentication schemes for CDP. Introducing the notion of a trained reference codebook, the Chapter shows how to use the PLL Score and the notion of reliable patterns to strengthen the authentication of CDP. This framework is then tested on various datasets of CDP with different acquisition devices and we show that this model is powerful enough to compete against the best attacks known as of today.

In Chapter 6, we further investigate the properties of the BPC model and its application for authentication purposes. Indeed, the authentication schemes presented in Chapter 5 called for a more in-depth study of their theoretical optimality. The authentication based on reliable patterns is thus studied in a more systematic way and linked to well-known statistical tests in order to obtain theoretical guarantees for the authentication of CDP. In

this regard, we deepen the statistical study of the mathematical model presented in Chapter 4 and develop a theoretical tool that allows to optimally discriminate between originals and fakes for each type of pattern that appears in the CDP. We also introduce a terminology to describe different linear aggregation strategies to arrive to a final decision score. These theoretical considerations are then tested on a real dataset of CDP and a comparison is done between theoretical predictions and empirical results.

Finally, Chapter 7 presents the conclusions that we draw from this study and proposes various directions of work for further developing the security of the CDP printing-imaging scheme.

1.6 Contributions

The work presented in this thesis constitutes the basis of corresponding publications:

- Chapter 3 contributed to the submission of a patent [60], four conference papers [61, 62, 63, 64] and a journal paper [65];
- Chapters 4 and 5 formed a base for a patent [66], a conference paper [67], which got the best paper award at IEEE WIFS 2022, and a journal paper [68];
- Chapter 6 is based on the work [69], which got the best paper award at IEEE WIFS 2024.

The details on all papers and patents produced in the scope of this PhD thesis can be found in Appendix A. The main contributions of this thesis to the field of Physical Object Security based on CDP technology are:

Chapter 3

- Applying a Deep-Learning based encoder-decoder architecture to describe and mimick a complex printing-imaging channel.
- Using the encoder-decoder architecture to strengthen authentication schemes based on multimetric aggregation.
- Generating synthetic augmentations of CDP based on various datasets of original and fake CDP.
- Assessing the quality of the generated synthetic augmentations by introducing weighted training objectives and hand-crafted functions for synthetic twins detection.
- Proposing a one-class authentication scheme based on two-class classifiers trained on original and synthetic samples.

Chapter 4

- Introducing a mathematical model which describes the printing-imaging process of CDP authentication.
- Validating the proposed model on an extensive amount of experimental data acquired by various imaging devices.
- Introducing the concept of pattern-based reliability codebooks drawn from a local characterization of CDP.
- Introducing a novel score for CDP called the Posterior-Log-Likelihodd (PLL) to evaluate the quality of a given CDP.
- Demonstrating the validity of the pattern-based reliability concept and the PLL score experimentally.

Chapter 5

- Proposing a new authentication scheme for CDP based on a reference codebook.
- Developing an authentication scheme based on the PLL metric.
- Developing an authentication scheme based on reliable patterns.
- Testing the authentication schemes on a real dataset and demonstrating the superiority of these new approach compared to classical authentication approaches.

Chapter 6

- Investigating the optimal authentication performance of Hamming distance per pattern in the BPC model.
- Developing a theoretical optimality objective based on a likelihood maximization approach.
- Deriving an alternative description of the PLL metric as a cross-entropy.
- Describing the optimality criterion using the Neyman-Pearson ratio test.
- Demonstrating the equivalence of the two approaches both theoretically and empirically.
- Proposing a generic description of channel aggregations to compute a final decision score for CDP authentication.
- Testing the different aggregations techniques on challenging datasets of CDP.

Chapter 2

Background Mathematical Knowledge

This chapter serves as a reference for the notations and concepts used throughout the Thesis. Each section introduces the fundamental notions that are used to describe the theorems, algorithms and tools highlighted in the title.

Section 2.1 presents the fundamentals of probability theory with key notions such as conditional probability, random variables and standard probability distributions. These notions are then used to formulate the rigorous notions of statistical estimators, binary hypothesis testing, bayesian approach and Receiver Operating Characteristic (ROC) curve and recall some well-known results, such as the law of large numbers, Chebyshev's inequality and Neymann-Pearson lemma. Section 2.3 describes a few notions drawn from information theory and Markov chains mostly following the material presented in [70].

2.1 Basics of Probability Theory

2.1.1 Probability Space, Events, Conditional Probability and Independance

A *probability space* is a triplet $(\mathcal{S}, \mathcal{E}, \mathbb{P})$ where:

- $\mathcal{S}$ is a set, called the *universe*. Elements $s \in \mathcal{S}$ are called *outcomes*;
- $\mathcal{E} \subset \mathcal{P}(\mathcal{S})$ is a set of subsets of $\mathcal{S}$, called the *set of events*;
- $\mathbb{P} : \mathcal{E} \to [0, 1]$ is a function, called the *probability measure*.

For a given event $E \in \mathcal{E}$, we call $\mathbb{P}(E) \in [0, 1]$ the *probability of event* E. For these definitions to make sense, we require additional properties of these objects namely:[1]

[1]The definitions and properties presented in this section are not fully formal. The goal is to give a general idea of the definitions and concepts that are used throughout the thesis. A rigorous basis for probability theory was introduced by Kolmogorov in [71].

1. *Total probability:* $\mathbb{P}(\mathcal{S}) = 1$;
2. *Disjoint events:* $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ if $A \cap B = \emptyset$.

Given two events $A, B \in \mathcal{E}$, we say that these events are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B). \tag{2.1}$$

If $\mathbb{P}(B) \neq 0$, we can introduce the *conditional probability* of event A knowing event B. It is given by :

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \tag{2.2}$$

In particular, if A and B are independent, $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

Baye's theorem, which is at the core of the bayesian approach presented in Section 2.2, allows to switch the role of A and B in the conditional probability:

Theorem 2.1.1 (Baye's theorem)**.** *Let $A, B \in \mathcal{E}$ be two events with $\mathbb{P}(A)$, $\mathbb{P}(B) \neq 0$. Then,*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)}{\mathbb{P}(B)}. \tag{2.3}$$

Example : Handwritten Digits

A popular dataset heavily studied in Machine Learning is the so called Modified National Institute of Standards and Technology (MNIST) database introduced in [72]. It consists of $70'000$, 28×28 pixels, 8-bit, grayscale images representing handwritten digits between 0 and 9. The universe describing these images can naturally be taken as

$$\mathcal{S}_{\text{MNIST}} = \{0, 1, ..., 255\}^{28 \times 28}$$

with most of the outcomes $s \in \mathcal{S}_{\text{MNIST}}$ representing random noise and having probability zero. The question of how to optimally design a probability distribution $\mathbb{P} : \mathcal{P}(\mathcal{S}_{\text{MNIST}}) \to [0, 1]$ remains largely an open question. A related question, asks whether the events

$$A = \{s \in \mathcal{S} : s \text{ represents a } 1\};$$

and

$$B = \{s \in \mathcal{S} : s \text{ represents a } 7\};$$

are disjoint. Intuitively, this question is unclear as handwritten 1's and 7's can be confusingly close.

2.1.2 Random Variables, Expectation, Variance

A *random variable* is a function

$$X : \mathcal{S} \to \mathbb{R} \tag{2.4}$$

which associates to each outcome $s \in \mathcal{S}$, a real value $X(s)$. Conceptually, a random variable maps a (potentially) complex probability space into a single real-valued measure. As such, a random variable has the effect of reducing the complexity and, if chosen correctly, extracts valuable insights on the random process underlying its distribution. A *random vector* is a tuple $\mathbf{X} = (X_1, ..., X_n)$, where each X_i is a random variable.

A random variable X is called *discrete* if the subset $X(\mathcal{S}) \subset \mathbb{R}$ of possible values taken by X is a discrete subset of $\mathbb{R}$. In this case, to each discrete value $x_1, ..., x_n \in X(\mathcal{S})$, we associate the probability value

$$p_i = \mathbb{P}(X = x_i) = \mathbb{P}(\{s \in \mathcal{S} : X(s) = x_i\}) \in [0, 1] \tag{2.5}$$

and we define the *probability mass function (PMF)* $p_X : \mathbb{R} \to [0, 1]$:

$$p_X(x) = \begin{cases} p_i & \text{if } x = x_i, \\ 0 & \text{otherwise.} \end{cases} \tag{2.6}$$

The random variable X is called *continuous* if $X(\mathcal{S})$ is a sub-interval of $\mathbb{R}$ and there exists a function $f_X : \mathbb{R} \to \mathbb{R}_+$ such that the following equation holds for every $a, b \in \mathbb{R}$ with $a < b$:

$$\mathbb{P}(X \in [a, b]) = \int_a^b f_X(x)dx \tag{2.7}$$

The function f_X is called the *probability density function (PDF)* of X.

In practice, it is often useful to further simplify a random variable X to a single mean value $\mathbb{E}[X] \in \mathbb{R}$ called the *mathematical expectation*. It is defined by the formula:

$$\mathbb{E}[X] = \sum_i x_i \cdot p_i, \tag{2.8}$$

if X is discrete and,

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)dx, \tag{2.9}$$

if X is continuous. Similarly, one introduces the notion of *mathematical variance*:

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \in \mathbb{R}_+, \tag{2.10}$$

and *standard deviation*:

$$\sigma_X = \sqrt{\mathrm{Var}(X)}. \tag{2.11}$$

Finally, given two random variables X and Y, one can measure their *covariance* and *Pearson's correlation*:

$$\mathrm{Cov}(X,Y) = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right], \tag{2.12}$$

$$\mathrm{pcor}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}. \tag{2.13}$$

Example : MNIST Digit Authentication

1. Going back to the MNIST database example discussed above, we can define a discrete random variable
$$X : \mathcal{S}_{\mathrm{MNIST}} \to \{0,1,2,3,4,5,6,7,8,9\}$$
mapping each image to the number it represents (its *label* using the machine-learning vocabulary). The question of building such a mapper has been the topic of many researches in the past few years and this problem has proven to be mostly solvable using deep-learning classifiers. In general all the models make the assumption that the distribution of X is uniform:
$$p_X(i) = 1/10, \text{ for } i = 0,1,...,9$$

2. In practice, the mapping is often implemented by a continuous random vector $\mathbf{Y} \in [0,1]^{10}$ with the additional condition that $\sum_i Y_i = 1$. In this model, $Y_i(s) \in [0,1]$ measures the probability that the handwritten digit $s \in \mathcal{S}$ represents the label i.

2.1.3 Common Probability Distributions

In the zoo of all possible distributions of random variables $X : \mathcal{S} \to \mathbb{R}$, we describe a few of them that arise very often in statistical studies and will prove useful in our models:

- *Bernoulli distribution:* A binary-valued random variable $X : \mathcal{S} \to \{0,1\}$ is said to follow a Bernoulli distribution of parameter $p \in [0,1]$, written $X \sim Bern(p)$. The parameter p is determined by the relation $p = \mathbb{P}(X = 1)$.
- *Binomial distribution:* A discrete random variable $X : \mathcal{S} \to \{0,1,...,n\}$ is said to follow a binomial distribution $X \sim Binom(n,p)$ if $X = X_1 + ... + X_n$, where each X_i is independent and follows a Bernoulli distribution of parameter p. Its probability mass function is given by
$$p_X(k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}. \tag{2.14}$$

- *Normal distribution:* A continuous random variable X is said to follow a normal distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 \in \mathbb{R}_+$ if its PDF is given by:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \tag{2.15}$$

2.2 Statistical Hypothesis Testing

2.2.1 Statistics, Parameters, Estimators

Mathematical statistics appear to be the inverse process to what is described in section 2.1. Probability theory studies well-defined models based on a given probability space $(\mathcal{S}, \mathcal{E}, \mathbb{P})$ and demonstrates properties of this model, through the definition of useful random variables. Statistics, on the other hand, start with:

- a set of *samples* or *observations* $x_1, ..., x_L \in \mathbb{R}$;
- a parametric probability space $(\mathcal{S}, \mathcal{E}, \mathbb{P}_\theta)$ whose definition relies on (yet unknown) *parameters* $\theta \in \mathbb{R}^\Lambda$;
- a random variable $X : \mathcal{S} \to \mathbb{R}$.

The problem statistics is trying to solve is the following: Find optimal parameters θ such that, if we were to sample $X_1, ..., X_L$ according to the law $\mathbb{P}_\theta$ independently, the quantity $\mathbb{P}_\theta(X_i = x_i)$ (or, in the continuous case, $f_{X_i}(x_i)$) would be jointly maximized for all samples x_i. An illustrative example is given in Fig. 2.1 where the parameters (μ, σ^2) of a normal distribution are being estimated from empirical samples.

Based on this motivation we give the following definitions:

- Given a probability space $(\mathcal{S}, \mathcal{E}, \mathbb{P}_\theta)$ and identical and independent random variables X_1, ..., X_L, a *statistic* for this model is any function $\Sigma : \mathbb{R}^L \to \mathbb{R}$.
- A *parameter* of the probability space is any fixed vector $\theta \in \mathbb{R}^\Lambda$ that is used to define the model.
- An *estimator* for a given parameter θ is any statistic Σ such that

$$\theta \approx \hat{\theta}_L = \Sigma(X_1, ..., X_L) \tag{2.16}$$

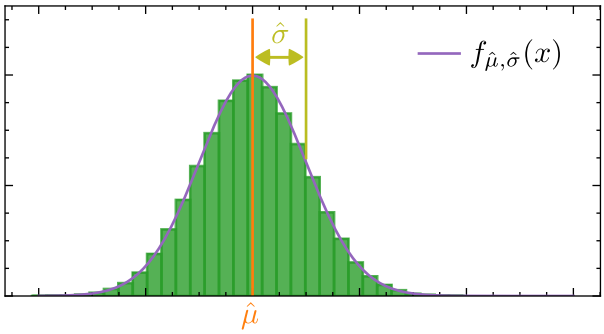


Figure 2.1. Illustration of a statistical estimator for a gaussian distribution (in violet) based on random samples visualized as a histogram (in green).

The precise meaning of the approximation sign in Eq. (2.16) is not as trivial as it may first seem. Indeed, $\hat{\theta}_L : \mathcal{S} \to \mathbb{R}^\Lambda$, is itself a random vector and the way we measure its convergence to a fixed value θ actually gives rise to distinct definitions in statistics. We introduce a few of the terminology here:[2]

- An estimator $\hat{\theta}_L$ is *strongly consistent* if it converges *almost-surely*:

$$\mathbb{P}\left(\lim_{L\to+\infty} \hat{\theta}_L = \theta\right) = 1. \tag{2.17}$$

- An estimator $\hat{\theta}_L$ is *unbiased* if it converges in *mean*:

$$\lim_{L\to+\infty} \mathbb{E}[\hat{\theta}_L] = \theta. \tag{2.18}$$

We now describe two different ways of producing statistical estimators. We start similarly with a probability space $(\mathcal{S}, \mathcal{E}, \mathbb{P}_\theta)$, where the probability distribution depends on a fixed parameter $\theta \in \mathbb{R}^\Lambda$. We consider iid random variables $X_1, ..., X_L$ corresponding to independent observations $x_1, ..., x_L$.

The *maximum likelihood estimator* (MLE) for the parameter θ is defined by :

$$\hat{\theta}_L^{MLE} = \underset{\theta\in\mathbb{R}^\Lambda}{\operatorname{argmax}} \prod_{i=1}^{L} \mathbb{P}_\theta(X_i = x_i). \tag{2.19}$$

The *Bayesian approach* for the estimation of the parameter θ is a little bit different. It starts of by considering the parameter $\theta \in \mathbb{R}^\Lambda$ as a random parameter, following a *prior distribution* given by a probability density function $f_\theta(s)$. It then uses Baye's Formula to derive the posterior likelihood pdf:

$$f_{\theta|X_1,...,X_L}(s|x_1, ..., x_L) = \frac{f_{X_1,...,X_L|\theta}(x_1, ..., x_L|s) \cdot f_\theta(s)}{f_{X_1,...,X_L}(x_1, ..., x_L)} = \frac{\prod_{i=1}^{L} f_{X_i|\theta}(x_i|s) \cdot f_\theta(s)}{f_{X_1,...,X_L}(x_1, ..., x_L)}.$$

[2] We refer to [73] for a detailed study of these notions.

The Bayesian approach then consists in maximizing the nominator:

$$\hat{\theta}_L^{Bayes} = \underset{s\in\mathbb{R}^\Lambda}{\operatorname{argmax}} \prod_{i=1}^{L} f_{X_i|\theta}(x_i|s) \cdot f_\theta(s). \tag{2.20}$$

There are many results in probability that refer to the convergence of sequence of random variables and give asymptotic results for the distribution of their limit. These results are particularly useful in statistics studies as they help justify the convergence of statistical estimators to their expected limits. We cite here a few of these results that will be useful for the proof of the consistency of various estimators that we'll meet later in this work. The first one tells us that most of the distribution of a random variable X is concentrated around its mean with a radius controlled by its variance:

Theorem 2.2.1 (Chebyshev's Inequality)**.** *Let X be a random variable with mean $\mu = \mathbb{E}[X]$ and finite variance $Var(X)$. Then, for every real positive number $a > 0$,*

$$\mathbb{P}(|X - \mu| > a) \leq \frac{Var(X)}{a^2}. \tag{2.21}$$

The second one is a famous result of convergence for a sequence of iid random variables:

Theorem 2.2.2 (Strong Law of Large Numbers)**.** *Let $(X_n)_{n=1}^{\infty}$ be a sequence of iid random variables with common finite mean $\mu = \mathbb{E}[X_1]$. Then, the sequence*

$$\mu_L = \frac{1}{L}\sum_{n=1}^{L} X_n, \tag{2.22}$$

converges almost surely to μ.

This result has a direct consequence that we will be using a lot:

Corollary 2.2.1. *μ_L is a consistent estimator for μ.*

Example: Binary Random Codes

The model for this example is based on random binary codes $\mathbf{T}$ of size L, with each entry of $\mathbf{T}$ being sampled independently according to a Bernoulli distribution with parameter p. We can describe the probability space: $\mathcal{S} = \{0,1\}^L$, $\mathcal{E} = \mathcal{P}(\mathcal{S})$ and,

$$\mathbb{P}_p(\mathbf{T}) = p^{L_+} \cdot (1-p)^{L_-}, \text{ where } L_+ = \sum_{i=1}^{L} T_i \text{ and } L_+ + L_- = L.$$

L_+ (respectively L_-), counts the number of symbol equal to 1 in $\mathbf{T}$ (respectively equal to 0 in $\mathbf{T}$). In this model, p is a parameter. Using the MLE approach, we can prove that

$$\hat{p}_L^{MLE} = \Sigma(T_1, ..., T_L) = \frac{1}{L}\sum_{i=1}^{L} T_i; \tag{2.23}$$

and this estimator is consistent by corollary 2.2.1 since $p = \mathbb{E}[T_1]$.

2.2.2 Statistical Test, ROC Curves, Neymann-Pearson Lemma

The point of view of statistical hypothesis testing is a little bit different from the discussion done in the previous section. Its goal is to introduce tools and terminology in order to decide between different hypotheses by comparing their respective likelihood. The departure point is similar to the preceding section with:

- a set of *samples* or *observations* $x_1, ..., x_L \in \mathbb{R}$;
- a parametric probability space $(\mathcal{S}, \mathcal{E}, \mathbb{P}_\theta)$ depending on *parameters* $\theta \in \mathbb{R}^\Lambda$;
- a random variable $X : \mathcal{S} \to \mathbb{R}$.

Introducing some terminology, we call a *(statistical) hypothesis*, any subset $\mathbb{H} \subset \mathbb{R}^\Lambda$. In most scenarios, we will test two mutually disjoint hypotheses:

- a *null hypothesis* $\mathbb{H}_0 \subset \mathbb{R}^\Lambda$, representing our initial guess of parameters;
- an *alternative hypothesis* $\mathbb{H}_1 \subset \mathbb{R}^\Lambda$ representing an alternate explanation.

A *statistical test* is a couple (Σ_0, R), where $\Sigma_0(X_1, ..., X_L) \in \mathbb{R}$ is a statistic and $R \subset \mathbb{R}$ is a subset called the *rejection region* for the null hypothesis. The choice of the couple (Σ_0, R) is paramount as they serve as indicators for the choice of hypothesis:

- if $\Sigma_0(x_1, ..., x_L) \notin R$, we say that we accept hypothesis $\mathbb{H}_0$;
- otherwise, if $\Sigma_0(x_1, ..., x_L) \in R$ we reject hypothesis $\mathbb{H}_0$ and accept hypothesis $\mathbb{H}_1$.

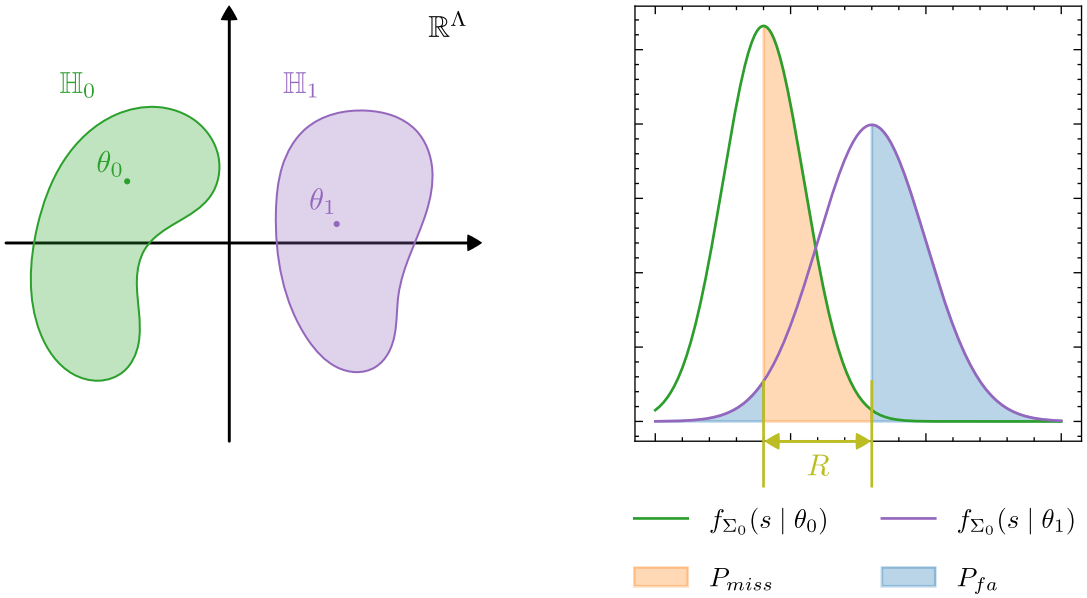


Figure 2.2. Binary hypothesis testing illustration for a statistical test Σ_0 under two competing hypotheses $\mathbb{H}_0$ and $\mathbb{H}_1$. The rejection region R is illustrated together with the associated probability of miss and probability of false acceptance.

Intuitively, this statistical test works as an indicator which tells us what choice of parameter best explains our set of observations $x_1, ..., x_L$. All this theory naturally drives to the following question: Given a statistical test (Σ_0, R), how good does it perform when trying to decide between two competing hypotheses $\mathbb{H}_0$ and $\mathbb{H}_1$ for a given set of observations ? Once again, as we are working with random processes, this question is less trivial than it first seems. We define:

- the *type I error*, also called the *probability of false acceptance* which is the probability that samples following hypothesis $\mathbb{H}_1$ get accepted by the test:

$$\epsilon^{\mathrm{I}} : \begin{array}{l} \mathbb{H}_1 \to [0,1] \\ \theta_1 \mapsto \mathbb{P}_{\theta_1}\left(\Sigma_0(X_1, ..., X_L) \notin R\right) =: P_{fa}(\theta_1, R); \end{array} \tag{2.24}$$

- the *type II error*, also called the *probability of miss*, which is the probability that samples following hypothesis $\mathbb{H}_0$ get rejected by the test:

$$\epsilon^{\mathrm{II}} : \begin{array}{l} \mathbb{H}_0 \to [0,1] \\ \theta_0 \mapsto \mathbb{P}_{\theta_0}\left(\Sigma_0(X_1, ..., X_L) \in R\right) =: P_{miss}(\theta_0, R). \end{array} \tag{2.25}$$

A good statistical test should minimize both errors simultaneously. These notions are illustrated in Fig. 2.2.

In order to measure the quality of a statistical test, we present here the celebrated Neymann-Pearson framework introduced in [74]. As described above, the goal is to minimize

both ϵ^{I} and ϵ^{II} by choosing the best performing statistical test (Σ_0, R). As demonstrated in their article, minimizing both errors actually leads to a dead-end as minimizing one error will eventually lead to increase the other error. In order to alleviate this apparent contradiction, they introduce an imbalance between both hypotheses so that type I error is far more harmful than type II error and should be controlled independently.[3] In this regard, they fix the type I error to stay below a certain threshold and say that a statistical test respects a *significance level* $\alpha \in [0, 1]$ if:

$$\sup_{\theta_1 \in \mathbb{H}_1} \epsilon^{\mathrm{I}}(\theta_1) \leq \alpha. \tag{2.26}$$

The new formulation for an optimal statistical test then reads: Given a significance level α for the type I error, which statistical test $(\Sigma_0^\alpha, R^\alpha)$ minimizes the type II error ? The answer for simple tests is given by the celebrated Neymann-Pearson ratio test: Let $(\mathcal{S}, \mathcal{E}, \mathbb{P}_\theta)$ be a parametric probability space with $\theta \in \mathbb{R}^\Lambda$, $X_1, ..., X_L$ independent and identical random variables and $\mathbb{H}_0 = \{\theta_0\}$, $\mathbb{H}_1 = \{\theta_1\}$ two competing hypotheses and $\alpha \in [0, 1]$ a fixed significance level. Writing $f_\theta(x)$ the PDF of the random variable X_i, the optimal test is given by the ratio of likelihoods:

$$\mathcal{R}(X_1, ..., X_L) = \prod_{i=1}^{L} \frac{f_{\theta_0}(X_i)}{f_{\theta_1}(X_i)}, \tag{2.27}$$

with a rejection region $R = (\tau, +\infty)$, where $\tau \in \mathbb{R}$ is chosen so that

$$\epsilon^{\mathrm{I}}(\theta_0) = \mathbb{P}_{\theta_0}(\mathcal{R}(X_1, ..., X_L) \in R) \leq \alpha.$$

The optimality is given by the Neymann-Pearson lemma:

Lemma 2.2.1 (Neymann-Pearson Lemma)**.** *Let (Σ_0, R) be any statistical test with a significance level α for the binary hypothesis testing problem described above. Then,*

$$\epsilon^{II}(\theta_0, \Sigma_0) \geq \epsilon^{II}(\theta_0, \mathcal{R}) \tag{2.28}$$

where $\mathcal{R}$ is the likelihood ratio test described above. This means that the likelihood ratio test is the most powerful test for a given significance level α.

Example: The Receiver Operating Characteristic (ROC) Curve

In the case of simple tests, where both hypotheses are singletons: $\mathbb{H}_0 = \{\theta_0\}$ and $\mathbb{H}_1 = \{\theta_1\}$ and the rejection region is an open interval $R = (\tau, +\infty)$, depending on a threshold $\tau \in \mathbb{R}$,

[3] Note that in practice, it is very common to have one type of error much more harmful than the other, so that both hypotheses are naturally imbalanced. For instance, when dealing with drug counterfeiting, you really want to control the false acceptance rate, so that no fake drugs is actually given to the consumer (type I error), while throwing away a few genuine drugs is acceptable in comparison (type II error).

the type I and type II error become simple functions of the threshold τ and they can be represented on a 2D graph called the Receiver Operating Characteristic (ROC) curves [75]:

$$\mathrm{ROC}(\tau) = (P_{fa}(\tau), 1 - P_{miss}(\tau)). \tag{2.29}$$

This curve has the property to be a non-decreasing function of τ, passing through point $(0, 0)$ when $\tau \to -\infty$ and point $(1, 1)$ when $\tau \to +\infty$. In the worst case scenario, both distributions $\mathbb{P}_{\theta_0}$ and $\mathbb{P}_{\theta_1}$ are identical, in which case $P_{fa}(\tau) = 1 - P_{miss}(\tau)$ and the ROC curve coincides with the diagonal line $x = y$. In the best case scenario, the support of both distributions is completely disjoint with distribution $\mathbb{P}_{\theta_0}$ having its support to the left of $\mathbb{P}_{\theta_1}$. In this case the ROC curve corresponds to the union of two segments $\{0\} \times [0, 1] \cup [0, 1] \times \{1\}$.

To further simplify the comparison of results based on ROC curves, there is two standard metrics associated. Given a ROC curve $ROC(\tau)$:

1. *AUC score:* The Area Under Curve (AUC) is the measure of the area under the ROC curve. This area is $1/2$ in the worst case and 1 in the best case. A bigger AUC score is generally the sign of a better separability.
2. P_{err}^{min} *score:* For each fixed threshold τ, we can compute an average probability of error:

$$P_{err}(\tau) := \frac{P_{miss}(\tau) + P_{fa}(\tau)}{2}, \tag{2.30}$$

which measures the average amount of misclassified samples. The minimal value of this function is a good indicator of the actual capacity of a given statistical test at separating two hypotheses:

$$P_{err}^{min} := \min_{\tau} P_{err}(\tau). \tag{2.31}$$

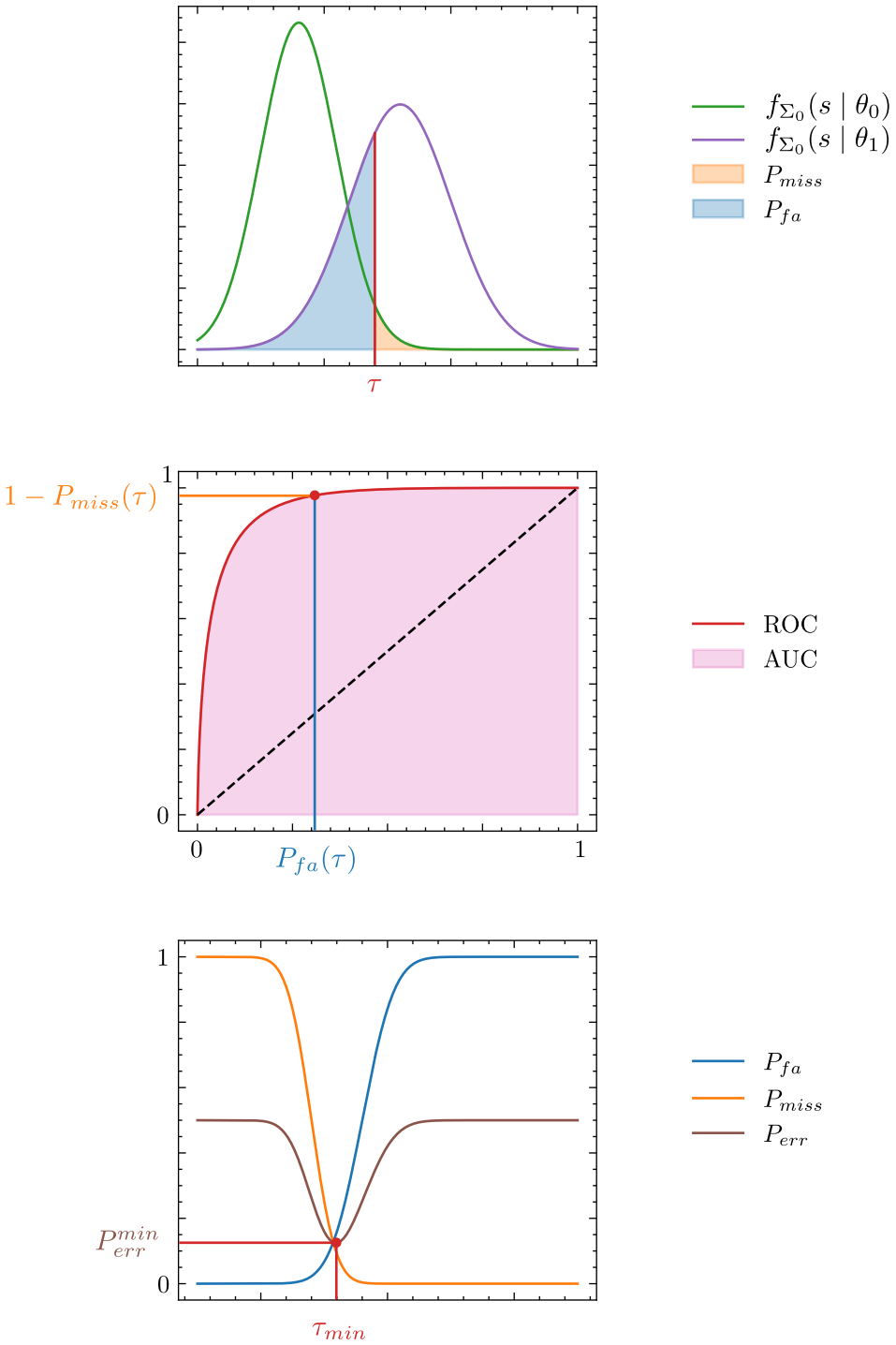


Figure 2.3. Illustration of the ROC curve construction. The first plot shows the pdf of two distributions and the measure of the two probabilities of error for a fixed threshold τ. The second plot shows the ROC curve and its associated AUC score. The third plot shows the computation of the average error and its minimization with respect to τ.

2.3 Information Theory and Markov Chains

2.3.1 Entropy, Mutual Information, KL-Divergence

We present in this section some concepts drawn from information theory which can be seen as an alternative viewpoint on statistics and allows to measure the quality of a random system in terms of information preservation.

Given a probability space $(\mathcal{S}, \mathcal{E}, \mathbb{P})$ and a discrete random variable $X : \mathcal{S} \to \{x_1, ..., x_n\}$, we define the *entropy* of X by:

$$\mathcal{H}(X) = -\sum_{i=1}^{n} p_i \cdot \log p_i, \tag{2.32}$$

where, $p_i = \mathbb{P}(X = x_i)$. This quantity measures the amount of randomness of the random variable X as shown by the following observations:

1. $0 \leq \mathcal{H}(X)$, with equality if and only if X is constant;
2. $\mathcal{H}(X) \leq \log(n)$ with equality if and only if X is uniformly distributed;
3. If $Y = f(X)$, for any function $f : \mathbb{R} \to \mathbb{R}$, then $\mathcal{H}(Y) \leq \mathcal{H}(X)$, with equality if and only if f is a bijection.

In this sense, $\mathcal{H}(X)$ is a measure of the average amount of information contained in the events $\{X = x_i\}$. The last result shows that any processing of the random variable X can only decrease the amount of information it contains.

The question we would like to adress now is the following: Given two random variables X, Y, which one contains more information and how much of the information is shared between both variables ? In order to answer this question, we introduce the *joint entropy*:

$$\mathcal{H}(X, Y) = -\sum_{i,j} p_{X,Y}(x_i, y_j) \cdot \log p_{X,Y}(x_i, y_j), \tag{2.33}$$

and the *conditional entropy*:

$$\mathcal{H}(X \mid Y) = -\sum_{i,j} p_{X,Y}(x_i, y_j) \cdot \log p_{X|Y}(x_i \mid y_j). \tag{2.34}$$

These two quantities measure respectively the total amount of information contained in both random variables and the amount of information left in X when we already have informations on Y. These quantities have the following properties:

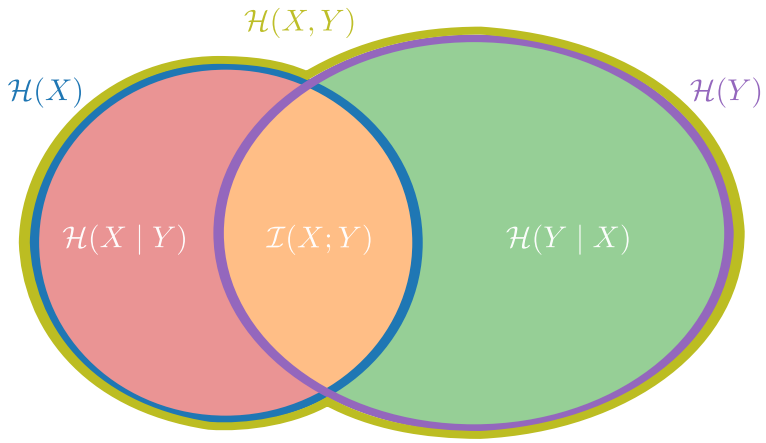


Figure 2.4. Venn diagram illustration of the relations between entropy, conditional entropy and mutual information.

1. $\mathcal{H}(X,Y) \leq \mathcal{H}(X) + \mathcal{H}(Y)$ with equality if and only if X and Y are independent.
2. $0 \leq \mathcal{H}(X \mid Y)$ with equality if and only if $X = f(Y)$ for a fixed function f.
3. $\mathcal{H}(X \mid Y) \leq \mathcal{H}(X)$ with equality if and only if X and Y are independent.

The last inequality can be seen as a measure of how much information is contained in X that is not contained in Y. Following this intuition, we define the *mutual information* between X and Y by:

$$\mathcal{I}(X;Y) = \mathcal{H}(X) - \mathcal{H}(X \mid Y). \tag{2.35}$$

All these quantities can be nicely illustrated using a classical Venn Diagramm like in Fig. 2.4 where each sets represents the amount of randomness contained in each random variable. For instance, the entropy of X is reduced (on average) to $\mathcal{H}(X \mid Y)$ once we have information about the actual value of Y (the size of the red set is strictly smaller than the blue circle).

Yet another way of measuring the difference between two random variables X and Y is the *Kullback-Leibler divergence*. Its definition relies on the probability mass functions p_X and p_Y:

$$\mathcal{D}_{KL}(p_X \parallel p_Y) = \sum_i p_X(i) \cdot \log \frac{p_X(i)}{p_Y(i)}. \tag{2.36}$$

This definition is closely related to the *cross-entropy*:

$$\mathcal{H}(X;Y) = -\sum_i p_X(i) \cdot \log p_Y(i). \tag{2.37}$$

For these definitions to work, we need the distribution p_Y to be *absolutely continuous* with respect to p_X that is, if $p_X(i) = 0$, then $p_Y(i) = 0$. Note that we use the convention: $0 \cdot \log 0 = 0$. The Kullback-Leibler divergence has the following properties:

1. $0 \leq \mathcal{D}_{KL}(p_X \parallel p_Y)$ with equality if and only if $p_X = p_Y$;

2. $\mathcal{D}_{KL}(p_X \parallel p_Y) = \mathcal{H}(X;Y) - \mathcal{H}(X)$;
3. $\mathcal{D}_{KL}(p_{X,Y} \parallel p_X \cdot p_Y) = \mathcal{I}(X;Y)$.

2.3.2 Markov Chains, Data Processing Inequality

A particular case that will be useful is the following setup involving three random variables $X, Y, Z : \mathcal{S} \to \mathbb{R}$. We say that X, Y, Z form a *Markov chain* written:

$$X \to Y \to Z, \tag{2.38}$$

if the random variables X and Z are independent, conditionally on Y:

$$p_{X,Z|Y}(x, z \mid y) = p_{X|Y}(x \mid y) \cdot p_{Z|Y}(z \mid y). \tag{2.39}$$

Another way of writing this is:

$$p_{Z|X,Y}(z \mid x, y) = p_{Z|Y}(z \mid y), \tag{2.40}$$

that is, we don't need any knowledge about X to determine the randomness of Z once we have information about Y (we also call this a *memoryless process*). Intuitively, a Markov chain can be seen as succession of processing steps:

- $Y = f(X, \delta)$,
- $Z = g(Y, \epsilon)$,

where f, g are deterministic functions and δ, ϵ are random variables which are mutually independent and independent of X, Y and Z.

A celebrated result in information theory called the *data processing inequality* states the following:

Theorem 2.3.1 (Data Processing Inequality)**.** *Let $X \to Y \to Z$ be a Markov chain. Then,*

$$\mathcal{I}(X;Y) \geq \mathcal{I}(X;Z) \tag{2.41}$$

This means that given a source of information X, any successive processing Y, Z can only decrease the amount of information that is shared with the source.

Chapter 3

Deep Learning Model of Printing-Imaging Channel

In this chapter, we study the problem of describing the Printing-Imaging framework of CDP using a deep learning encoder-decoder architecture and its applications to the problem of CDP authentication. Reliable authentication of CDP requires a statistical understanding of the full printing–imaging channel, including both genuine and fake distributions. Ideally, this would involve modeling the distributions of observed CDP data conditioned on original and fake CDP, enabling optimal hypothesis testing via a Neyman–Pearson framework. However, due to the complexity and variability of the mobile phone acquisition and printing processes, an explicit analytical model is intractable. Therefore, this chapter investigates machine learning-based approaches to approximate this channel and enable robust CDP authentication. Inspired by the multiple advances in deep learning models generation [56, 76] and their ability to produce high quality images mimicking actual real life photography and perform image domain translation, this chapter explores their ability to mimick the process of printing and acquiring digital CDP as well as inverting this process using deep learning based encoders and decoders. We study the stability of such systems and their reliability when used as a part of an authentication scheme. In a second part, we study the use of digital twins obtained via deep architecture as adversarial examples to train deep binary classifiers without having access to fake samples during training.

Parts of this work have been used as a basis in the following research articles [46, 64, 63, 43] and in the patent [60]. However, in contrast to these other researches, this chapter explores the full Turbo setup for the understanding and robustification of the CDP authentication against machine-learning based attacks.

3.1 Introduction

Modelling the full pipeline of CDP from printing to mobile phone image acquisition presents a great challenge due to the complexity and variability inherent in both the printing and imaging processes.

3.1.1 Complexity of the Printing and Imaging Pipeline

First, the process of deriving an exact mathematical model of the printing and image acquisition pipeline is particularly involved. The printing process itself relies on a variety of technologies, including inkjet, laser, offset, thermal, and other specialized mechanisms, each governed by different physical and chemical principles. Efforts to model printing technologies analytically have been ongoing since the 1980s, with limited success in fully capturing all contributing factors in a unified framework [19, 20, 23, 36].

Even assuming a theoretically complete model, the physical realization of the printing process remains fundamentally stochastic. There exists a high degree of randomness in the interaction between ink or toner and substrate (paper, polymer, metal, etc.). This randomness is influenced by numerous variables such as ink viscosity, substrate roughness, drying conditions, and capillary interactions. Capturing this stochasticity in a mathematically precise way is itself a grand challenge in statistical modelling and simulation.

The image acquisition stage introduces an additional layer of complexity. The behavior of mobile phone imaging systems depends on a large set of technologies such as:

- *Environmental conditions:* ambient light, shadows, specular reflections;
- *Technology of the imaging sensor:* CCD, CMOS;
- *Signal processing pipeline:* denoising, sharpening, tone mapping;
- *Post-processing operations:* demosaicing, histogram equalization, color correction;
- *Compression schemes:* JPEG or others.

Each of these factors significantly alter the final captured image, and their combination makes the imaging channel highly non-deterministic and device-dependent. Developing a generalized and transferable mathematical model for such a channel represents an independent field of research, and indeed has formed the basis of multiple research articles in signal and image processing [24, 25].

3.1.2 Data-Driven and Approximate Modelling: The TURBO Paradigm

In response to the difficulty of developping a fully analytical model, several research groups have turned to data-driven modeling strategies that approximate the printing–imaging channel using simulations informed by empirical observations. One influential line of work in this direction is the TURBO model [60]. This approach provides a practical, modular

approximation of the printing–imaging pipeline, and serves as a surrogate channel model in learning-based systems.

The TURBO model was subsequently extended and analyzed in a simulation-based framework in the IEEE Transactions on Information Forensics and Security (TIFS) article [65]. In this paper, the authors developed a learnable surrogate of the imaging pipeline, capturing essential transformations and distortions without explicitly modeling each hardware or physical step. This enables training of robust machine learning models that remain effective despite acquisition variability. However, this paper was limited in scope to demonstrating the viability of such simulation-based modeling. It did not provide real-world examples where this framework was developed further and applied to downstream applications such as CDP authentication.

3.1.3 Our Contributions

In this work, we explore this gap. Building upon the foundational idea of the learnable digital twin introduced in [65], we propose a concrete implementation of an encoder-decoder model for the entire printing–imaging channel. Our model retains the autoencoder structure but is tailored to support practical authentication tasks. Specifically, we evaluate its effectiveness in the context of CDP-based security systems, where the physical instance must be linked to its original design via robust and discriminative features, despite the distortions introduced by printing and mobile phone imaging.

3.2 Research Questions

Can the printing-imaging in a CDP authentication framework process be modelled using a deep learning based encoder-decoder architecture ?

Training a deep learning encoder-decoder architecture on a fixed dataset of paired digital template $\mathbf{t}$ and printed CDP $\mathbf{y}$, we use various metrics to evaluate the quality of the digital twins and the influence of various training losses. The results demonstrate the strengths and the limitations of this approach.

Can we use a trained encoder-decoder architecture to effectively enhance the authentication capability of CDP ?

Using a trained deep learning based encoder-decoder architecture on a reference dataset of CDP $\mathbf{x}_{ref}$, we study the application of this architecture to authentication of unknown probe CDP $\mathbf{y}$. By examining multiple metrics based on latent and reconstructed signals, we demonstrate the possible gain in performance of such an authentication scheme while also pointing out the great variability in model performance depending on the training initialisation.

Can synthetic augmentations of CDP be used to train deep classifiers at the task of discriminating between original and fake CDP ?

Using a trained deep learning based encoder-decoder architecture on a reference dataset $\mathbf{x}_{ref}$ to produce digital augmentations $\tilde{\mathbf{x}}$, we use these as adversarial examples to train deep binary classifiers and test them on a real dataset of probe original $\mathbf{x}$ and fake $\mathbf{f}$ CDP. Comparing this approach with a standard supervised training of binary classifiers, we show that synthetic augmentations offer an interesting solution to the training of classifiers without actual fakes. The performance of such classifiers is studied for various qualities of digital twins as well as the stability of the training process.

3.3 CDP Security as Encoder and Decoder

3.3.1 Binary Communication Channel Point of View

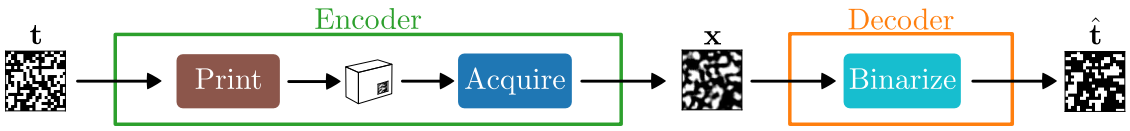


Figure 3.1. The CDP framework seen from an information theoretic viewpoint. A secret message $\mathbf{t}$ is encoded into a printed-scanned (noisy) version $\mathbf{x}$ and decoded back into an estimate message $\hat{\mathbf{t}}$.

The CDP security framework, as described in Section 1.3, uses a secret key based *template* that is represented by a binary image $\mathbf{t} \in \{0,1\}^{L\times L}$, where L^2 is the number of symbols in $\mathbf{t}$. The printing-imaging process, which relies on a pair of printer and scanner $(Print, Scan)$, produces a random CDP represented by a grayscale image $\mathbf{x} \in [0,1]^{kL\times kL}$ with $k \in \mathbb{N}^*$, a magnification factor depending on the resolution of the printing-imaging technology being used. The CDP $\mathbf{x}$ is a distorted and lossy representation of $\mathbf{t}$. The measure of the distortion from $\mathbf{t}$ to $\mathbf{x}$ is classically performed using standard metrics such as ℓ_1, ℓ_2, or Pearson correlation. Another classical approach consists in using a binarization algorithm such as Otsu's threshold [77] and measuring a Hamming distance between the binary template $\mathbf{t}$ and the estimated binarized version $\hat{\mathbf{t}}_{\mathbf{x}}$ obtained from the printed counterpart $\mathbf{x}$. We give here, for reference, the formulas that we use for the various metrics:[1]

1. Normalized ℓ_1 or ℓ_2 distance between the probe $\mathbf{y}$ and the template $\mathbf{t}$:

$$\ell_1(\mathbf{t},\mathbf{y}) = \frac{1}{L^2}\|\mathbf{t}-\mathbf{y}\|_1 = \frac{1}{L^2}\sum_{i,j}^{L}|t_{ij}-y_{ij}| \in [0,1] \tag{3.1}$$

$$\ell_2(\mathbf{t},\mathbf{y}) = \frac{1}{L^2}\|\mathbf{t}-\mathbf{y}\|_2 = \frac{1}{L^2}\sum_{i,j}^{L}(t_{ij}-y_{ij})^2 \in [0,1] \tag{3.2}$$

2. Pearson correlation between $\mathbf{t}$ and $\mathbf{y}$:

$$\mathrm{pcor}(\mathbf{t},\mathbf{y}) = \frac{\sum_{i,j}^{L} t_{ij}\cdot y_{ij} - \left(\sum_{i,j}^{L} t_{ij}\right)\cdot\left(\sum_{i,j}^{L} y_{ij}\right)}{\sqrt{\sum_{i,j}^{L} t_{ij}^2 - \left(\sum_{i,j}^{L} t_{ij}\right)^2}\cdot\sqrt{\sum_{i,j}^{L} y_{ij}^2 - \left(\sum_{i,j}^{L} y_{ij}\right)^2}} \in [-1,1] \tag{3.3}$$

[1]Note that if $k > 1$, we first upsample $\mathbf{t}$ before computing the metrics so that both $\mathbf{t}$ and $\mathbf{y}$ have the same dimensions.

3. Hamming distance between the template $\mathbf{t}$ and an estimation $\tilde{\mathbf{t}}$ of the template, based on the probe $\mathbf{y}$:

$$d_H(\mathbf{t}, \tilde{\mathbf{t}}) = \frac{1}{L^2} \sum_{i,j}^{L} |t_{ij} - \tilde{t}_{ij}| \in [0, 1] \tag{3.4}$$

In real situations, the size and orientation of the probe $\mathbf{y}$ and the digital template $\mathbf{t}$ might be different due to the complex printing and imaging processes. To proceed with the estimation of $\tilde{\mathbf{t}}$ from $\mathbf{y}$, the image $\mathbf{y}$ is first geometrically aligned with $\mathbf{t}$ using common template matching techniques. In our case, we have used the synchronization markers around the printed codes to align $\mathbf{y}$ and $\mathbf{t}$. After the geometrical alignment, the binarization based on Otsu's algorithm is applied to the aligned image.[2] As a result, the image $\mathbf{y}$ is transformed to a binary counterpart $\tilde{\mathbf{t}}$ aligned in size and orientation to $\mathbf{t}$.

Using the terminology of binary communication from information theory, we describe the printing-imaging pair as an encoding of the binary message $\mathbf{t}$ and refer to $\mathbf{x}$ as an encoded (and noisy) version of $\mathbf{t}$. The binarization of the encoded message $\mathbf{x}$ can then be interpreted as decoding and produces $\hat{\mathbf{t}}_{\mathbf{x}}$ an estimate of the binary message $\mathbf{t}$ (see Fig. 3.1 for an illustration of this concept).

3.3.2 Deep Learning Encoder-Decoder Architecture

Inspired by the approach done in [56] for image to image translation, we introduce a deep learning architecture, composed of two Unet deep neural networks refered to as the encoder and the decoder with matching input and output dimensions. The encoder is trained at learning a mapping:

$$\begin{aligned} enc\colon \{0,1\}^{L\times L} &\longrightarrow [0,1]^{kL\times kL\times C} \\ \mathbf{t} &\longmapsto \mathbf{x}, \end{aligned} \tag{3.5}$$

while the decoder is trained at learning a mapping:

$$\begin{aligned} dec\colon [0,1]^{kL\times kL\times C} &\longrightarrow \{0,1\}^{L\times L} \\ \mathbf{x} &\longmapsto \mathbf{t}. \end{aligned} \tag{3.6}$$

These networks are being used as pairs and produce latent and reconstructed images.[3] We write

- $\tilde{\mathbf{x}} = enc(\mathbf{t})$, the latent representation of the printed CDP,
- $\tilde{\mathbf{t}} = dec(\mathbf{x})$, the latent representation of the digital template,

[2] The algorithm being used for the binarization has been heavily studied in the past few years with the possibility of training a deep neural network on this task. See [41, 43, 52] for more details on this matter.

[3] From a technical point of view, the input and output images should share the same dimensions. Thus, the digital template $\mathbf{t}$ is upsampled with a factor k in order to match the dimensions of the printed CDP $\mathbf{x}$ and the number $C = 3$ of color channels for RGB images is also matched.

- $\hat{\mathbf{x}} = enc(\tilde{\mathbf{t}})$, the reconstructed representation of the printed CDP,
- $\hat{\mathbf{t}} = dec(\tilde{\mathbf{x}})$, the reconstructed representation of the digital template.

An illustration of these latent and reconstructed representations is shown in Fig. 3.2.

Given a training set of pairs $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{N_{train}}$, both the encoder and the decoder can be trained jointly with a cyclic loss condition:

$$\begin{aligned} \mathcal{L}_{DE}(\mathbf{t}^n, \mathbf{x}^n) = &\quad \lambda_1 \cdot \ell_1(\tilde{\mathbf{t}}^n, \mathbf{t}^n) + \lambda_2 \cdot \ell_1(\hat{\mathbf{x}}^n, \mathbf{x}^n), \\ \mathcal{L}_{ED}(\mathbf{t}^n, \mathbf{x}^n) = &\quad \lambda_3 \cdot \ell_1(\tilde{\mathbf{x}}^n, \mathbf{x}^n) + \lambda_4 \cdot \ell_1(\hat{\mathbf{t}}^n, \mathbf{t}^n), \end{aligned} \tag{3.7}$$

where ℓ_1 measures the normalized Manhattan distance (3.1) and both losses are computed successively by switching the encoder and decoder on each batch (see Fig. 3.3 for a visual diagram of the training process). The coefficients $\lambda_i \in \mathbb{R}_+$ in the training losses have been introduced in order to study the influence of each loss term on the overall performance of the trained networks.

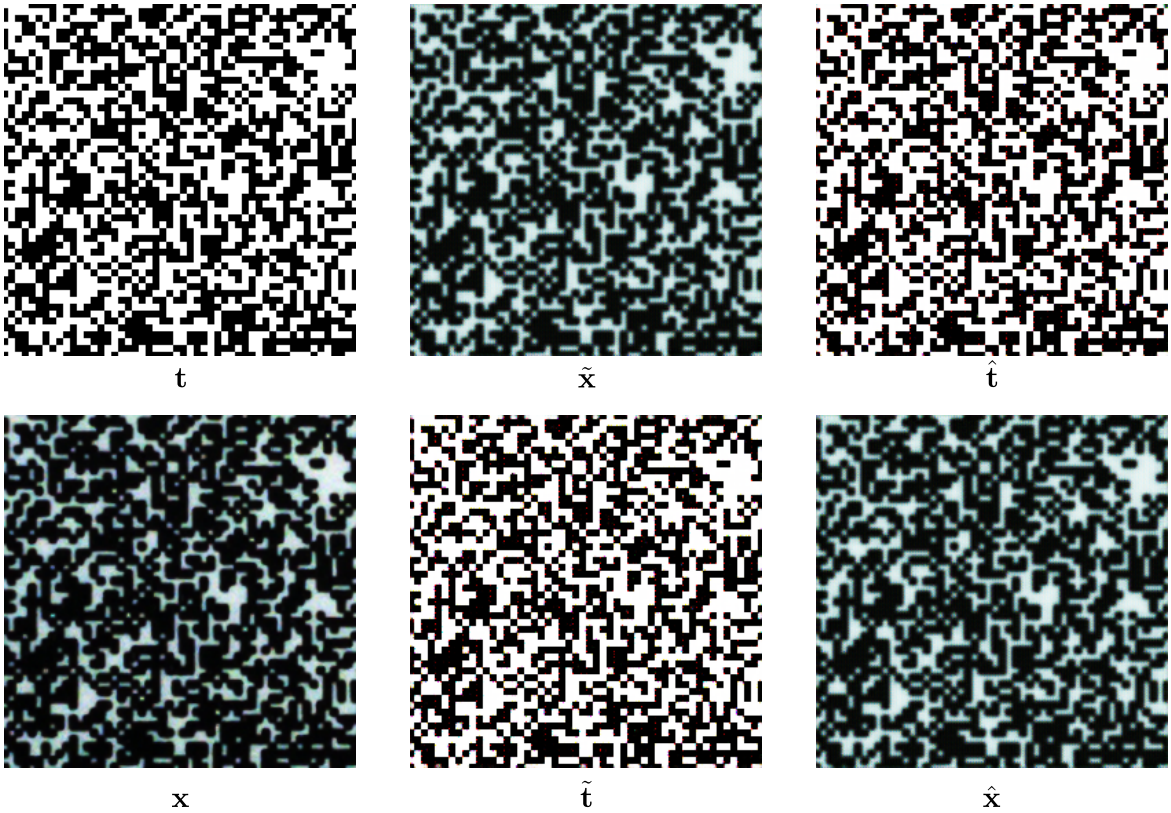


Figure 3.2. An illustration of the various inputs, latents and outputs images produced by a trained encoder-decoder architecture. The top row corresponds to an encoder-decoder forward pass and the bottom row to a decoder-encoder forward pass.

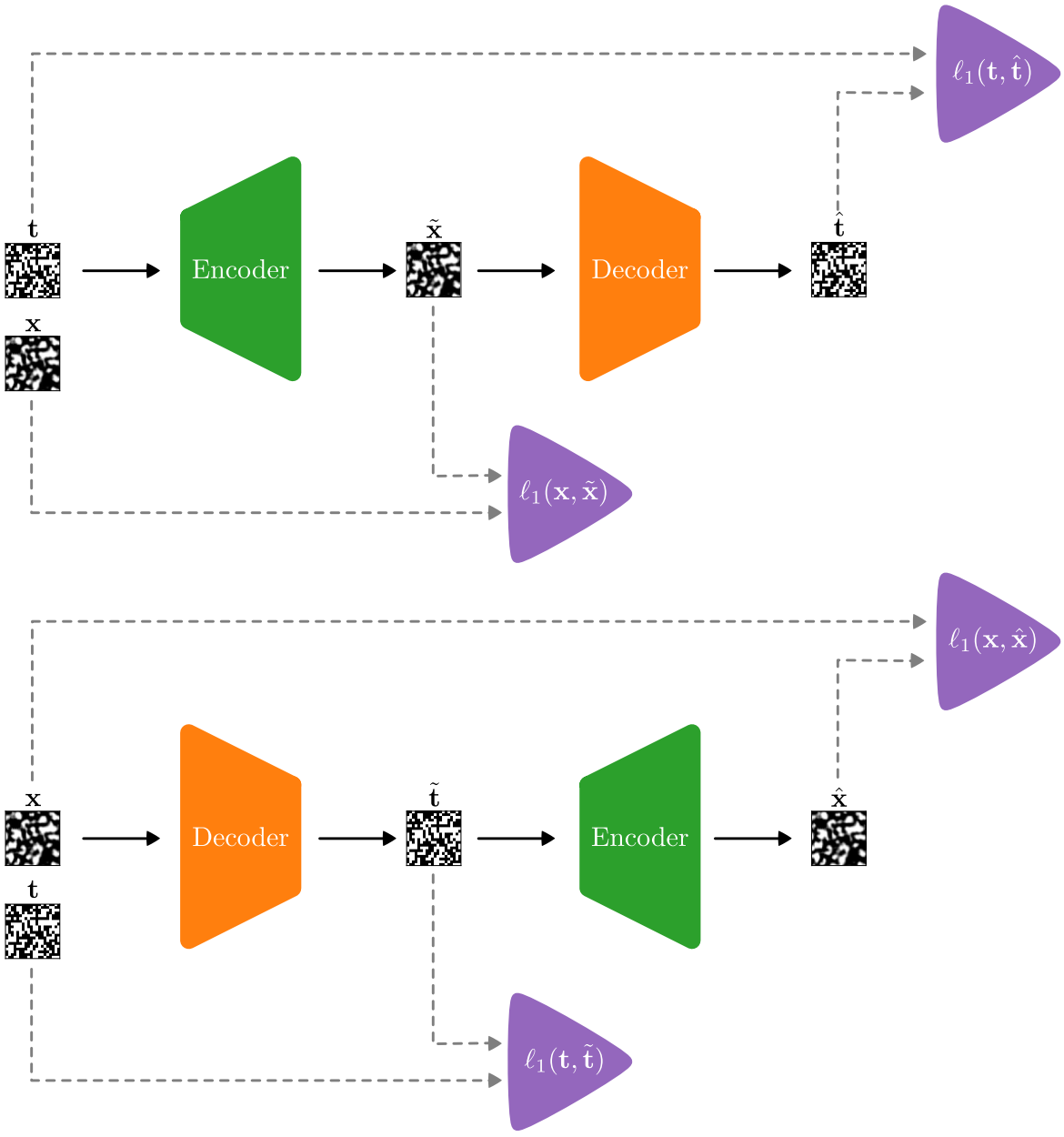


Figure 3.3. A schematic representation of the training process for the encoder-decoder architecture. A training pair $(\mathbf{t}, \mathbf{x})$ is fed to the encoder-decoder pair, producing two latent representations $\tilde{\mathbf{t}}$, $\tilde{\mathbf{x}}$ and two reconstructed representations $\hat{\mathbf{t}}, \hat{\mathbf{x}}$. These latent and reconstructed images are then used for the computation of a global loss based on ℓ_1 metric (violet triangles).

3.3.3 CDP Authentication Based on Multimetric Vectors

Standard metric approaches to CDP authentication have proven to be ineffective against the new generation of deep learning attacks [43, 46]. We describe here the use of the encoder-decoder architecture introduced above to strengthen the authentication scheme. The CDP authentication based on multimetrics scheme can be decomposed in the following steps:

1. *Digital Clone Model Training:* The Defender generates a reference dataset of printed-acquired CDP $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{N_{train}}$. This dataset is then used to train the encoder-decoder architecture. The outcome of this process is:
   - A reference dataset $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{N_{train}}$,
   - A trained encoder and decoder pair (enc, dec).

   Both networks as well as all reference templates $\mathbf{t}^n$ are then stored on a secured server controlled by the Defender. This server works as an oracle for authentication when given a probe CDP $\mathbf{y}$.
2. *Probe Augmentation Generation:* At the authentication step, the Verifier is given an unidentified probe $\mathbf{y}$ together with an identification code that allows him to identify the corresponding digital template $\mathbf{t}$. The Defender then sends the probe to the server, which in turn feeds it to the decoder-encoder pair and generate digital twins $\tilde{\mathbf{t}}_{\mathbf{y}}$ and $\hat{\mathbf{y}}$ and, feeding the reference digital template $\mathbf{t}$ to the encoder-decoder pair, generate $\tilde{\mathbf{x}}$ and $\hat{\mathbf{t}}$. As such, at this step, given a probe $\mathbf{y}$ and an identification code the server generates:
   - the reference pair $(\mathbf{t}, \mathbf{y})$;
   - the estimate pair based on decoder-encoder applied to $\mathbf{y}$: $(\tilde{\mathbf{t}}_{\mathbf{y}}, \hat{\mathbf{y}})$;
   - the estimate pair based on encoder-decoder applied to $\mathbf{t}$: $(\tilde{\mathbf{x}}, \hat{\mathbf{t}})$.
3. *Multimetric Measure:* Given all 6 samples produced in the preceding step, the server uses them to measure metrics between pairs. The benefit of this approach is the dimension reduction, going from high dimensional datasets of digital twins to a small number of relevant metrics $D \ll L^2$. From the point of view of the interpretation, metrics of particular interest are
   - *Deep Learning Based Binarization Metric:* $d(\tilde{\mathbf{t}}_{\mathbf{y}}, \mathbf{t})$ which is expected to enhance the results obtained by simply measuring $d(\mathbf{t}, \mathbf{y})$.
   - *Digital Twin Generation Metric:* $d(\tilde{\mathbf{x}}, \mathbf{y})$ which is expected to be similar to the measure $d(\mathbf{x}, \mathbf{y})$. This measure should be an answer to the problem posed by the unfeasability of enrolling every printed CDP as a physical reference for authentication.
   - *Reconstruction Metric:* $d(\mathbf{y}, \hat{\mathbf{y}})$. The distortion introduced by the decoder-encoder network should be minimal when applied to the original CDP $\mathbf{y} = \mathbf{x}$ and thus this metric is expected to help separate original and fake CDP.

Note that the metric $d(\cdot,\cdot)$ described above can be any of the well-known metric usually used in a classical authentication framework for CDP (ℓ_1, ℓ_2, Hamming d_H, Pearson correlation *pcor*). The outcome of this step is a multimetric vector $\underline{d}_{\mathbf{y}} \in \mathbb{R}^D$.

4. *One-class Based Decision Boundary:* Based on the multimetric vector $\underline{d}_{\mathbf{y}}$, the server finally needs to take a decision on whether the probe under examination is an original CDP or a counterfeited one. To this end, the Defender may use the reference training set $\{(\mathbf{t}^n, \mathbf{x}^n_{ref})\}_{n=1}^{N_{train}}$ to generate a set of reference multimetric vectors $\underline{d}^n \in \mathbb{R}^D$.
   Given that $D \ll L^2$ is small and N_{train} is sufficiently large, the Defender may train a one-class classifier algorithm (OC-SVM, k-mean, random forest, etc.) in order to design a rejection region $R \subset \mathbb{R}^D$ for the null hypothesis (see Section 2.2.2 for more details on statistical hypothesis testing). The server then uses this rejection region to decide whether the probe $\mathbf{y}$ is genuine or not.

These four steps are illustrated in Fig. 3.4 and are investigated in details in the experimental part of the chapter. The quality of the training process in terms of image generation and reconstruction together with the quality of augmentations on various types of fakes are studied in Sections 3.5.1 and 3.5.2. The correlation of the various metrics for different types of attacks and the separability of multimetric vectors are studied in Section 3.5.3.

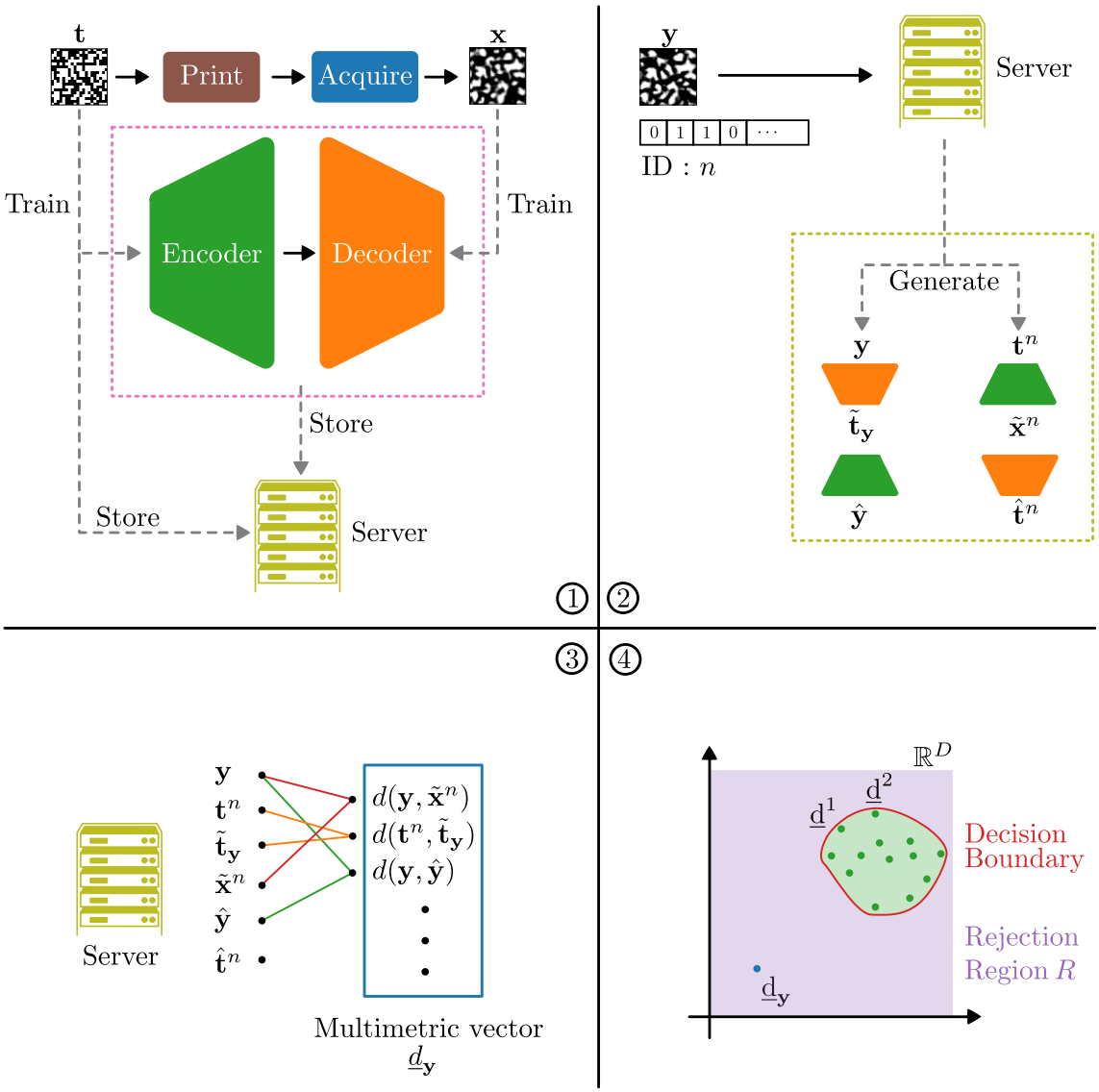


Figure 3.4. An illustration of the four steps of the authentication of a probe CDP based on the multimetric vector scheme. The first step illustrates the training of the encoder-decoder architecture using the reference training set. The second step shows an unidentified CDP probe $\mathbf{y}$ being fed to the trained model and the generation of the digital twins. Step three is the computation of the multimetric vector. Step 4 illustrates the one class based decision boundary and rejection region for the probe multimetric vector $\underline{d}_{\mathbf{y}}$.

3.3.4 Deep Classifiers Authentication Trained on Digital Twins

The topic of deep classifiers trained on the task of fake CDP detection has been the subject of multiple studies in the past few years. Indeed, deep classifiers have demonstrated unmatched performances when compared with more traditional non-deep computer vision algorithms [52, 61, 62]. One main issue with deep classifiers lies in the fact that these networks are being trained in a supervised way where instances of both inlier class and outlier class need to be available at training time. In the case of CDP-based object security, this is an unrealistic scenario since in real life applications, fake CDP are not available to the Defender when designing the authentication scheme. As such, we propose in this research to use synthetically generated samples $\tilde{\mathbf{x}} = enc(\mathbf{t})$ that serve as an outlier class for the training of the binary deep classifier against a reference inlier class $\mathbf{x}_{ref}$.

Based on this approach, we propose a novel authentication procedure for the detection of fake CDP based on deep binary classifiers which does not require the knowledge of fake for training and calibration of the model. This authentication scheme can be decomposed in the following steps:

1. *Reference dataset generation:* The Defender creates a reference dataset of CDP $\{\mathbf{x}_{ref}^n\}_{n=1}^{N_{train}}$ by printing a set of secretly generated digital templates $\{\mathbf{t}^n\}_{n=1}^{N_{train}}$ and acquiring them with the same device that will be used for authentication (typically a smartphone).
2. *Synthetic augmentation production:* Similarly to the multimetric authentication scheme described in Section 3.3.3, the Defender trains an encoder-decoder architecture on the pair $\{(\mathbf{t}^n, \mathbf{x}_{ref}^n)\}_{n=1}^{N_{train}}$ and uses this architecture to produce synthetic augmentations $\{\tilde{\mathbf{x}}^n\}_{n=1}^{N_{train}}$.
3. *Deep classifier training:* This step consists in training a deep classifier at discriminating between the reference class $\{\mathbf{x}_{ref}^n\}_{n=1}^{N_{train}}$ and the synthetic class $\{\tilde{\mathbf{x}}_{ref}^n\}_{n=1}^{N_{train}}$ in a typical supervised fashion. Note that a part of the training set can be reserved as a validation set in order to fix the decision thresholds on the outcome of the classifier.
4. *Authentication of unknown probe:* Given an unknown probe CDP $\mathbf{y}$, the Verifier uses the trained classifier to decide whether the sample is an original or a fake CDP based on the output score.

This authentication scheme is studied in the experimental part where we test the capacity of this model to actually discriminate original and fake CDP on a real dataset. We test the model performance and stability for various training regime and synthetic sample types.

3.4 Experimental Setup

3.4.1 Indigo Mobile CDP Dataset

The dataset that we use for this research is the Indigo Mobile dataset [62] created by Olga Taran et al. in December 2021. At the time, it was the first publicly available dataset of CDP captured using a handheld mobile phone. This dataset consists of:

- 300 digital templates $\mathbf{t}$ with a size of 66×66 binary symbols generated randomly with equal and independent probability.
- 300 reference printed CDP $\mathbf{x}_{ref}$ using an industrial printer[4] with a resolution of 812dpi, captured using an iPhone XS and aligned with the digital template using synchromarkers. The final resolution of the enrolled CDP is increased by a factor of 5, thus achieving a 5×5 pixel symbol-size for a full resolution of 330×330 pixels.
- 300 printed CDP $\mathbf{x}$ which represent a second round of capture session for the same dataset as the reference printed CDP. This second capture allows to account for the intrinsic variability of the capture device and compare it with the variability introduced by copy machine fakes.
- 1200 fake $\mathbf{f}$ CDP created based on the printed CDP by copy machine attack[5] and reprinted on white or gray paper. Thus for each genuine CDP, we have four corresponding fakes of various quality:
  - $\mathbf{f}_{1w}$: Fake printed using printer 1 on white paper;
  - $\mathbf{f}_{1g}$: Fake printed using printer 1 on gray paper;
  - $\mathbf{f}_{2w}$: Fake printed using printer 2 on white paper;
  - $\mathbf{f}_{2g}$: Fake printed using printer 2 on gray paper.

The dataset samples are illustrated in Fig. 3.5. More details on the creation of this dataset can be found in the original paper [62].

[4] HP Indigo 5500
[5] *Printer 1:* RICOH MP C307 and *Printer 2:* Samsung CLX-6220FX



Printed reference $\mathbf{x}_{ref}$ Digital template $\mathbf{t}$ Printed original $\mathbf{x}$



Fake type 1 white $\mathbf{f}_{1w}$ Fake type 1 gray $\mathbf{f}_{1g}$



Fake type 2 white $\mathbf{f}_{2w}$ Fake type 2 gray $\mathbf{f}_{2g}$

Figure 3.5. The Indigo Mobile dataset introduced in [62]. It consists of 7-tuples composed of a digital template, a reference original CDP captured two times and 4 types of fake CDP based on copy-machine attacks printed on white and gray paper.

Training dataset pre-processing

In order to compensate for the relative small amount of available samples to train deep neural networks, the dataset is processed before being fed to the neural network and random augmentations are performed:

- Data preparation:
  - *cropping:* The central 256×256 square is cropped out of the CDP;
  - *normalization:* The image is normalized by mean and standard deviation;
- Random augmentation: one or several augmentations are randomly performed on the input image (see Fig. 3.6 for an illustration):
  - *rotation:* A rotation of the image by $90°, 180°, 270°$;
  - *flipping:* The image is flipped vertically or horizontally;
  - *gamma correction:* A gamma correction is applied to the printed CDP $\mathbf{y}$ with $\gamma \in \{0.5, 0.6, \ldots, 1.1, 1.2\}$.



Reference | Rotation 90° | Horizontal flip



$\gamma = 0.5$ | $\gamma = 0.8$ | $\gamma = 1.2$

Figure 3.6. Examples of augmentations applied to the dataset of CDP during the training of the encoder-decoder framework.

3.4.2 Encoder-Decoder Model Details

Unet architecture parameters

The encoder-decoder architecture is composed of two Unet models trained jointly. Each Unet can be decomposed into:

- *Down blocks:* A composition of 2D convolutions with 4×4 kernels, batch normalization and leaky ReLU activation function;
- *Up blocks:* Composed with transposed 2D convolutions with 4×4 kernels, batch normalization and ReLU activation;
- *Skip connections:* The output of the kth downblock is connected to the input of the following $(k+1)^{th}$ downblock as well as to the input of the matching $(n-k)^{th}$ upblock;
- *Tanh final activation:* The last activation function of the network is a renormalized Tanh function: $0.5 \cdot (\tanh(x) + 1) \in [0, 1]$.

The number of filters used for each downblock and upblock is listed here:

- *Downblock filters:* $[64, 128, 256, 512, 512, 512, 512, 512]$;
- *Upblock filters:* $[512, 1024, 1024, 1024, 1024, 512, 256, 128, 1]$.[6]

A diagrammatic representation of the architecture of the Unet model is given in Fig. 3.7 and details of the Downblock and Upblock are shown in Fig. 3.8.

Training regime

The encoder-decoder architecture is trained for 180 epochs, with a batch size of 20 using the losses described in Eq. (3.7). The strategy for choosing the weights during the training of the model varies across experiments and is summarized in Table 3.1. The general idea is to measure the impact of the various weights on the objective of each experiment. The three different strategies can be described as follows:

(S1) *Reconstruction loss:* Study how the reconstruction losses $\ell_1(\mathbf{t}, \hat{\mathbf{t}})$ and $\ell_1(\mathbf{x}, \hat{\mathbf{x}})$ impact the results;

(S2) *Synthetic latent loss:* Study how putting a higher weight on the latent loss $\ell_1(\mathbf{x}, \tilde{\mathbf{x}})$ impacts the performance;

(S3) *Printed reconstruction loss:* We discovered empirically that putting a heavy weight on the reconstruction loss $\ell_1(\mathbf{x}, \hat{\mathbf{x}})$ gives the best result for multimetric vector authentication.

[6] Note that due to the skip connections, the number of filters in upblocks is twice as big as those in the downblock.

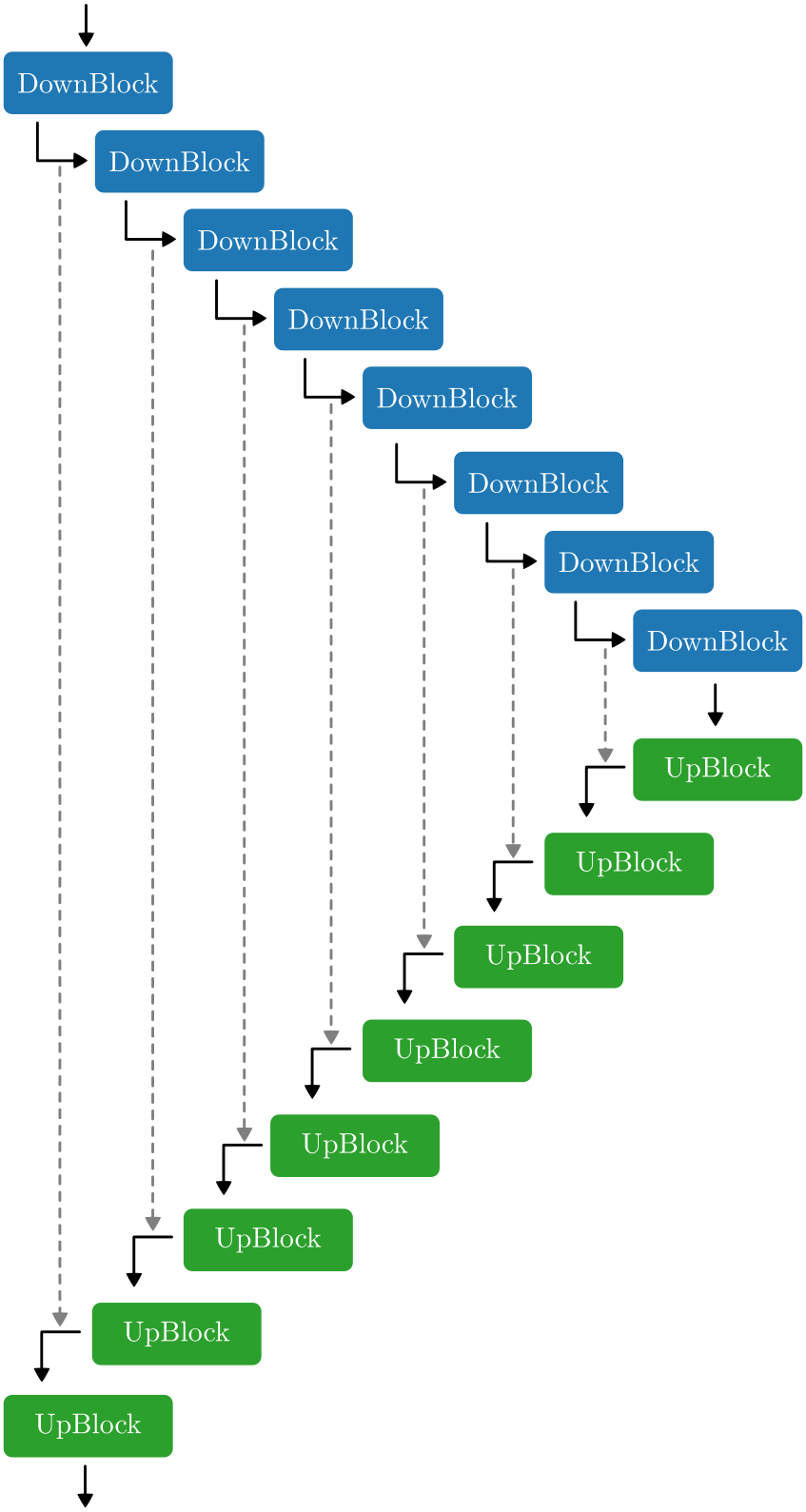


Figure 3.7. A diagrammatic representation of the Unet architecture. The input of the Unet is processed from the top to the bottom. The skip connections are represented in dashed gray lines.

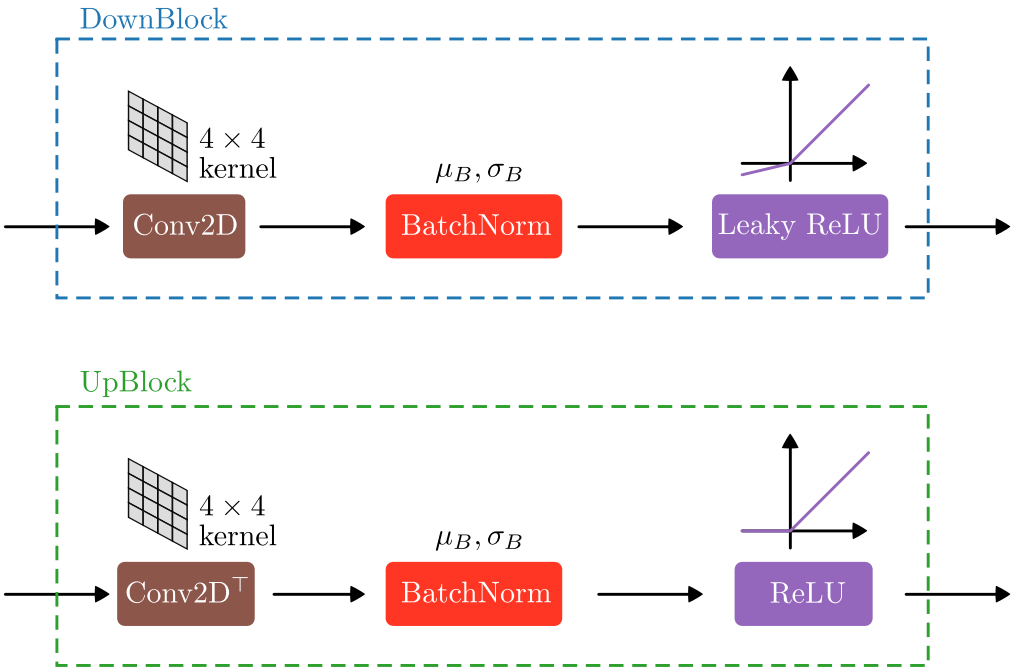


Figure 3.8. Details of the architecture of Downblocks and Upblocks used in the Unet architecture.

Table 3.1. A Table describing the various strategies for the weight selection when training the encoder-decoder architecture.

Training strategy	Weight values	Experiments
Reconstruction loss $\rho = 0.1, 1.0, 25.0$	$\lambda_1 = \lambda_3 = 1,$ $\lambda_2 = \lambda_4 = \rho$	3.5.1, 3.5.2
Synthetic latent loss $\lambda = 1, 5, 10, 25$	$\lambda_1 = \lambda_2 = \lambda_4 = 1,$ $\lambda_3 = \lambda$	3.5.1, 3.5.4
Printed reconstruction loss	$\lambda_1 = \lambda_3 = \lambda_4 = 1,$ $\lambda_2 = 25$	3.5.3

3.4.3 Classifier Model Details

We describe here the details of the standard convolutional binary classifier used to discriminate between a reference class $\mathbf{x}_{ref}$ and an outlier class $\mathbf{y}$.

Architecture

The classifier can be decomposed into two parts:

- *Feature extractor:* A Sequential model with
  - Convolution2D with kernel size 5×5, stride 1, no-padding and filter dimensions $[32, 16, 8]$;
  - ReLU activation function;
  - Average Pooling 2D with kernel 3×3.
- *Fully connected block:* Five fully-connected layers with ReLU activations (except for the last where we use a Sigmoid).
  - layers dimensions : $[392, 256, 128, 64, 16, 2]$.

Both parts of the classifier are illustrated in Figures 3.9 and 3.10.

Training losses

Training of the network to discriminate between two classes uses BinaryCrossEntropy Loss on the output of the classifier with the following hyperparameters:

- *Optimizer:* Adam
  - learning rate: 10^{-4};
  - betas: $(0.5, 0.999)$;
- *Training parameters:*
  - number of epochs: 100;
  - batch size: 21.

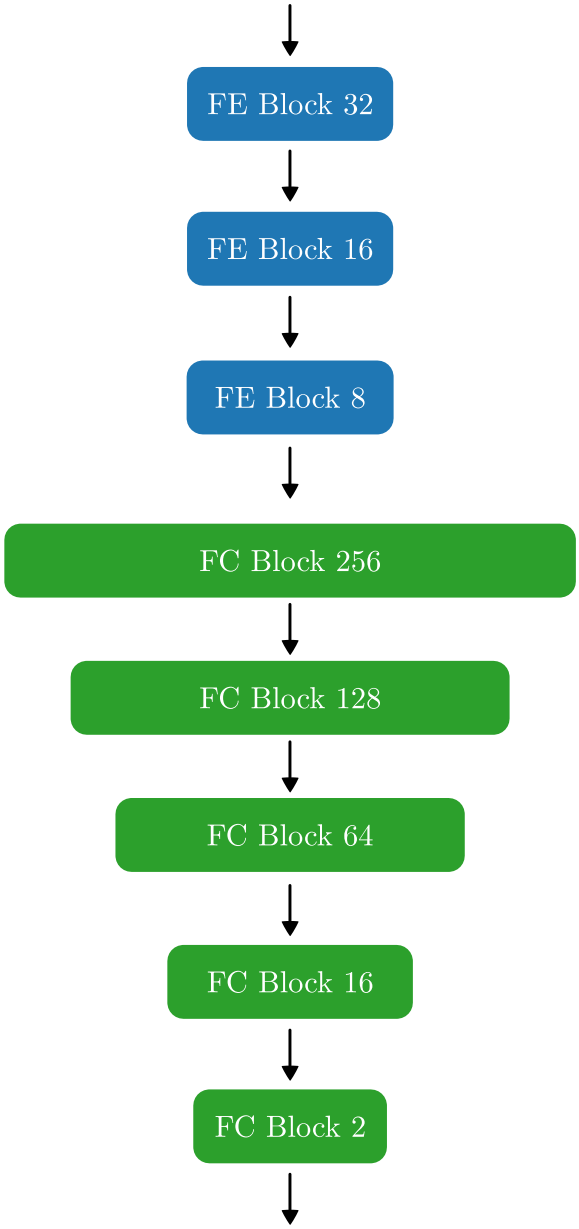


Figure 3.9. Global architecture of the classifier composed of Feature Extractor Blocks and Fully Connected Blocks.

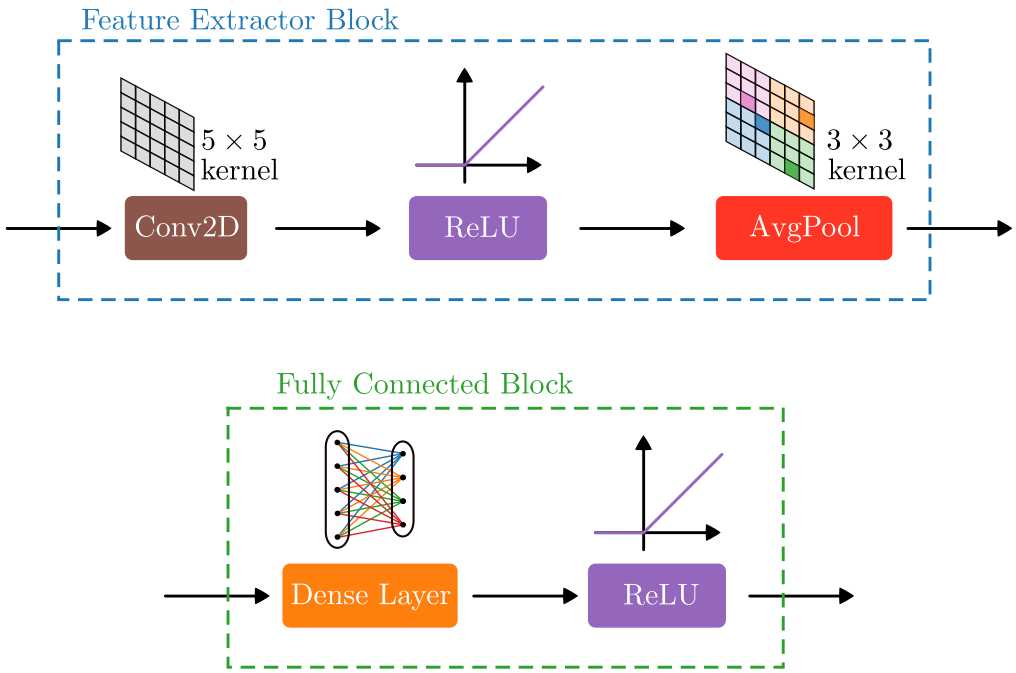


Figure 3.10. Details of the Feature Extractor Block and the Fully Connected Block used in the classifier.

3.5 Experimental Results

3.5.1 Quality Assessment of Digital Twins Using Standard Metrics

Impact of reconstruction loss weighting factor

The goal of this first experiment is to test the capability of the encoder-decoder architecture to accurately reproduce the printing-imaging process of the CDP. We thus perform the training of the architecture on different training pairs: $\{(\mathbf{t}^n, \mathbf{y}^n)\}_{n=1}^{120}$ where $\mathbf{y}$ is a chosen type of CDP in the set $\{\mathbf{x}_{ref}, \mathbf{f}_{1w}, \mathbf{f}_{2w}, \mathbf{f}_{1g}, \mathbf{f}_{2g}\}$. We compare different training losses with fixed latent loss $\lambda_1 = \lambda_3 = 1$ and varying importance given to the reconstruction loss $\lambda_2 = \lambda_4 = \rho \in \{0.1, 1.0, 25.0\}$.

Table 3.2 shows a measure of the quality of the digital twins produced using such a system and compares them to the performance of real printed CDP. The measure of the quality is performed by computing the ℓ_1 distance between the produced probes $\tilde{\mathbf{y}} = dec_\rho(\mathbf{t})$ and either the digital template $\mathbf{t}$, the reference original CDP $\mathbf{x}_{ref}$ or the printed probe itself $\mathbf{y}$. A first general observation of the second column shows that digital twins $\tilde{\mathbf{y}}$ demonstrate very similar behaviour with their printed counterparts $\mathbf{y}$ in terms of standard metric $\ell_1(\mathbf{t}, \mathbf{y})$. Indeed, synthetic samples metrics are distributed within the variability range of their printed references. In addition, synthetic samples based on originals $\mathbf{x}$ are the best in terms of distance to originals $\ell_1(\mathbf{x}, \tilde{\mathbf{y}})$. This results thus confirms that synthetic samples $\tilde{\mathbf{x}}$ can be used as references for authentication tasks.

The third column, which measures the distance to the original printed CDP $\mathbf{x}$, seems to indicate that a small weight on the reconstruction loss tends to produce synthetic samples $\tilde{\mathbf{y}}$ that are closer to $\mathbf{x}$. Finally, the fourth column, which measures the quality of the synthetic sample when compared to its printed CDP target $\mathbf{y}$, indicates that this architecture seems to be working best for imbalanced weights $\rho \neq 1$ and the best quality of synthetic samples is achieved for the dataset of original CDP $\mathbf{y} = \mathbf{x}$. This is understandable since original CDP $\mathbf{x}$ introduce less distorsion when compared to the digital template $\mathbf{t}$ than fake CDP $\mathbf{f}$.

Another visualisation of the distribution of these metrics is shown in Fig. 3.11. These histograms show the distribution of the ℓ_1 metric $\ell_1(\mathbf{y}, \tilde{\mathbf{y}})$ for the various types of training datasets matching the average values shown in Table 3.2. From the point of view of these histograms, it is hard to pinpoint an optimal choice of weighting factor ρ to enhance the quality of the produced synthetic samples $\tilde{\mathbf{y}}$. In fact, we see that all architectures produce results whose in-class variance is greater than cross-class variance. Still, the architecture trained on originals (first row) have a smaller ℓ_1 distance on average than all other datasets and the best choice for this dataset is $\rho = 25$.

Table 3.2. Average ℓ_1 metric for various dataset of digital twin samples and comparison with the actual real performance for printed CDP. Each block of four rows represents an experiment on a different training dataset pair $(\mathbf{t}, \mathbf{y})$ with different ρ weight on reconstruction loss. The second column measures the deviation with respect to the digital template $\mathbf{t}$, the third column the deviation with respect to the original CDP $\mathbf{x}$ and the fourth column the distance to the printed probe $\mathbf{y}$.

Sample type	$\ell_1(\mathbf{t}, \cdot)$	$\ell_1(\mathbf{x}, \cdot)$	$\ell_1(\mathbf{y}, \cdot)$
$\mathbf{x}$	0.260 ± 0.020		
$\tilde{\mathbf{x}}$ with $\rho = 0.1$	0.244 ± 0.005	0.059 ± 0.012	
$\tilde{\mathbf{x}}$ with $\rho = 1.0$	$\mathbf{0.226 \pm 0.004}$	0.084 ± 0.021	
$\tilde{\mathbf{x}}$ with $\rho = 25.0$	0.246 ± 0.005	$\mathbf{0.056 \pm 0.012}$	
$\mathbf{f}_{1w}$	0.249 ± 0.019	0.074 ± 0.006	
$\tilde{\mathbf{f}}_{1w}$ with $\rho = 0.1$	0.247 ± 0.005	$\mathbf{0.069 \pm 0.012}$	$\mathbf{0.066 \pm 0.006}$
$\tilde{\mathbf{f}}_{1w}$ with $\rho = 1.0$	$\mathbf{0.228 \pm 0.004}$	0.097 ± 0.021	0.076 ± 0.015
$\tilde{\mathbf{f}}_{1w}$ with $\rho = 25.0$	0.229 ± 0.004	0.090 ± 0.020	0.073 ± 0.013
$\mathbf{f}_{1g}$	0.239 ± 0.021	$\mathbf{0.080 \pm 0.009}$	
$\tilde{\mathbf{f}}_{1g}$ with $\rho = 0.1$	$\mathbf{0.214 \pm 0.003}$	0.123 ± 0.023	0.092 ± 0.021
$\tilde{\mathbf{f}}_{1g}$ with $\rho = 1.0$	0.224 ± 0.004	$\mathbf{0.080 \pm 0.019}$	$\mathbf{0.071 \pm 0.011}$
$\tilde{\mathbf{f}}_{1g}$ with $\rho = 25.0$	0.221 ± 0.004	0.090 ± 0.021	0.075 ± 0.014
$\mathbf{f}_{2w}$	0.299 ± 0.018	0.084 ± 0.009	
$\tilde{\mathbf{f}}_{2w}$ with $\rho = 0.1$	$\mathbf{0.282 \pm 0.005}$	$\mathbf{0.082 \pm 0.007}$	0.078 ± 0.014
$\tilde{\mathbf{f}}_{2w}$ with $\rho = 1.0$	0.287 ± 0.005	0.083 ± 0.006	0.073 ± 0.014
$\tilde{\mathbf{f}}_{2w}$ with $\rho = 25.0$	0.292 ± 0.006	0.075 ± 0.005	$\mathbf{0.059 \pm 0.009}$
$\mathbf{f}_{2g}$	0.290 ± 0.017	0.092 ± 0.011	
$\tilde{\mathbf{f}}_{2g}$ with $\rho = 0.1$	0.292 ± 0.006	$\mathbf{0.076 \pm 0.005}$	$\mathbf{0.070 \pm 0.011}$
$\tilde{\mathbf{f}}_{2g}$ with $\rho = 1.0$	0.280 ± 0.005	0.084 ± 0.007	0.080 ± 0.015
$\tilde{\mathbf{f}}_{2g}$ with $\rho = 25.0$	$\mathbf{0.278 \pm 0.005}$	0.081 ± 0.006	0.073 ± 0.013

Table 3.3. Comparison of the latent metrics $d(\mathbf{x}, \tilde{\mathbf{x}})$ between printed original CDP and synthetic augmentations produced with the encoder-decoder architecture for various balance weights λ on the training latent loss $d(\mathbf{x}, \tilde{\mathbf{x}})$ and various deviation measure ℓ_1, ℓ_2 and Pearson correlation.

Sample type	$\ell_1(\mathbf{x}, \cdot)$	$\ell_2(\mathbf{x}, \cdot)$	$\mathrm{pcor}(\mathbf{x}, \cdot)$
$\tilde{\mathbf{x}}$ with $\lambda = 1$	**0.056**	**0.083**	**0.970**
$\tilde{\mathbf{x}}$ with $\lambda = 5$	0.057	0.084	**0.970**
$\tilde{\mathbf{x}}$ with $\lambda = 10$	0.061	0.085	**0.970**
$\tilde{\mathbf{x}}$ with $\lambda = 25$	0.067	0.094	0.968

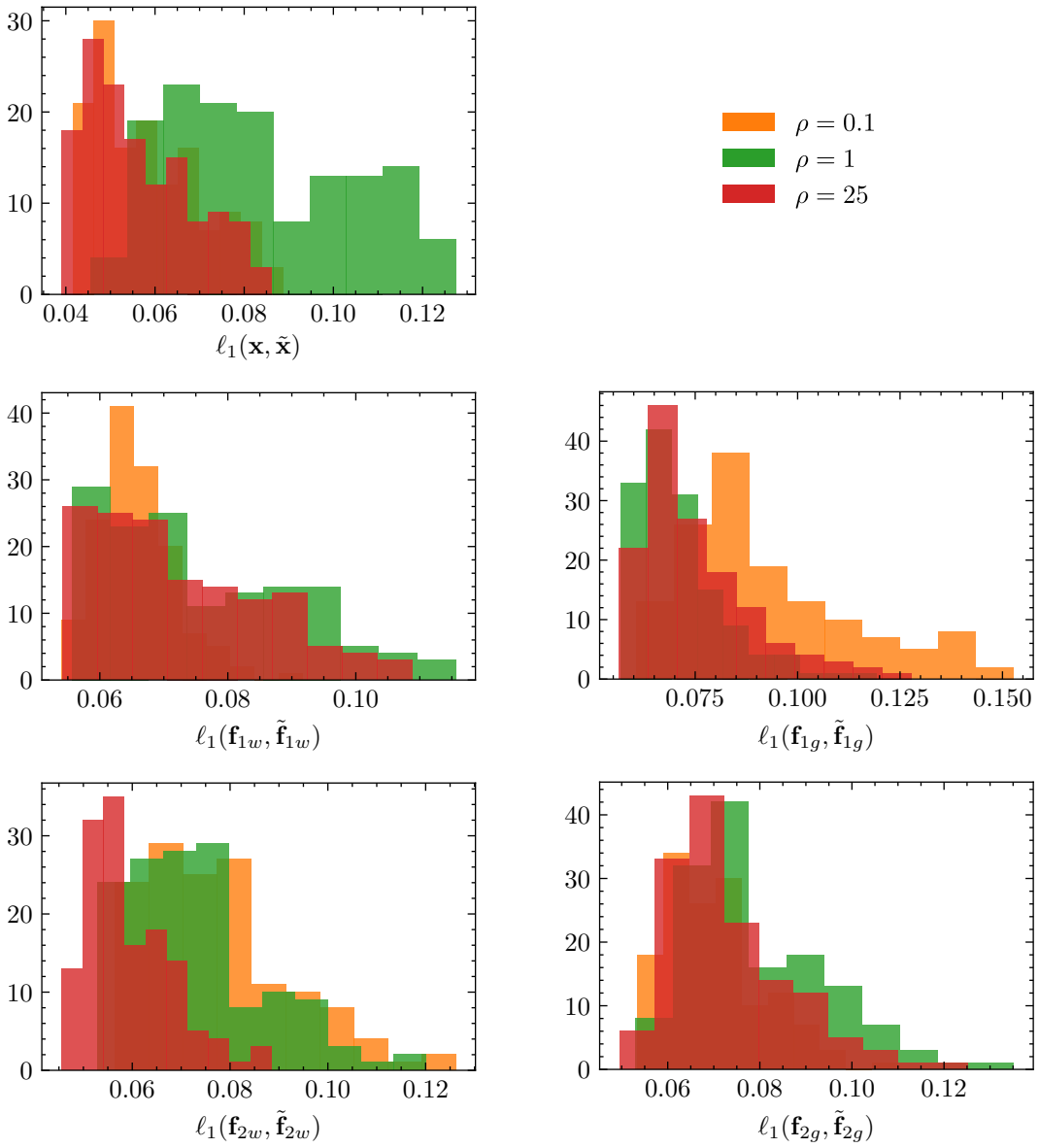


Figure 3.11. Histograms of ℓ_1 distances between printed CDP samples $\mathbf{y}$ and their synthetic twins $\tilde{\mathbf{y}}$. The first row shows the results for an architecture trained on originals $\mathbf{x}$, the second row on the two type 1 fakes and the third row on the two type 2 fakes. The various colors account for various choices of the ρ weighting factor in training losses.

Impact of synthetic augmentation latent loss weighting factor

We conduct a second experiment where a heavy weight is put on the latent loss $\ell_1(\mathbf{x}, \tilde{\mathbf{x}})$ with $\lambda_3 = \lambda \in \{1, 5, 10, 25\}$ and $\lambda_1 = \lambda_2 = \lambda_4 = 1$. This second experiment is performed only on the dataset of original CDP $\mathbf{y} = \mathbf{x}$ with the goal of further improving the quality of synthetic augmentations for this dataset. As Table 3.3 shows, the best synthetic augmentations are obtained with equal weights on all losses ($\lambda = 1$) and all synthetics augmentations are closer to the original printed CDP $\mathbf{x}$.

3.5.2 Quality Assessment of Digital Twins Using Hand-Crafted Metrics

In the study of the synthetic samples $\tilde{\mathbf{y}}$ obtained with a trained encoder-decoder architecture, we have been noticing that the model is unable to mimick what we interpreted as a Photon Counting Noise [78] when trained with RGB captured images (see Fig. 3.12 for an illustration). Indeed, the synthetically generated sample $\tilde{\mathbf{x}}$ on the right, shows a white texture that is much more uniform than the original printed CDP $\mathbf{x}$ captured on mobile phone.

In order to measure this discrepancy, we define a measure of this disagreement by taking the ℓ_1 distance between pairs of color RGB channels and summing up the scores. We introduce the *RGB photon counting noise deviation measure*:

$$\Delta(\mathbf{y}) = \ell_1(\mathbf{y}_R, \mathbf{y}_G) + \ell_1(\mathbf{y}_R, \mathbf{y}_B) + \ell_1(\mathbf{y}_B, \mathbf{y}_G), \tag{3.8}$$

where $\mathbf{y}_R, \mathbf{y}_G, \mathbf{y}_B$ represent the three color channels of the RGB image $\mathbf{y}$. This function is zero when all color channels are identical (the image appears grayscale on standard monitors) and maximal when the color channels disagree.

The results are shown in Fig. 3.13 and demonstrate a very distinctive behaviour between physically enrolled probes $\mathbf{y}$ and synthetically generated samples $\tilde{\mathbf{y}}$. Indeed, the histograms clearly confirm what we observed intuitively in Fig. 3.12, that synthetic samples tend to produce color channels that are much more correlated than real captured CDP with a histogram distribution closer to zero for synthetic CDP and further away for printed CDP. This experiment thus shows that despite performing very optimally with respect to standard metrics, the synthetic samples still have very distinctive statistics when compared with physically enrolled printed CDP.



Figure 3.12. Cropped visualisation of a CDP illustrating the photon counting noise in mobile phone captured images. The top image represents the digital template $\mathbf{t}$, the central image the original printed CDP $\mathbf{x}$ and the bottom image a synthetically generated CDP $\tilde{\mathbf{x}}$.

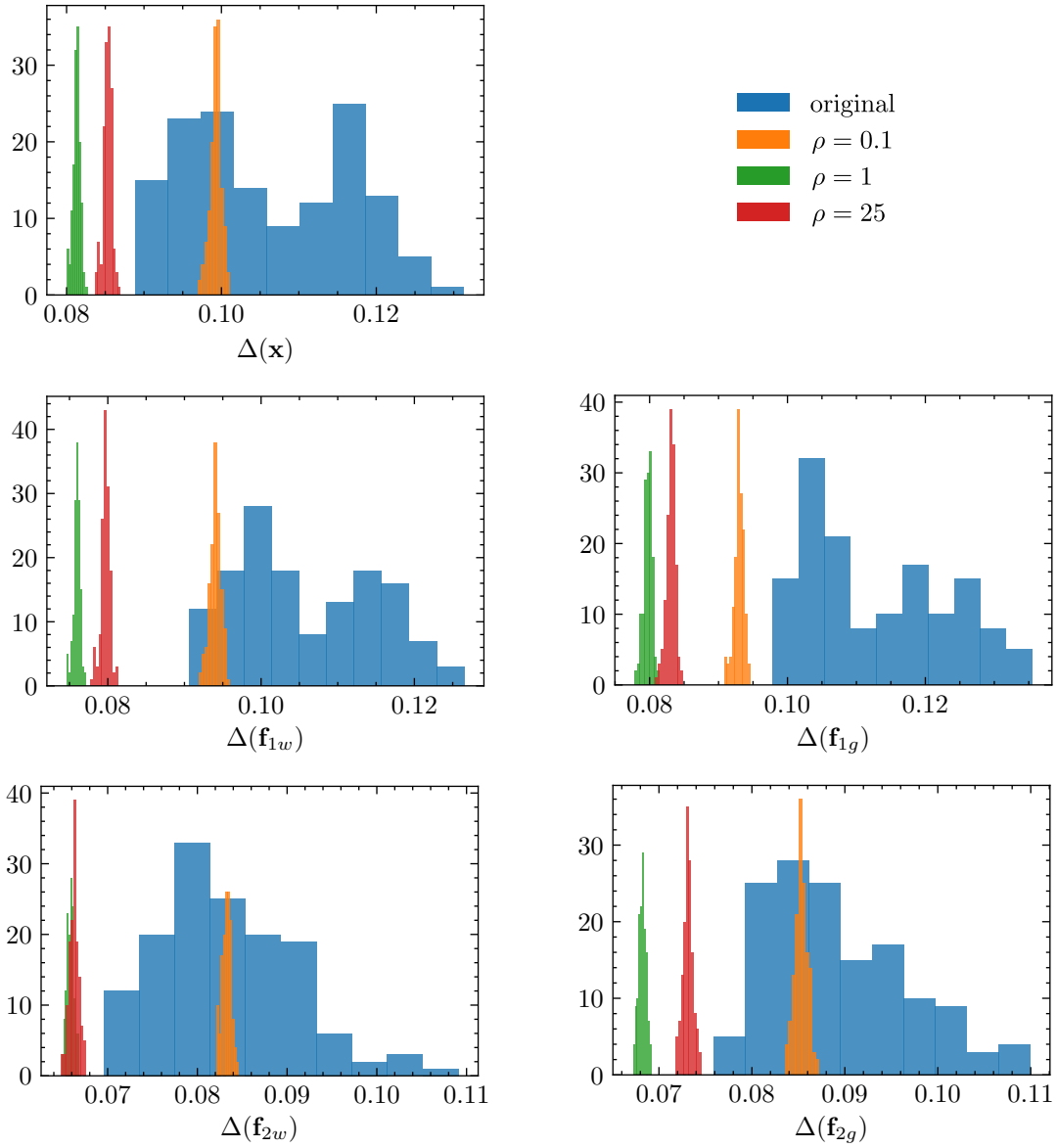


Figure 3.13. Histogram visualisation of the values of the RGB photon counting noise deviation measure $\Delta(\mathbf{y})$ applied to printed CDP (in blue). This histogram is compared with the values obtained with the same function on synthetic samples $\Delta(\tilde{\mathbf{y}})$ for various choices of weighting loss ρ (in orange, green and red). The top row shows the results for originals $\mathbf{y} = \mathbf{x}$, the second row for type 1 fake and the third row for type 2 fake.

3.5.3 Authentication Based on Turbo Architecture

In this experiment, we test the ability of the multimetric vector defined in the authentication scheme to effectively discriminate between original CDP and fake CDP. In order to visualize their separability, we follow the steps 1. , 2. and 3. described in Section 3.3.3, training the encoder-decoder architecture on $\{(\mathbf{t}^n, \mathbf{x}^n_{ref})\}_{n=1}^{120}$ and producing synthetic augmentations based on a probe $\mathbf{y} \in \{\mathbf{x}, \mathbf{f}_{1w}, \mathbf{f}_{1g}, \mathbf{f}_{2w}, \mathbf{f}_{2g}\}$. The multimetric vector $\underline{d}_{\mathbf{y}} \in \mathbb{R}^{14}$ is composed of the following metrics:

1. *Non-deep metrics*
   - $\ell_1(\mathbf{t}, \mathbf{y})$, $\ell_2(\mathbf{t}, \mathbf{y})$, $d_H(\mathbf{t}, \mathbf{y}_{tresh})$[7]
   - $\ell_1(\mathbf{x}_{ref}, \mathbf{y})$, $\ell_2(\mathbf{x}_{ref}, \mathbf{y})$
2. *Latent metrics*
   - $d_H(\mathbf{t}, \tilde{\mathbf{t}}_{\mathbf{y}})$
   - $\ell_1(\tilde{\mathbf{x}}_{ref}, \mathbf{y})$, $\ell_2(\tilde{\mathbf{x}}_{ref}, \mathbf{y})$
3. *Reconstruction metrics*
   - $\ell_1(\mathbf{y}, \hat{\mathbf{y}})$, $\ell_2(\mathbf{y}, \hat{\mathbf{y}})$
   - $\ell_1(\mathbf{x}_{ref}, \hat{\mathbf{y}})$, $\ell_2(\mathbf{x}_{ref}, \hat{\mathbf{y}})$
   - $\ell_1(\tilde{\mathbf{x}}_{ref}, \hat{\mathbf{y}})$, $\ell_2(\tilde{\mathbf{x}}_{ref}, \hat{\mathbf{y}})$

Non-deep baseline experiment

As a baseline, we first visualize the separability of the different datasets based only on non-deep metrics $\underline{d}^{base}_{\mathbf{y}} \in \mathbb{R}^5$. Fig. 3.14 shows three 2D representation of the separability. The first plot, based simply on standard metrics shows that the separability is not sufficient for type 1 fakes to distinguish them from originals. The second plot applies PCA to the full batch of baseline non-deep metrics and projects the result on the two first eigenvectors. This again shows that adding more types of metric does not help discriminating between originals and type 1 fakes. Finally, even t-SNE shown on the third row is unable to fully separate originals and type 1 fakes.

Full multimetric vector setup experiment

As a second experiment, we perform the same visualisations but applied this time to the full multimetric vector $\underline{d}_{\mathbf{y}} \in \mathbb{R}^{14}$ obtained from a trained encoder-decoder architecture. The best results were obtained for unbalanced weighting factors in Eq. (3.7), with heavy weight put on printed reconstruction $\lambda_2 = 25$ and equal weights otherwise $\lambda_1 = \lambda_3 = \lambda_4 = 1$. Fig. 3.15 shows the corresponding results. Comparatively, we see a net improvement in separability

[7] Note that the Hamming distance is computed by first applying a binarization algorithm (Otsu's algorithm) to the probe $\mathbf{y}$.

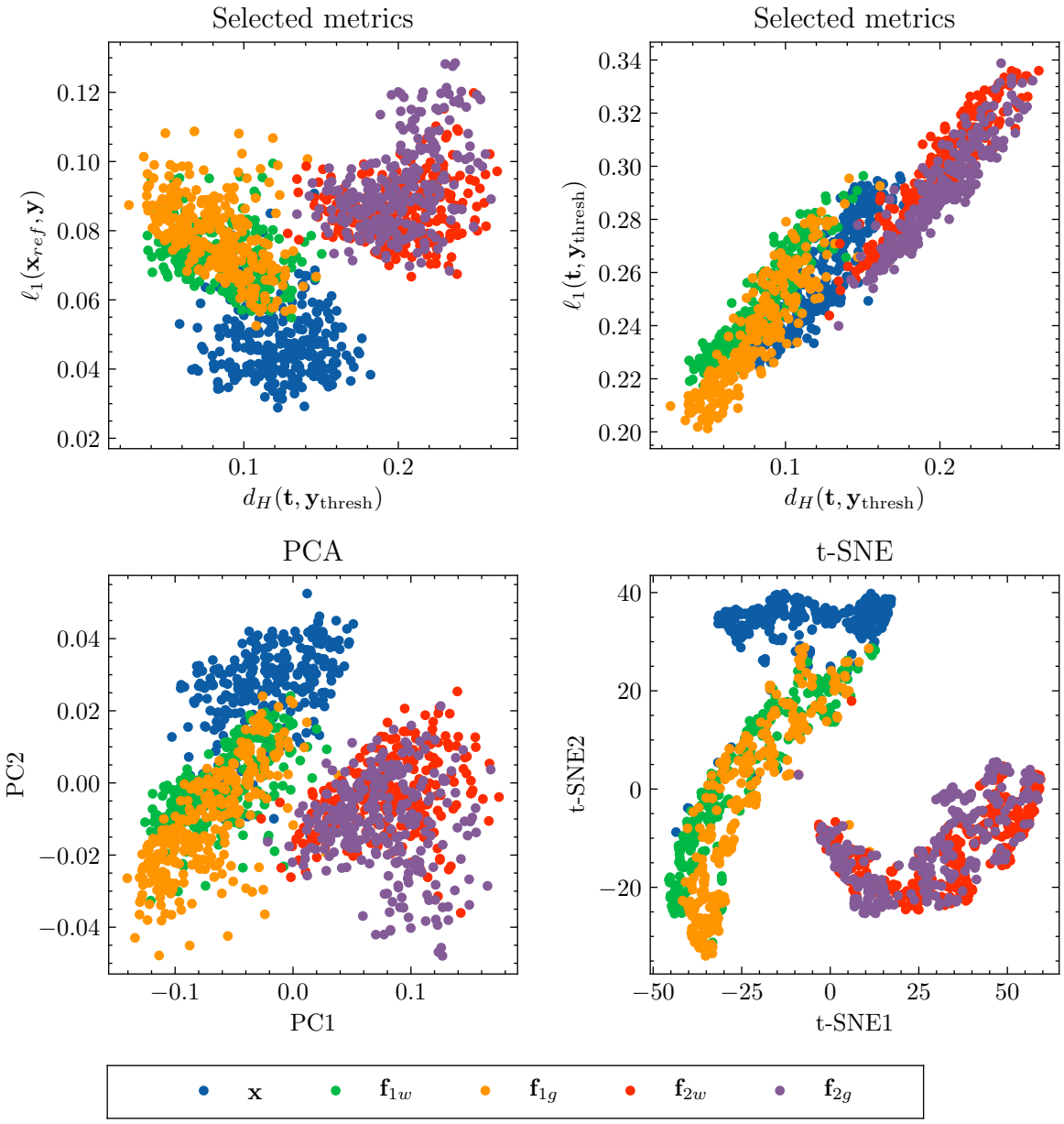


Figure 3.14. Separability of multimetric vectors $\underline{d}_{\mathbf{y}}^{base}$ based only on the 5 standard metrics for various types of original and fake CDP. The first row shows two different selections of pair metrics and the second row shows different projections of $\underline{d}_{\mathbf{y}}^{base}$ based on PCA and t-SNE algorithms. The second scatterplot is obtained after applying PCA to $\underline{d}_{\mathbf{y}}^{base}$ and projecting on the first two eigenvectors. The third scatterplot is obtained by applying t-SNE to the dataset with an output dimension of 2.

when compared with the results on non-deep metrics. Indeed, the couple of metrics shown in the top two plots already show a perfect separability between original and fake CDP, actually achieving comparable results to those obtained in [46] but using the encoder-decoder swap training loss setup instead of training in the decoder-encoder one-direction only. In view of building authentication schemes based on multimetric vectors, these result demonstrate that adding these synthetic-based metrics allows for a linear separability between original samples and fake samples. The PCA plot also demonstrates a much greater separability between originals and fakes. Finally, t-SNE is this time able to almost perfectly separate originals, fake type 1 and fake type 2 into three clear distinct groups, thus showing great promise for building an efficient authentication scheme based on multimetric vectors.

Sensitivity of results to training procedure

As a third experiment on the authentication scheme, we test the stability of results for various random initialization seed for the training batches and dataset augmentation. As can be seen in Fig. 3.16, the ability for the reconstruction loss $\ell_1(\mathbf{y}, \hat{\mathbf{y}})$ to effectively separate originals and fakes varies a lot across different training sessions. Indeed, the two plots on the right hand side show a clear overlap for originals $\mathbf{x}$ and type 1 fakes $\mathbf{f}_{1w}$ and $\mathbf{f}_{1g}$, while the two plots on the left hand side effectively separates them.

This result raises concern on the reliability of approaches based on deep learning architectures. Indeed, this experiment shows that, despite being very powerful and actually producing state-of-the-art performances when carefully selected, deep learning based schemes act as black boxes and give no warranty on their performance or explainability on poor performing results. In a context of building security schemes to protect critical products, this observation comes as a big drawback for deep learning based approaches.

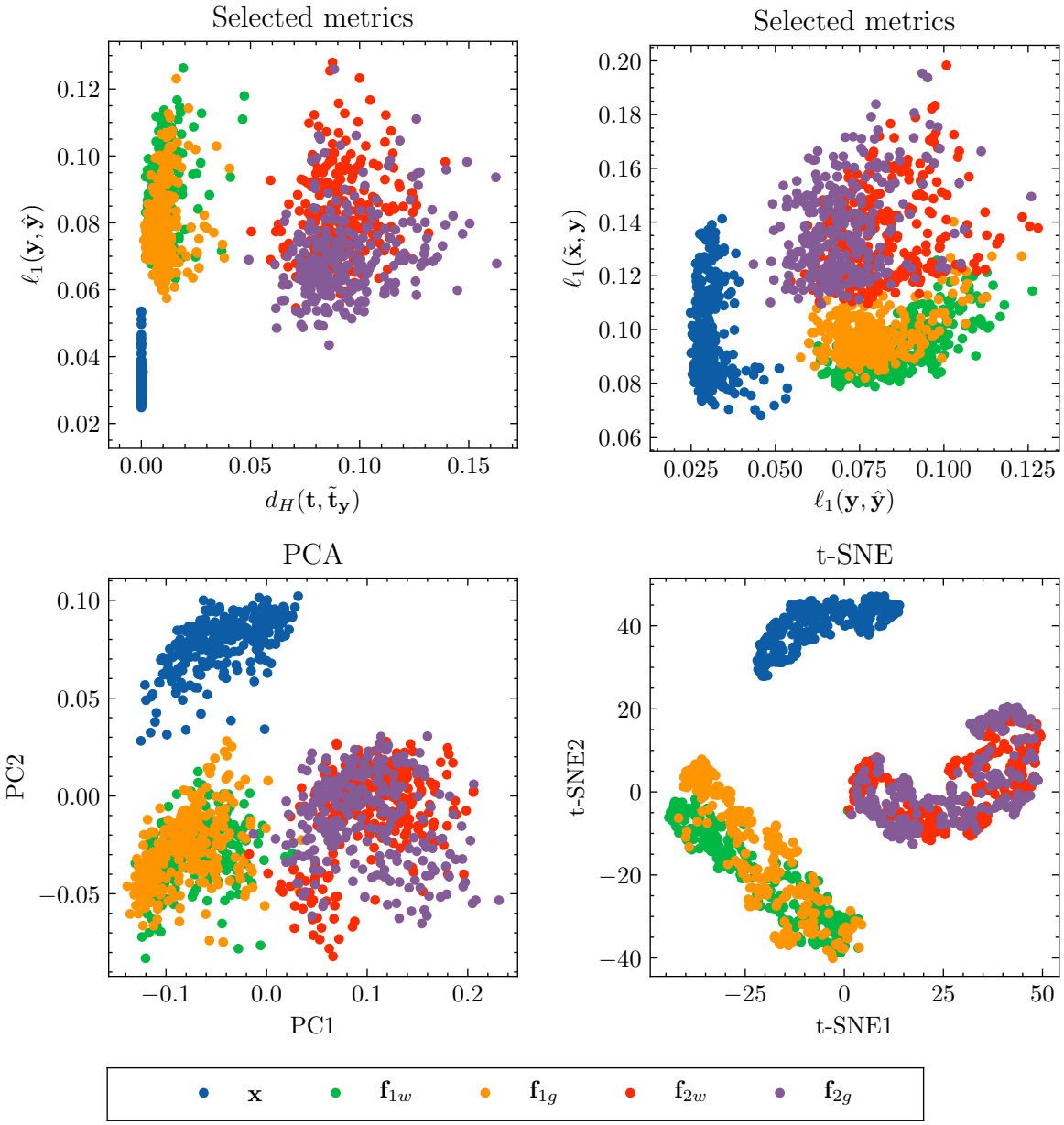


Figure 3.15. Separability of multimetric vectors $\underline{d}_\mathbf{y}$ based on standard and deep metrics for various types of original and fake CDP. The first row shows a selection of pair of deep metrics. The second row shows a projection in 2 dimension based on the PCA and t-SNE algorithm.

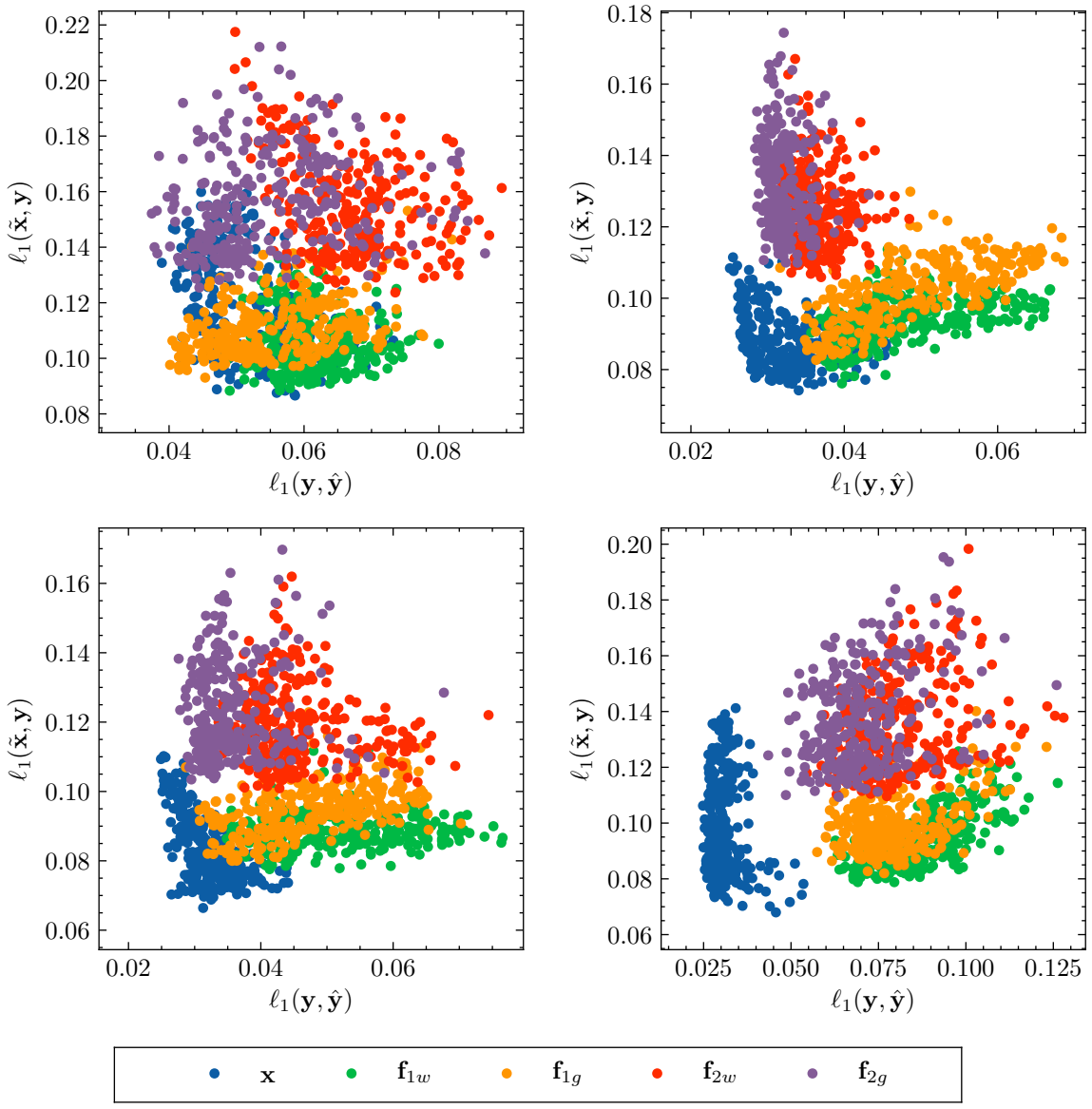


Figure 3.16. Sensitivity of multimetric vectors $\underline{d}_{\mathbf{y}}$ distribution for four different random seeds training initialization for various types of original and fake CDP. The scatterplots show a projection of $\underline{d}_{\mathbf{y}}$ on two deep metrics: the ℓ_1 reconstruction distance between $\mathbf{y}$ and $\hat{\mathbf{y}}$ and the latent Hamming distance $\ell_1(\tilde{\mathbf{x}}, \mathbf{y})$.

3.5.4 Deep Classifiers Trained on Synthetic Augmentations

The goal of this experiment is to test whether synthetic datasets can be used as outliers to train deep classifier at effectively discriminating between real printed original and fake CDP. In all following experiments, we train a deep classifier as described in Section 3.4.3 to discriminate between a reference class $\{\mathbf{x}_{ref}^n\}_{n=1}^{N_{train}}$ and an outlier class $\{\mathbf{y}_{ref}^n\}_{n=1}^{N_{train}}$ where $\mathbf{y}^n$ is either a printed fake from the dataset or a synthetic augmentation $\tilde{\mathbf{x}}^n = enc(\mathbf{t}^n)$.

The encoder-decoder architecture used to produce digital augmentations is trained with the same loss as described in Section 3.5.1 that is with a heavy weight put on the latent loss $d_1(\mathbf{x}, \tilde{\mathbf{x}})$ with $\lambda_3 = \lambda \in \{1, 5, 10, 25\}$ and $\lambda_1 = \lambda_2 = \lambda_4 = 1$. The training set size is $N_{train} = 120$ samples and a the testing set size is 180 samples.

Baseline classifier results

As a baseline result to ensure the capacity of the classifier, we first train the classifier at the task of discriminating real samples $\mathbf{x}$ and printed fakes $\mathbf{f}$ where $\mathbf{f}$ is one of the four types of fake in the dataset.

Table 3.4 shows the result of the training procedure for the different type of fakes and for an optimal choice of threshold on the raw output of the classifier for each type of tested probes (each row corresponds to one type of probe under test). As expected and confirming previous researches, the results demonstrate that the deep classifier is indeed able to perfectly separate original and fake samples when the fake presented at testing time is the same type as the one used for training. Indeed, the average probability of classification error is zero for matching type of fakes during training and testing. When trying to generalize the approach to unseen fake type during training, we see that results start to deviate from perfect separability. Results indicate that Fakes type 1 and 2 ($\mathbf{f}_{1w}$ and $\mathbf{f}_{1g}$) are higher quality fakes than Fakes type 3 and 4 ($\mathbf{f}_{2w}$ and $\mathbf{f}_{2g}$) in the sense that a classifier trained against type 1 or type 2 fake are able to generalize to other types of fakes (false acceptance is zero for type 3 and 4) while the converse is not true (Fake 1 and Fake 2 have a non-zero false acceptance rate when training is performed on Fake 3 and 4). Another observation that can be made is the high error classification score of synthetic samples. We interprete these results as one more demonstration that synthetic augmentations are closer to the reference dataset $\mathbf{x}_{ref}$ than any type of real printed fakes. The test on synthetic samples also illustrates the difference of quality between fakes 1 and 2 versus fakes 3 and 4 with an error rate 5 to 10 times bigger on the latter.

Table 3.4. Minimal average probability of classification error P_{err}^{min} (in %) of a binary class classifier trained at discriminating a reference class $\mathbf{x}_{ref}$ and an outlier class $\mathbf{f}$ for four different types of fakes. The outlier classes used for training correspond to the four rows of the Table. The minimal average probability of error is calculated as the average between the probability of miss and the probability of false acceptance for an optimal threshold between the reference inlier class and the tested outlier class.

	Train outlier class			
Test outlier class	printed $\mathbf{f}_{1w}$	printed $\mathbf{f}_{1g}$	printed $\mathbf{f}_{2w}$	printed $\mathbf{f}_{2g}$
Printed $\mathbf{f}_{1w}$	0.00	0.00	0.00	0.36
Printed $\mathbf{f}_{1g}$	0.00	0.00	1.07	0.71
Printed $\mathbf{f}_{2w}$	0.00	0.00	0.00	0.00
Printed $\mathbf{f}_{2g}$	0.00	0.00	0.00	0.00
Synthetic $\lambda = 1$	3.21	5.36	**21.79**	**25.00**
Synthetic $\lambda = 5$	3.57	6.43	23.57	25.36
Synthetic $\lambda = 10$	7.14	9.29	28.21	30.36
Synthetic $\lambda = 25$	**1.07**	**2.86**	22.86	28.21

These results are further confirmed in Fig. 3.17 where the ROC curves in the first two rows (corresponding to fakes type 1 and type 2 used during training) show a perfect separability for all type of fakes, while the ROC curve on the two bottom rows (corresponding to fakes type 3 and type 4 used during training) do not perform optimally for fakes $\mathbf{f}_{1w}$ and $\mathbf{f}_{1g}$. Yet another insightful illustration of this fact is obtained by applying a t-SNE 2D projection to the second to last 16 dimensional layer of the classifier. In these plots, we see that original reference and original printed are perfectly separated from fakes when the classifier is trained on type 1 and type 2 fakes, while they are getting slightly overlapped with type 1 and 2 fakes, when the classifier is trained on type 3 or type 4 fakes.

Deep classifier trained on synthetic samples

In this second experiment, we actually train a deep classifier at discriminating a reference class $\mathbf{x}_{ref}$ and an outlier class that consists of synthetic augmentations $\tilde{\mathbf{x}}$. The training is done similarly to the process described in the baseline experiment and test is performed both on synthetic samples $\tilde{\mathbf{x}}$ and actual printed fakes $\mathbf{f}$. Table 3.5 shows the average probability of error of classification for different probes used during tests on each row. The classifier has been trained with four different types of outlier samples produced using the encoder-decoder architecture described in Section 3.3.2 for four different weights λ on the latent loss $\ell_1(\mathbf{x}_{ref}, \tilde{\mathbf{x}})$ and the best performing classifier has been selected for each training set.

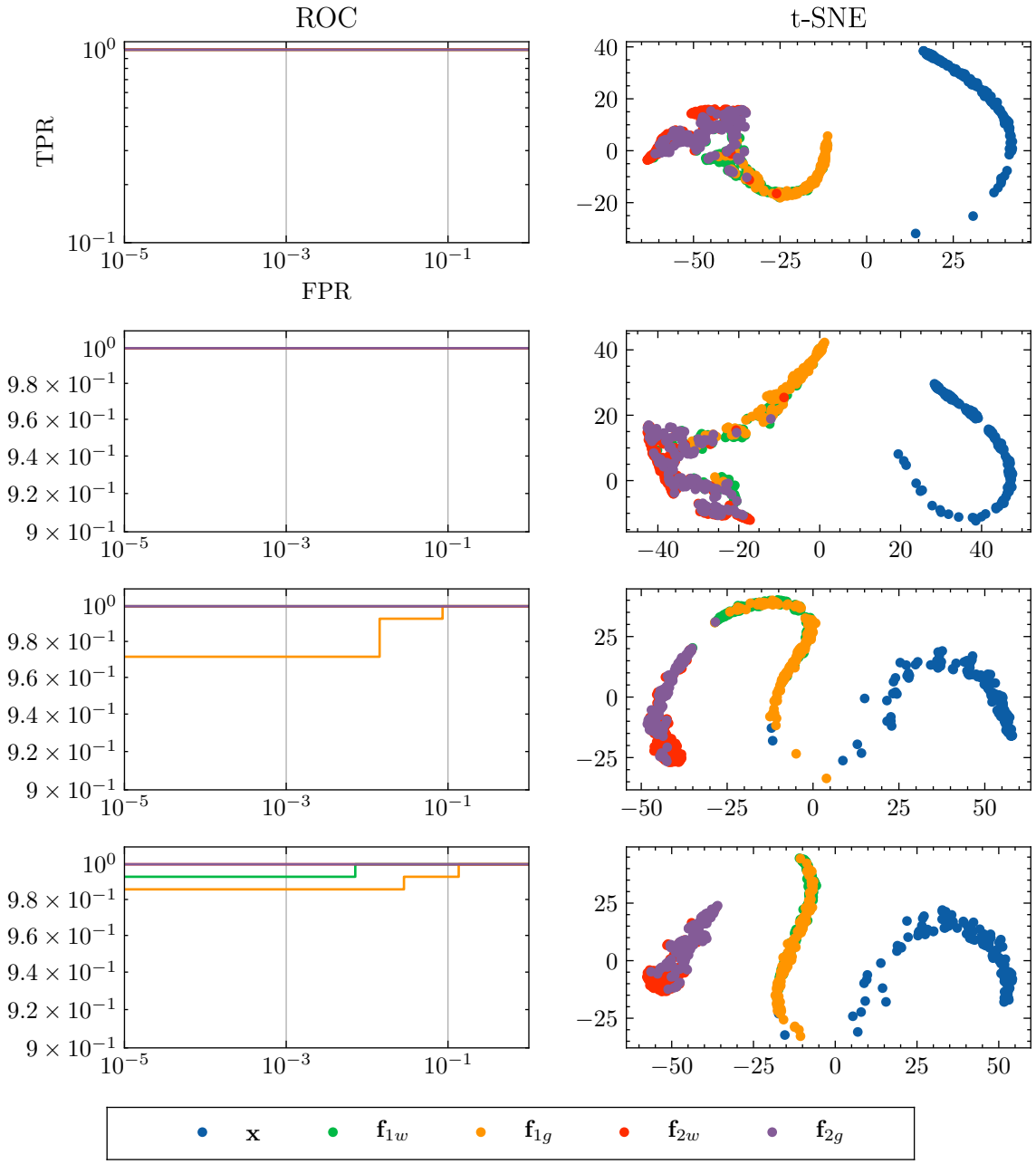


Figure 3.17. ROC curves (left) and t-SNE visualisation of the 16D second to last layer of the classifier (right) for a two classifier trained at discriminating between a reference dataset $\mathbf{x}_{ref}$ and a fake dataset $\mathbf{f}$ for four different types of fakes (row 1: Fake type 1, row 2: Fake type 2, etc.). The trained classifier is then tested on all types of fakes.

Table 3.5. Minimal average probability of classification error P_{err}^{min} (in %) of a binary class classifier trained at discriminating a reference class $\mathbf{x}_{ref}$ and an outlier synthetic class $\tilde{\mathbf{x}}$ for four different generation of synthetic augmentations. The outlier classes used for training correspond to the four rows of the Table. The average probability of error is calculated as the average between the probability of miss and the probability of false acceptance for an optimal threshold between the reference class and the tested fake class.

	Train outlier class			
Test outlier class	$\tilde{\mathbf{x}}$ with $\lambda = 1$	$\tilde{\mathbf{x}}$ with $\lambda = 5$	$\tilde{\mathbf{x}}$ with $\lambda = 10$	$\tilde{\mathbf{x}}$ with $\lambda = 25$
Printed $\mathbf{f}_{1w}$	0.71	1.43	0.36	**0.36**
Printed $\mathbf{f}_{1g}$	2.86	6.43	8.21	**2.50**
Printed $\mathbf{f}_{2w}$	1.43	14.29	10.71	**0.71**
Printed $\mathbf{f}_{2g}$	3.93	14.64	15.00	**1.43**
Synthetic $\lambda = 1$	0.36	1.07	0.36	0.00
Synthetic $\lambda = 5$	0.36	0.00	0.00	0.00
Synthetic $\lambda = 10$	0.36	0.00	0.00	0.00
Synthetic $\lambda = 25$	0.36	0.00	0.36	0.00

The most striking observation when looking at Table 3.5 is the outstanding performance of the classifier on all class of real printed fakes in most cases. Indeed, although being trained on synthetic samples, that is without actual reference to real printed fakes, the classifier is able to separate original CDP from all types of real fakes with an average probability of classification error not greater than 2.5% (last column based on synthetic samples with $\lambda = 25$). These kind of results and experiments have, up to our best knowledge, never been studied before and show a very promising approach to CDP authentication based on deep classifiers without the need of real fakes for training.

Fig. 3.18 illustrate the results from Table 3.5 by showing the corresponding ROC curve (left column) and the t-SNE based on the second to last activation layer of the classifier. We find the t-SNE scatterplot for $\lambda = 1$ (first row) to be very informative about the way the synthetic augmentations (in gray) are actually pushing the type 1 and 2 fakes (in orange and green) away from the reference original CDP (in blue). Results for $\lambda = 5$ and $\lambda = 10$ (second and third row) illustrate how the classifier may fail to properly separate originals and fakes. Finally, results for $\lambda = 25$ show how the training may result in good separability.

All these results should however be taken with caution since, as described above, we selected the best performing classifiers to demonstrate the power of this approach. Despite achieving spectacular performances, the stability of results raises serious concerns that requires even deeper investigation of involved factors.

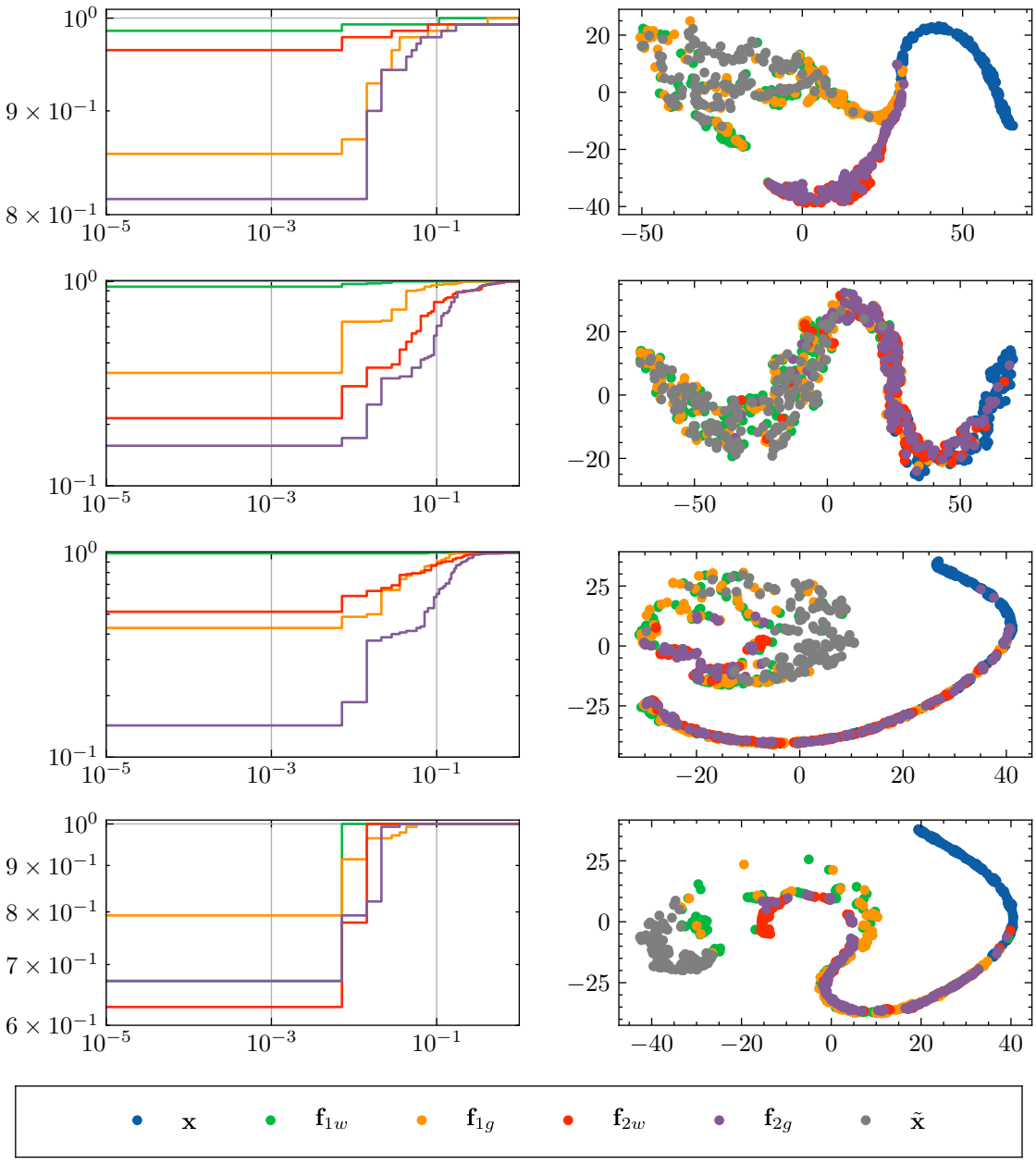


Figure 3.18. ROC curves (left) and t-SNE visualisation of the 16D second to last layer of the classifier (right) for a two classifier trained at discriminating between a reference dataset $\mathbf{x}_{ref}$ and a synthetic dataset $\tilde{x}$ of digital augmentations for four different type of synthetic samples (row 1: $\lambda = 1$, row 2: $\lambda = 5$, row 3: $\lambda = 10$, row 4: $\lambda = 25$). The trained classifier is then tested on four types of fakes.

Classifier training stability

The last experiment that we conduct is concerned with the variability of the results with respect to the training random initialization. To this end, we conduct the exact same training of deep binary classifier based on reference dataset $\mathbf{x}_{ref}$ and synthetic augmentation $\tilde{\mathbf{x}}$, and only change the random seed for the splitting of training-validation-testing set and the augmentations and batch selections during training. The results are shown in Table 3.6 for three different seeds and for a fixed type of outlier synthetic class $\lambda = 1$. The results in the Table show that the stability of results is quite poor, with the classifier performing well on either type 1 and 2 fakes (seed 1 & 3) or type 3 and 4 fakes (seed 2). The t-SNE plots in Fig. 3.19 correspond to each training session and are another illustration of this phenomenon where we see that for seed 1, the classifier learns features on the synthetic dataset that are mostly irrelevant to separate actual printed CDP from fakes. On the contrary, seed 3 shows a situation where the synthetic samples push away the real fakes. Seed 2 also shows a similar behaviour but with less success in terms of overlap between original and fake samples.

We interprete these results as a warning on using deep neural networks at the heart of critical security frameworks. Indeed, as of today, we have very little understanding regarding why the exact same architecture, trained in the exact same way, can on one experiment perform outstandingly well and on another experiment, suffer from important deviations.

Table 3.6. Minimal average probability of classification error P_{err}^{min} (in %) of a binary class classifier trained at discriminating a reference class $\mathbf{x}_{ref}$ and the same outlier synthetic class $\tilde{\mathbf{x}}$ with $\lambda = 1$ for different choice of training seed. The different seeds used for training correspond to the three columns of the table. The average probability of error is calculated as the average between the probability of miss and the probability of false acceptance for an optimal threshold between the reference class and the tested fake class.

	Train outlier class $\tilde{\mathbf{x}}$		
Test outlier class	Seed 1	Seed 2	Seed 3
Printed $\mathbf{f}_{1w}$	0.36	6.07	0.71
Printed $\mathbf{f}_{1g}$	5.36	15.71	2.86
Printed $\mathbf{f}_{2w}$	14.29	0.00	1.43
Printed $\mathbf{f}_{2g}$	16.07	0.00	3.93

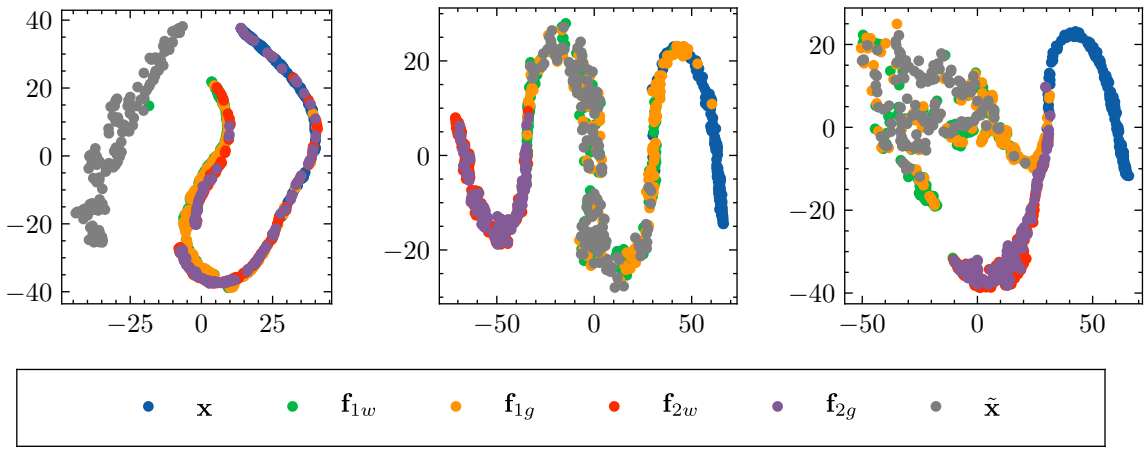


Figure 3.19. t-SNE visualisation of the 16D second to last layer of the classifier for a two-class classifier trained at discriminating between a reference dataset $\mathbf{x}_{ref}$ and a synthetic dataset $\tilde{x}$ of digital augmentations. The three columns correspond to three different training seed.

3.6 Conclusion

In this Chapter, we studied how the problem of secure CDP authentication can be solved using a deep learning based approach. By developing a deep learning clone of the printing imaging system based on an encoder-decoder architecture, we have shown that this model can be used to complement and enhance traditional authentication scheme based on standard metrics. However, several limitations regarding the quality of the synthetic digital twins produced and the stability of the training of such systems have also been pointed out.

In a second part of the Chapter, we have extended the use of digital twins to the training of deep classifiers without the need of actual fakes. We have demonstrated that this approach is very powerful and can actually detect unseen fakes if correctly trained and initialized. Despite these remarkable performances, we showed that this approach also suffers from many instabilities, making it hard to implement in a real-life scenario where actual protection guarantees are required.

As such, this Chapter, despite presenting state-of-the-art performances lacks solid grounds for the design of a reliable authentication scheme based on deep learning systems. Deep models acting, in essence, as black boxes prevent us from giving theoretical performance guarantees that are much needed in real life scenarios.

In this regard, in the following Chapters, we will focus our attention on a different model that provides theoretical guarantees and interpretability while still achieving comparable performances, without the need of resorting to deep learning models. This approach is introduced in Chapter 4 and Chapter 5 and theoretical guarantees are studied in Chapter 6.

Chapter 4

Mathematical Model of Printing-Imaging Channel

In this Chapter, we study the printing-imaging channel of CDP authentication from a statistical viewpoint. Chapter 3 concentrated on mimicking the printing-imaging channel using a deep learning based encoder-decoder model and demonstrated that despite achieving optimal performances in good situations, the price of relying on deep learning models was a lack of understanding of the underlying process and unreliable variability when used for authentication of CDP. This approach thus fundamentally lacks reliable foundations when trying to build reliable and understandable tools to strenghten CDP authentication.

In contrast to this situation, the mathematical model that we introduce in this Chapter provides fully interpretable and comprehensive tools for the authentication of CDP. In this Chapter, we concentrate on introducing formally the mathematical model and its key notions such as the pattern-based neighbourhood, bit-flipping probability and posterior log-likelihood probability measure. In Chapter 5, we study how this model can be applied to strengthen existing CDP authentication and demonstrate the superiority of this approach compared to standard authentication. Finally, Chapter 6 introduces new tools to optimally use this model for CDP authentication.

4.1 Introduction

In Chapter 3, we demonstrated the limits of deep learning based approach when studying the deep learning based encoder-decoder description of the printing-imaging channel. Indeed, despite providing impressive results on selected trained model, we demonstrated that such models lack stability and their performance are dramatically affected by random training factors (up to our best knowledge) and thus cannot be trusted for building a reliable security framework with critical consequences where actual performance guarantees are required. Therefore, there is a critical need in a one-class (OC) authentication scheme able to operate

in the generalized setup of the above printing-imaging channel without prior knowledge of the fakes.

4.1.1 Our Contributions

In view of these shortcomings, we present in this Chapter a theoretical mathematical model called the Binary Pattern-based Channel (BPC) model. This Chapter is based out on the conference paper [67] and the journal paper [68] and introduces the following notions:

- Propose a new mathematical stochastic model called the Binary Pattern-based Channel (BPC) model presenting it as an evolution of the stationary Binary Symmetric Channels (BSC) from information theory [25, 79] and linking it with previous related works addressing bit reliability in identification and retrieval applications [80, 81];
- Introduce the notion of neighbouring pattern for a given symbol in a digital template $\mathbf{t}$ and the notion of reliability for fixed patterns;
- Introduce a new measure of similarity called the Posterior Log-likelihood (PLL) for printed CDP which outperforms the standard metrics for fake CDP detection;
- Validate the proposed approach and explore the notions introduced above on a real dataset of CDP of originals and ML-based fakes based on codes designed with 1×1 symbol and produced on two industrial printers, acquired using a high-definition scanner and two popular smartphones from 2021 (see Section 4.4 for more details).

The Chapter is organized as follows: Section 4.3 introduces the classical framework of CDP-based authentication schemes and present the Binary Pattern-based Channel model and its properties that form the theoretical foundations on which the new framework is built. Section 4.4 and 4.5 describe the experiments that were performed on a real dataset of CDP enrolled with smartphones and discuss the results obtained. The final Section 4.6 concludes the Chapter. All mathematical notations used in this Chapter can be found in Table 4.1.

Table 4.1. Mathematical notations used in Chapter 4. Note that we are using bold font to denote vectors or matrices and capital letters to denote random variables in general.

	Mathematical notation	Meaning
CDP	$\mathbf{t}$	binary digital template
	$\mathbf{x}$	digital original printed from $\mathbf{t}$
	$\mathbf{f}$	digital fake version of $\mathbf{t}$
	$\mathbf{y}$	probe representing either $\mathbf{x}$ or $\mathbf{f}$
	$\tilde{\mathbf{t}}$	digital template estimated from $\mathbf{y}$
BPC	$\mathbf{T}$	binary random matrix for $\mathbf{t}$
	$\mathbf{X}$	random matrix for $\mathbf{x}$
	$\tilde{\mathbf{T}}$	binary random matrix for $\tilde{\mathbf{t}}$
	$\mathbf{\Omega}_{ij}$	random pattern in $\mathbf{T}$
	$p \in [0,1]$	probability of white symbol in $\mathbf{T}$
	$\omega \in \Pi$	set of all patterns
	$P_b(\omega)$	probability of bit-flipping at ω
	L^ω	number of time ω appears in $\mathbf{T}$
	$\mathcal{C}$	codebook of probabilities
	$\mathbb{P}$	probability measure for $(\mathbf{T}, \mathbf{X}, \tilde{\mathbf{T}})$
Numbers	$n = 1, ..., N$	index within the dataset
	(i,j) or (r,s)	coordinates of pixels in $\mathbf{t}$
	$B = L \times L$	size of $\mathbf{t}$
	$h = 1, 3, 5, ...$	integer defining the size of ω
	$k = 1, 2, 3, ...$	magnification factor from $\mathbf{t}$ to $\mathbf{x}$
	$\mu \in [0,1]$	threshold used for reliable patterns
	$\gamma \in \mathbb{R}$	decision threshold for measures

4.2 Research Questions

How can the Binary Symmetric Channel model from information theory be adapted to describe the printing-imaging channel of CDP authentication ?

By carefully studying the strengths and limitations of the Binary Symmetric Channel when applied to the CDP authentication scheme, we formulate a new binary model that is able to capture the geometric dependencies that appear during the printing-imaging process of CDP. To this end, we introduce the Binary Pattern-based Channel model.

Are there relevant statistics that can be defined based on the BPC model ?

Starting from the mathematical model introduced above, we define the key notion of pattern reliability and mathematically demonstrate its soundness. We also study the posterior distribution and define a measure of reliability based on this notion.

How well does the BPC model actually describe reality when tested on a real dataset of acquired CDP ?

Using a reference dataset of printed and attacked CDP channel, we provide experiments demonstrating that the model introduced in this Chapter properly describes the random behaviour observed in reality for varying printing and imaging modalities.

4.3 Mathematical Model of the Printing-Imaging Channel

4.3.1 Markov Chain and Communication Channels

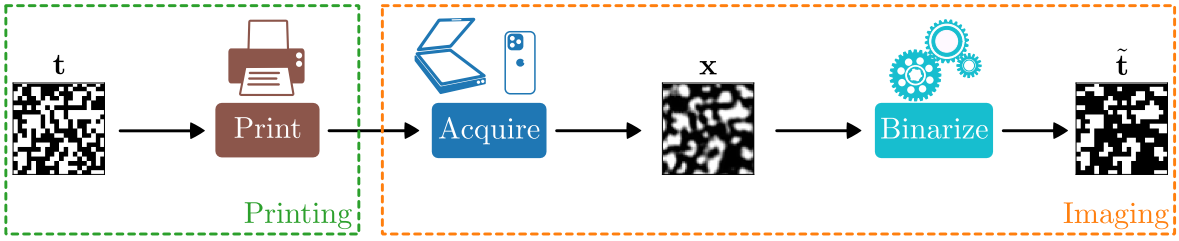


Figure 4.1. A visual representation of the printing-imaging process. Notice how the dot-gain effect at the printing stage creates dependencies on neighbouring symbols and thus implies non-uniformity for the probability of bit-flipping.

The proposed mathematical model describes the process of printing and imaging as a Markov Chain composed of three random matrices $\mathbf{T} \to \mathbf{X} \to \tilde{\mathbf{T}}$, where:

- $\mathbf{T}$ is a random binary matrix of size $B = L \times L$ sampled from i.i.d. Bernoulli distribution: $T_{ij} \sim Bern(p)$, $p \in [0, 1]$ is the probability of white symbol;
- $\mathbf{X}$ is a random matrix of size $kL \times kL$, $X_{ij} \in [0, 1]$ for some magnification factor[1] $k = 1, 2, 3, ...$;
- $\tilde{\mathbf{T}}$ is a random binary matrix of size $B = L \times L$.

The core idea of the model relies on information theory, where each symbol location (i, j) gives rise to a communication channel $T_{ij} \mapsto \tilde{T}_{ij}$. Thus, a very natural question arises about how reliable each channel is, which we measure in terms of posterior probability of bit-flip. For each symbol location (i, j), we define the posterior probability of bit-flipping given a fixed template $\mathbf{T} = \mathbf{t}$:

$$P_b(i, j \mid \mathbf{T} = \mathbf{t}) := \mathbb{P}(T_{ij} \neq \tilde{T}_{ij} \mid \mathbf{T} = \mathbf{t}) \quad \forall i, j = 1, \dots, L. \tag{4.1}$$

The underlying real process this model is trying to describe is the following: when a digital template $\mathbf{t}$ is being printed and acquired as an image $\mathbf{x}$, some distortions occur in $\mathbf{x}$ due to the dot-gain effect, printing-related natural randomness and acquisition conditions. Thus, when one tries to estimate $\tilde{\mathbf{t}}$ from $\mathbf{x}$, it will end up with some errors, dependant on the printer, type of paper, acquisition device, conditions and chosen estimator. Fig. 4.1 illustrates this degradation process.

A first model describing this phenomenon was studied in [25]. In this article, the authors model the probability of bit-flip as a Binary Symmetric Channel (BSC):

[1]The magnification factor is related to the resolution of printing and enrollment equipment. It reflects how many dots of printer are being used to represent one symbol of the template. At the same time, it also depends on how many samples of imaging device are allocated per printed dots.

$$P_b(i,j \mid \mathbf{T}=\mathbf{t}) = \mathbb{P}(\tilde{T}_{ij} \neq T_{ij}) =: \bar{P}_b \in [0,1] \quad \forall i,j. \tag{4.2}$$

This model assumes that all symbols T_{ij} of $\mathbf{T}$ have the same average probability of bit-flip $\bar{P}_b$. Although being very common in information theory, this model is too restrictive to be able to capture the random dependencies of the printing-imaging process. Indeed, the BSC model imposes the following assumptions:

1. *Symmetry:* the probability of bit-flipping from a black symbol to a white symbol is the same as from a white symbol to a black one;
2. *Stationarity and independence:* all channels are independent and identically distributed (iid), regardless of their particular location (i,j).

These assumptions are very strong and do not match the real behaviour of the printing-imaging process. Indeed, when looking at Fig. 4.1, one easily notices that the dot-gain effect breaks the symmetry, with black symbols having a lower probability of bit-flip than white symbols. Furthermore, the random deviations of the printing process creates a lot of local dependencies on neighbouring symbols. White symbols surrounded by black symbols tend to flip more easily than those surrounded by white pixels. As such, independency is not a realistic assumption neither. Stationarity however, if carefuly stated, should be an expected behaviour, as there is no physical reason for the printing process to differ from one location to another on the paper. Another related model with multilevel symbols was studied in [82].

4.3.2 Proposed Binary Pattern-Based Channel Model

Based on the limitations of the standard BSC model, we introduce the Binary Pattern-based Channel (BPC) model, a new stochastic model aimed at better describing the local dependencies of the printing-imaging process. To this end, we introduce the notion of pattern:

Definition 1. *A (random) pattern $\mathbf{\Omega}_{ij} \subset \mathbf{T}$, is a small neighbourhood surrounding symbol T_{ij} in $\mathbf{T}$. A (fixed) pattern is a corresponding small neighbourhood $\omega_{ij} \subset \mathbf{t}$ from a fixed template $\mathbf{t}$. The set of all possible fixed patterns ω is denoted by Π. In practice, we will restrict our study to square patterns centered around (i,j):*

$$\mathbf{\Omega}_{ij} = \{T_{i\pm a, j\pm b} \mid 0 \leq a,b < h/2\}, \tag{4.3}$$

where $h = 1,3,5,...$ is fixed and describes the size of the square.[2]

[2]Notice that, from a technical point of view, the pattern $\mathbf{\Omega}_{ij}$ written with a capital letter is itself a random binary matrix of size $h \times h$, while a pattern $\omega \in \Pi$, written with a lowercase letter, describes a fixed binary matrix of size $h \times h$.

We now introduce the assumptions of the proposed model:

1. *Locality:* the posterior probability of $\tilde{T}_{ij}$ at a particular symbol location (i,j) only depends on the local pattern $\mathbf{\Omega}_{ij}$ surrounding it:

$$\mathbb{P}(\tilde{T}_{ij} \mid \mathbf{T} = \mathbf{t}) = \mathbb{P}(\tilde{T}_{ij} \mid \mathbf{\Omega}_{ij} = \omega_{ij}). \tag{4.4}$$

2. *Stationarity:* the posterior probability does not depend on the location inside the image. Similar patterns in $\mathbf{T}$ lead to similar probability distributions:[3]

$$\mathbb{P}(\tilde{T}_{ij} \mid \mathbf{\Omega}_{ij} = \omega_{ij}) = \mathbb{P}(\tilde{T}_{rs} \mid \mathbf{\Omega}_{rs} = \omega_{rs}), \text{ if } \omega_{ij} = \omega_{rs}. \tag{4.5}$$

3. *Posterior independance:* the joint posterior probability factorizes as:

$$\mathbb{P}(\tilde{\mathbf{T}} \mid \mathbf{T} = \mathbf{t}) = \prod_{i,j} \mathbb{P}(\tilde{T}_{ij} \mid \mathbf{T} = \mathbf{t}). \tag{4.6}$$

These assumptions are illustrated in Fig. 4.2, where a similar pattern ω in $\mathbf{t}$ leads to various outcomes in the estimated template $\tilde{\mathbf{t}}$. Fig. 4.3 illustrates the fundamental difference between the BSC model and the BPC model.

4.3.3 Concept of Pattern Reliability

With assumptions (4.4) and (4.5), one can easily construct an unbiased estimator for the posterior distribution for any given pattern $\omega \in \Pi$:

$$\mathbb{P}(\tilde{T}_{ij} \mid \mathbf{\Omega}_{ij} = \omega) = \frac{1}{L^{\omega}} \sum_{\substack{r,s: \\ \mathbf{\Omega}_{rs} = \omega}} \tilde{T}_{rs}. \tag{4.7}$$

where the sum is taken over all patterns $\mathbf{\Omega}_{rs}$ in $\mathbf{T}$ that are similar to ω.

This formula is a key to the proposed model as it can be estimated directly using statistical inference on a training dataset. For each type of pattern $\omega \in \Pi$ (there can be at most $|\Pi| = 2^{h^2}$ such patterns), we learn the probability distribution which is highly related with the printing-imaging process on which it was trained. The natural measure associated with this distribution is the *probability of bit-flipping of the central symbol in pattern* ω which we define as:

$$P_b(\omega) := \frac{1}{L^{\omega}} \sum_{\substack{i,j: \\ \omega_{ij} = \omega}} 1_{\{\tilde{t}_{ij} \neq t_{ij}\}}, \tag{4.8}$$

[3]The printing and imaging process introduces a lot of variability. The goal of the model is not to learn the fingerprint of a particular realization but rather measure the average variability for each neighbourhood and to take advantage of this knowledge. (4.5) should be read as an equality in distribution, allowing every realisation of $\tilde{T}_{i,j}$ to be different while still following a common law, independent of the location (i,j).

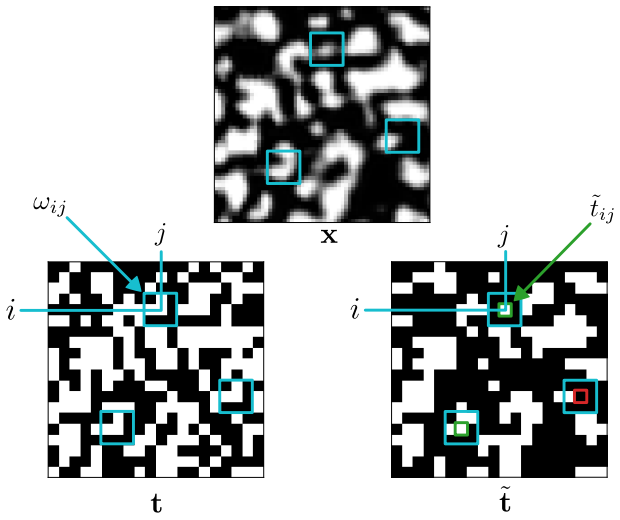


Figure 4.2. An illustration of the BPC model. The pattern ω_{ij} (in blue) appears three times in the template $\mathbf{t}$. This same pattern undergoes various random deformations in $\mathbf{x}$ leading to correct and incorrect symbol estimates in $\tilde{\mathbf{t}}$ (see the green and red symbols in $\tilde{\mathbf{t}}$. Measuring the average behaviour of the estimated symbol $\tilde{t}_{ij}$ is a relevant measure of the reliability of the pattern ω_{ij}.

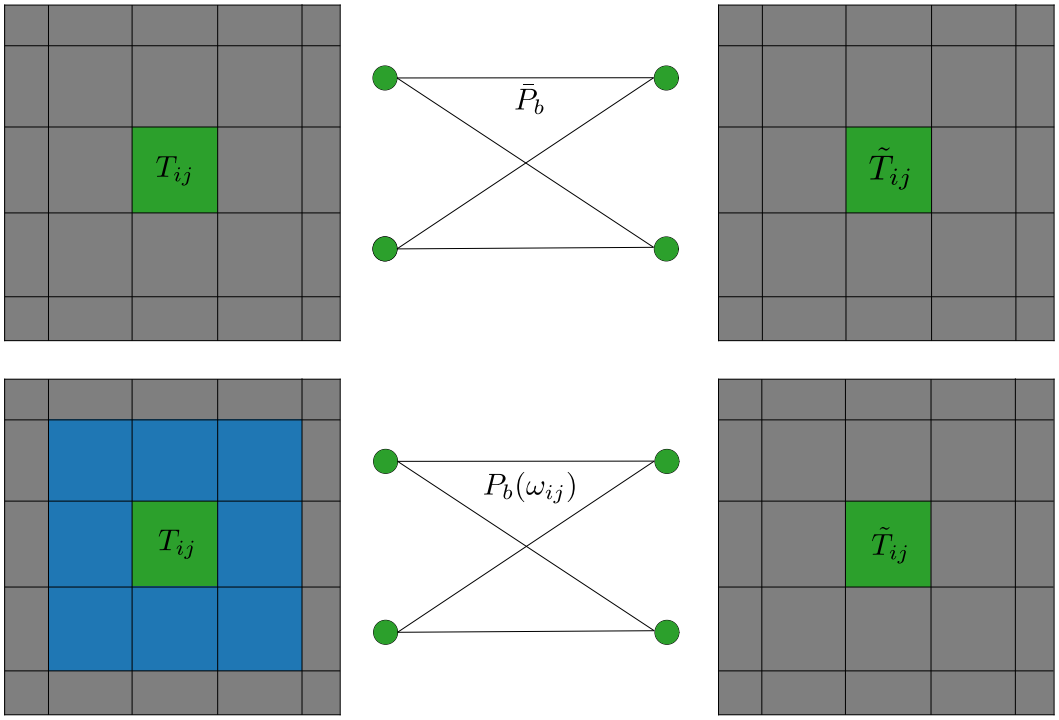


Figure 4.3. A visual comparison of the BSC model (on top) and the BPC model (on bottom). In the BSC model, each symbol T_{ij} has the same probability of bit-flip $\bar{P}_b$ independent of the neighbouring patterns (in gray). In the BPC model, the probability of bit-flip $P_b(\omega_{ij})$ depends on the local pattern ω_{ij} (in blue) surrounding the symbol T_{ij}.

where $1_{\{condition\}}$ denotes an indicator function, which is equal to 1 when *condition* is true and 0 otherwise. This equality is a statistical estimator of the conditional expectation with L^{ω} representing the number of times the pattern ω appears in the template $\mathbf{t}$. We can thus create a codebook $\mathcal{C}$ in which we store all these different probability values for each type of neighbourhood and use them as references in the authentication scheme.

4.3.4 Posterior Log-Likelihood Score

The BPC model described above gives us a theoretical tool to better understanding the process of printing and acquisition of CDP. In this subsection, we show that this model comes with a very natural metric that can be easily implemented and used for authentication. Indeed, given an estimated template $\tilde{\mathbf{t}}$, one can ask how likely it is that the BPC model produced such an outcome. The answer is given by the following lemma:

Lemma 4.3.1. *In the BPC model, the Posterior Log-Likelihood (PLL) can be computed as:*

$$PLL(\mathbf{t}, \mathbf{y} \mid \mathcal{C}) := \log \mathbb{P}(\tilde{\mathbf{T}} = \tilde{\mathbf{t}} \mid \mathbf{T} = \mathbf{t}) = \sum_{i,j} \log \pi_{ij}, \tag{4.9}$$

where,

$$\pi_{ij} = \begin{cases} 1 - P_b(\omega_{ij}) & \text{if } \tilde{t}_{ij} = t_{ij} \\ P_b(\omega_{ij}) & \text{if } \tilde{t}_{ij} \neq t_{ij}. \end{cases} \tag{4.10}$$

Proof. The proof relies on two steps. The first one is to use posterior independence of the symbols in $\tilde{\mathbf{T}}$ given $\mathbf{T}$ (4.6) and locality (4.4):

$$\begin{aligned} \log \mathbb{P}(\tilde{\mathbf{T}} = \tilde{\mathbf{t}} \mid \mathbf{T} = \mathbf{t}) &= \sum_{i,j} \log \mathbb{P}(\tilde{T}_{ij} = \tilde{t}_{ij} \mid \mathbf{T} = \mathbf{t}) \\ &= \sum_{i,j} \log \mathbb{P}(\tilde{T}_{ij} = \tilde{t}_{ij} \mid \mathbf{\Omega}_{ij} = \omega_{ij}). \end{aligned}$$

The last step is a simple case study for $t_{ij}, \tilde{t}_{ij} \in \{0, 1\}$:

$$\pi_{ij} = \mathbb{P}(\tilde{T}_{ij} = \tilde{t}_{ij} \mid \mathbf{\Omega}_{ij} = \omega_{ij}) = \begin{cases} 1 - P_b(\omega_{ij}) & \text{if } \tilde{t}_{ij} = t_{ij} \\ P_b(\omega_{ij}) & \text{if } \tilde{t}_{ij} \neq t_{ij}. \end{cases}$$

■

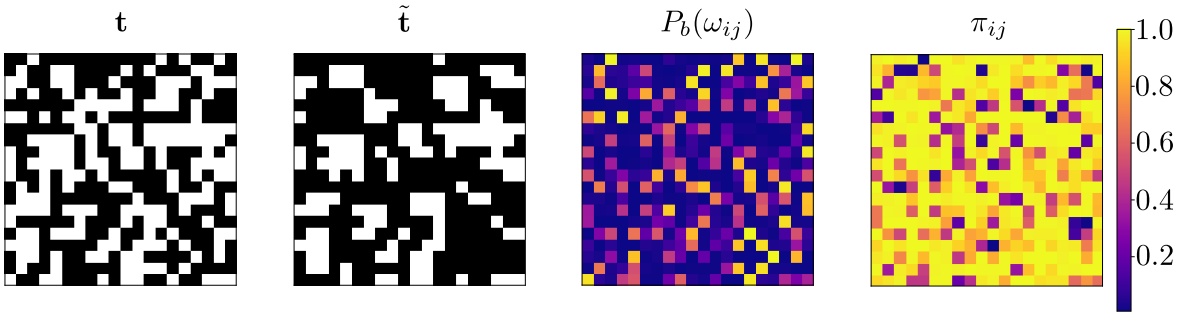


Figure 4.4. An illustration of the various quantities used in the definition of the PLL metric. From left to right, the first image is a binary template $\mathbf{t}$, the second image its estimated counterpart $\tilde{\mathbf{t}}$ after the printing-imaging-binarization process, the third image shows the probability of bit-flipping $P_b(\omega_{ij})$ of the central pixel for each pattern ω_{ij} appearing in $\mathbf{t}$ and the fourth image shows the actual probability π_{ij} of each pixel in $\tilde{\mathbf{t}}$ according to the BPC model.

4.4 Experimental Setup

4.4.1 Indigo 1x1 CDP Base Dataset

The dataset we use in this Chapter and the following Chapters is the Indigo 1×1 base dataset that was created in the SIP group and presented in the following articles [43, 44].[4] This dataset consists of 1440 unique digital templates $\mathbf{t}$ printed with two different printers: *HP Indigo 5500 DS* (HPI55) and *HP Indigo 7600 DS* (HPI76) at 812.8 dpi, and enrolled with three different devices: An *iPhone 12 Pro*, a *Samsung Galaxy Note 20 Ultra* and a high resolution scanner *Epson Perfection V850 Pro* at 2400 dpi.

The dataset also includes deep learning-based fakes obtained from originals by the process of scanning with the high resolution scanner, deepnet-based binarization, and reprinted using the same printers. These fakes are then acquired with the same mobile phones and scanner in the same conditions. In order to account for the random variability of the acquisition device, the capture by mobile phone is performed 6 times (hereafter referred to as runs). In summary, the dataset consists of the following samples:

1. *Reference Digital Templates* $\mathbf{t}$*:*
   1440 samples of size 228×228
2. *Acquisition Done with Scanner:*
   6×720 samples of size 684×684, magnification factor $k = 3$,
   - Original CDP printed with HPI55 and HPI76: $\mathbf{x}^{55}_{scanner}$ and $\mathbf{x}^{76}_{scanner}$;
   - Four different fakes:[5] $\mathbf{f}^{55/55}_{scanner}$, $\mathbf{f}^{76/55}_{scanner}$, $\mathbf{f}^{55/76}_{scanner}$ and $\mathbf{f}^{76/76}_{scanner}$.

[4] The dataset is publicly available and can be found here: https://sipcloud.unige.ch/index.php/s/tYKffnKNRgSwBAN

[5] Fake $\mathbf{f}^{mm/nn}$ is obtained from $\mathbf{x}^{nn}$ by the process of deepnet-based binarization, reprinted using HPImm and rescanned.

3. *Acquisition Done with Smartphone:*
   4×1440 samples of size 228×228, magnification factor $k = 1$,
   - Original CDP printed with HPI55: $\mathbf{x}_{iphone}$ and $\mathbf{x}_{samsung}$;
   - Two fake CDP based on HPI55: $\mathbf{f}_{iphone}$ and $\mathbf{f}_{samsung}$.

The whole dataset is illustrated in Figure 4.5 where the creation process is outlined. The CDP acquired in this way are then processed using histogram matching to a reference CDP with a uniform histogram in order to mitigate the variability of the acquisition process and a quality check is performed to discard poorly captured CDP suffering from an obvious blurring effect.

4.4.2 Codebook Estimation Parameters

In order to learn the codebook, we need to fix a certain number of parameters and choose an estimator $\mathbf{x} \mapsto \tilde{\mathbf{t}}$. As discussed in Section 3.3.1, there are many different approaches to this problem. We decide to stick with Otsu's algorithm [77] for binarization as deep binarization techniques tend to give unpredictable outcomes which are not consistent for authentication. Moreover, we prefer to keep a simple explainable algorithm that can also be executed on any device. Note that for the dataset captured with scanner, an extra step of 3×3 majority voting is performed for each symbol after binarization so that the output size of $\tilde{\mathbf{t}}$ matches the size of the digital template $\mathbf{t}$. We fix the size of patterns ω in $\mathbf{t}$ to be 3×3 for the following reasons:

- This brings the total number of possible patterns down to $|\Pi| = 2^9 = 512$ which is small enough in comparison to the total number of patterns in a single template: $B = L^2 = 228^2 = 51'984$. We can thus expect to see every pattern appear roughly $L^\omega = \frac{B}{|\Pi|} \approx 100$ times on average in each template.[6]
- The printing process produces random deviations as discussed in Section 4.3, but these deviations are local in the sense that they only affect neighbouring symbols in most cases in the current dataset. Thus, 3×3 patterns are sufficient to capture them (see Fig. 4.1 and 4.2 for an illustration).

[6]This claim is obviously a simplification as the distribution of patterns Ω in a random template $\mathbf{T}$ is non-uniform. For a detailed study of pattern distribution in random 1D sequences see [83]. In practice, however, this simplification does not seem to impact the performance of the proposed model.

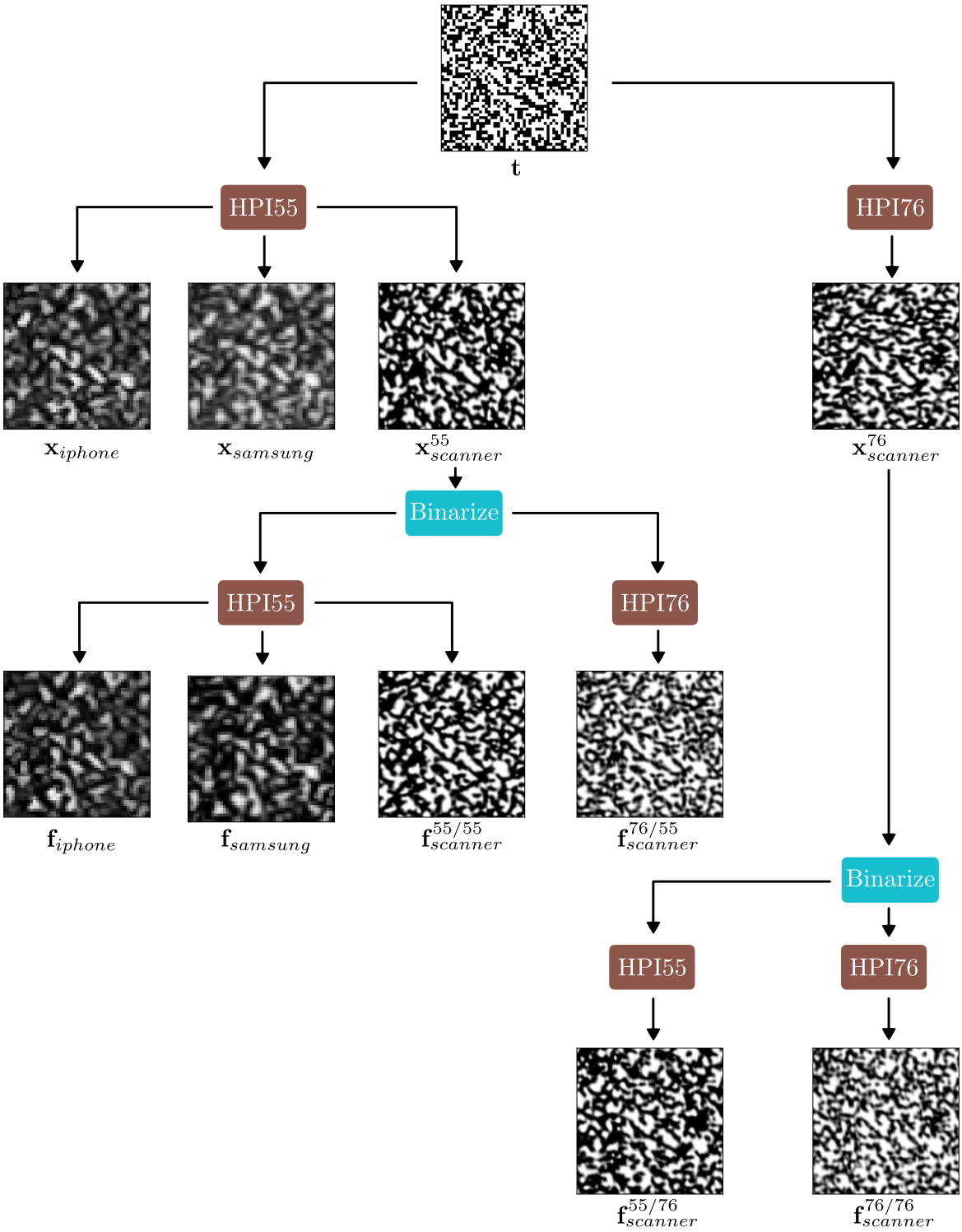


Figure 4.5. Visualisation of the Indigo 1×1 base dataset. Top-row is a digital template $\mathbf{t}$, second row are the original CDP $\mathbf{x}$ captured using three different devices and printed with two different printers. The third and fourth row are the deep learning based fake CDP $\mathbf{f}$ captured by the same devices and printed using the same two printers. Note that not all pairs of printer-acquisition are available in this dataset.

4.5 Experimental Results

4.5.1 Codebook Estimation and Pattern Reliability

In this first experiment, we measure the codebook introduced in section 4.3.3 and visualize the probability of bit-flipping for each pattern $\omega \in \Pi$. For simplicity of visualisation, each pattern is associated to a number $0, 1, ..., 511$, simply based on its flattened binary representation (see Fig. 4.6 for a visual example). In Fig. 4.7, we compare the reliability across different datasets of original CDP captured with iPhone, Samsung and Scanner and for the two different printers HPI55 and HPI76. The x-axis represents the 512 different patterns ω and the y-axis represents the probability of bit-flip $P_b(\omega)$ for each pattern in log-scale.

A first observation clearly indicates that some patterns are much more reliable than others and thus confirms that the locality hypothesis (4.4) of the BPC model makes sense. For instance, the all-white pattern 511 has a probability of bit-flip of zero. The other two patterns with a zero probability of bit-flip are pattern 255 (one black symbol in top-left corner) and pattern 447 (one black symbol in top-right corner). Moreover, we also see a clear correlation between the curves across the various acquisition device and printers. For instance the two codebooks based on scanner (first plot) show very similar behaviour, with the dataset based on printer HPI55 having sligthly smaller probability of bit-flipping. For smartphones (second plot), we see that the probability of bit-flipping is generally better for the iPhone dataset with the orange curve almost always below the Samsung green curve. Finally, comparing both smartphone and Scanner devices (third plot) shows that lower capture resolution leads to smaller bit-flipping probability in general. This stability across various capturing modalities is encouraging in view of testing cross-device performances where the codebook is trained on one device and the authentication is performed using other devices.

Although fake samples might not be available in real case scenarios, we decided to train the codebook algorithm both on originals and fakes in order to visualize the difference of behaviour between them. The results are shown in Fig. 4.8. A very important observation that we make is the striking difference in the performances between the original CDP and fake CDP. Indeed, as expected, fake samples have a higher probability of bit-flip than originals (the orange and green curves are always above the blue curves for all datasets). The strength of the authentication scheme of the BPC model relies on this difference and our ultimate goal is to select only the patterns where this difference is big. Notice also that this difference is bigger for the iPhone dataset than for the Samsung dataset which explains why the authentication scheme is less powerful on the latter. The best separability is obtained for the dataset printed with HPI55 and acquired using the high-resolution scanner.

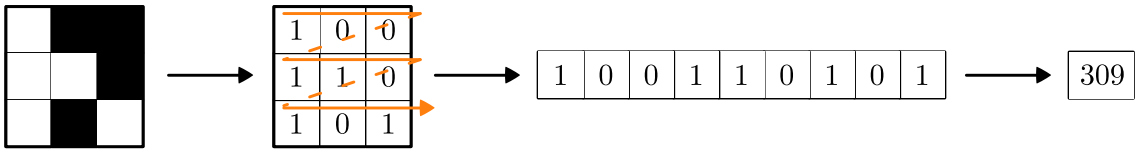


Figure 4.6. An example of translation from a binary 3×3 pattern to its decimal representation.

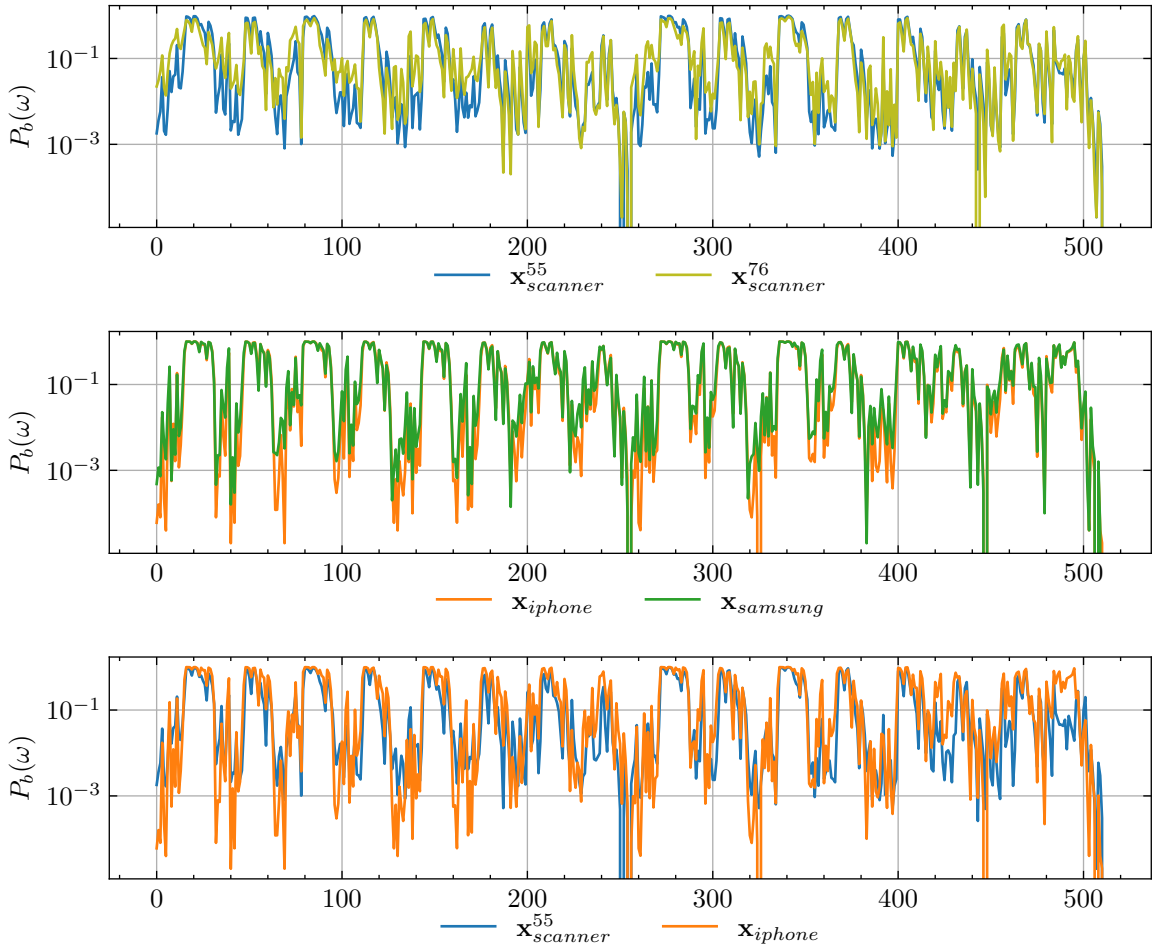


Figure 4.7. Comparison of codebooks using various acquisition devices and printers. The first row compares codebooks obtained using an HPI55 printer and an HPI76 printer. The second row compares two codebooks captured by iPhone and Samsung device. The third row compares two codebooks captured using a scanner and an iPhone.

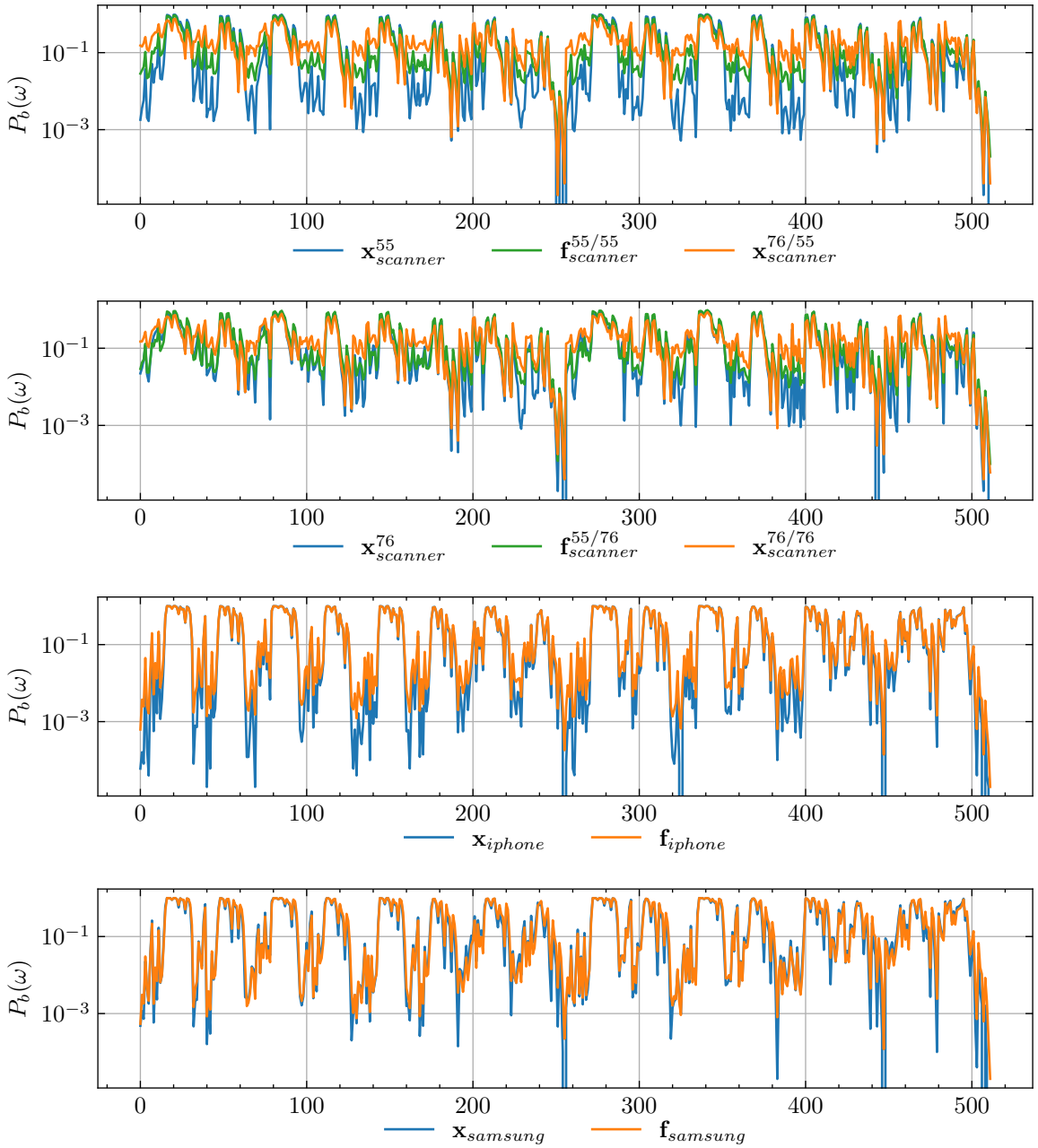


Figure 4.8. Comparison of codebooks for original dataset and fake dataset. The first and second row show the results for CDP acquired by scanner and for the attacks using two different printers. The third and fourth row show the results for CDP acquired by iPhone and Samsung.

4.5.2 Codebook Sensitivity to Training Set

The goal of this second experiment is to test the stability of the codebook estimation with respect to the number of training samples N_{train}. As discussed in Section 4.4.2, every pattern appears 100 times on average in each template, it thus makes sense to run the algorithm on very small training sets. In order to measure the performance of a codebook $\mathcal{C}$ learned on a training set $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{N_{train}}$, we compare it with a reference codebook $\mathcal{C}^{ref}$ learned on a big dataset of 500 pairs $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{500}$. The comparison is done by computing an average ℓ_1 distance between the codebooks:

$$\ell_1(\mathcal{C}, \mathcal{C}^{ref}) = \frac{1}{|\Pi|} \sum_{\omega \in \Pi} |P_b(\omega) - P_b^{ref}(\omega)|, \tag{4.11}$$

or a maximum ℓ_∞ distance:

$$\ell_\infty(\mathcal{C}, \mathcal{C}^{ref}) = \max_{\omega \in \Pi} |P_b(\omega) - P_b^{ref}(\omega)|. \tag{4.12}$$

Fig. 4.9 shows the results of this study for different training sets size with a number of samples going from 1 to 100 randomly selected. The curves show the average distance and standard deviation for 30 different random seeds. What we can see is that when using 50 samples, the probabilities in the codebook $\mathcal{C}$ differ with the reference by less than 1% on average (plot on the top) and the absolute error of estimation for $P_b(\omega)$ is less than 6% (plot on the bottom). This explains why we decided to use 50 training samples in our experiments as this represents a good tradeoff between training set size and stability of results. We also note that the Scanner datasets produce higher variability and less accurate estimations. However, taking the variability of results in terms of standard deviation, we see that the codebooks converge to their optimum at a similar rate for all devices.

Among the smartphone datasets, we note that the Samsung dataset has a higher variability, again pointing out the fact that this dataset is less stable than the iPhone dataset. This behaviour can be explained by the higher variability of images acquired by Samsung in terms of blur and focusing comparatively to the images acquired by iPhone. Overall, images captured with Samsung have a lower quality than those captured with iPhone.

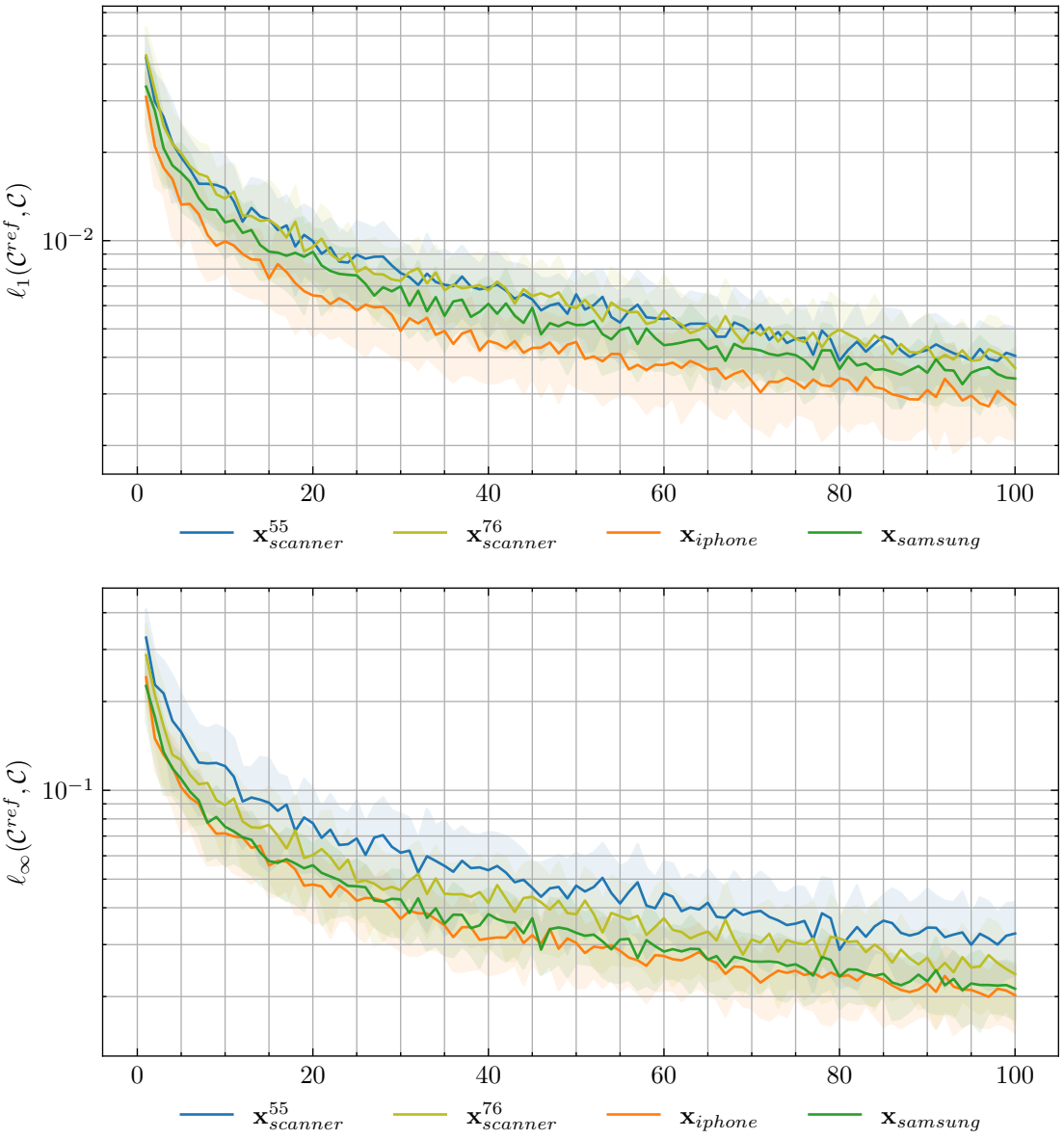


Figure 4.9. Measure of codebook sensitivity to training set size. The plot on the first row shows the average ℓ_1 measure to a reference codebook and the plot on the bottom the ℓ_∞ measure. The x-axis corresponds to the number of samples in the training set.

4.5.3 Performance of PLL Measure

In this experiment, we compare the performances of the Posterior Log-Likelihood (PLL) score introduced in Section 4.3.4 with other standard metrics. Fig. 4.10 shows the histograms of the different metrics on originals and fakes for the different datasets. In this experiment, we test four different setups:

- $\mathbf{x}_{scanner}^{55}$ vs $\mathbf{f}_{scanner}^{55/55}$;
- $\mathbf{x}_{scanner}^{76}$ vs $\mathbf{f}_{scanner}^{76/76}$;
- $\mathbf{x}_{iphone}$ vs $\mathbf{f}_{iphone}$;
- $\mathbf{x}_{samsung}$ vs $\mathbf{f}_{samsung}$;

and the PLL measure (first row) is compared to five standard metrics namely:

- the Hamming distance $d_H(\mathbf{t}, \tilde{\mathbf{t}})$ (second row);
- the Manhattan distance $\ell_1(\mathbf{t}, \mathbf{y})$ (third row);
- the mean-squared-error $\ell_2(\mathbf{t}, \mathbf{y})$ (fourth row);
- the Pearson correlation $\mathrm{pcor}(\mathbf{t}, \mathbf{y})$ (fifth row).

The first observation that we can make is that the standard metrics are very bad at separating originals and deep learning based fakes. Indeed both original histograms (in blue) and fake histograms (in orange) show a clear overlap. This result is further confirmed by examining the receiver operating characteristic (ROC) curves of these metrics in Fig. 4.11. In comparison, the PLL metric is performing very well in this setup, with an almost non-overlapping histogram for all datasets. We analyze this increase in performance by the fact that the PLL metric treats each pattern differently, effectively weighting the contribution of each pattern according to its reliability, whereas standard metrics assume equal contribution of all patterns indistinctively and compute an equally weighted average on all patterns. The only notable exception is the case of the Samsung dataset which shows poor performance, actually comparable to other standard metrics. Again a further confirmation that this dataset has a lower overall quality.

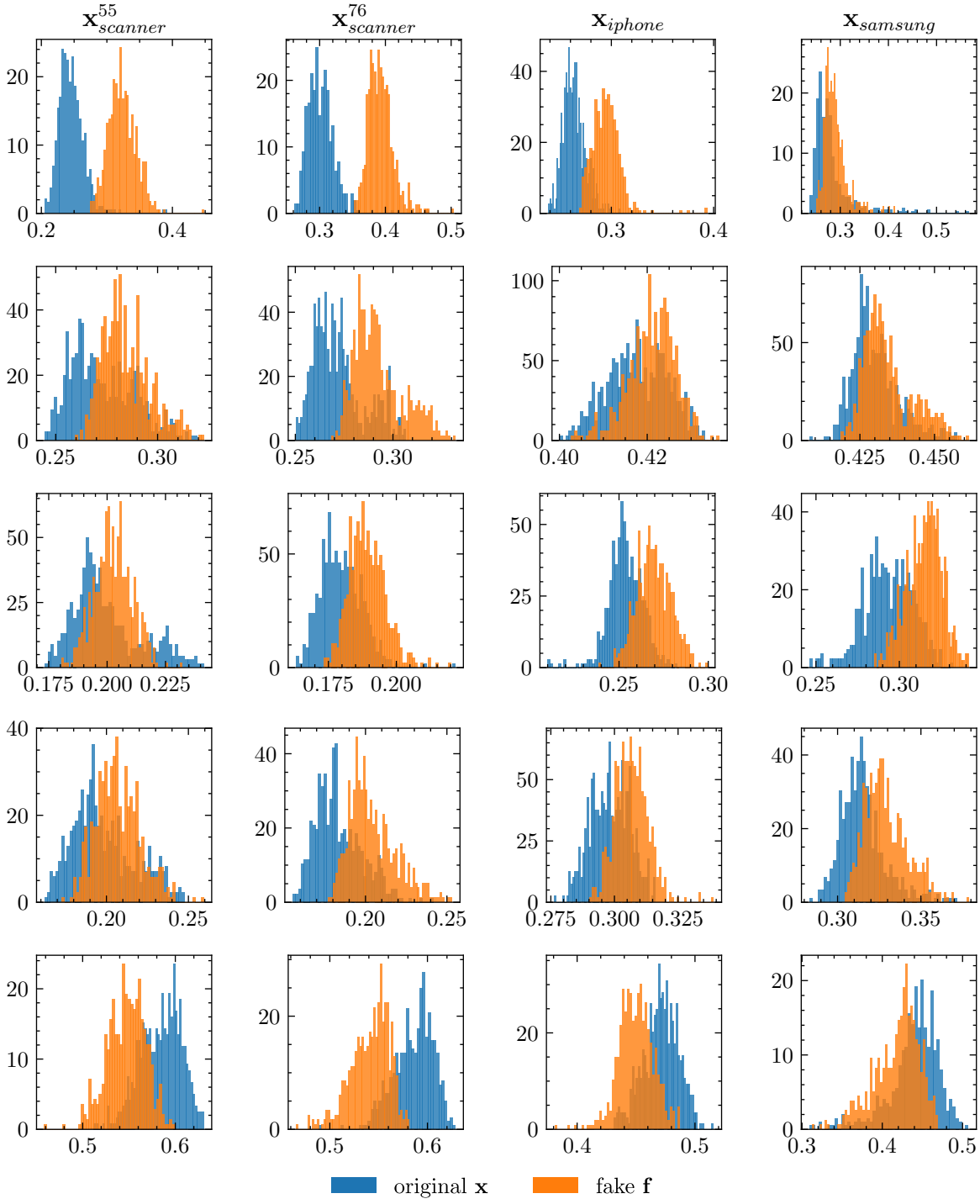


Figure 4.10. Histogram visualisation of separability between original (in blue) and fake (in orange) CDP for various metrics. The top row corresponds to the PLL metric. The following rows are respectively the ℓ_1 metric, ℓ_2 metric, Hamming distance d_H and the Pearson correlation pcor.

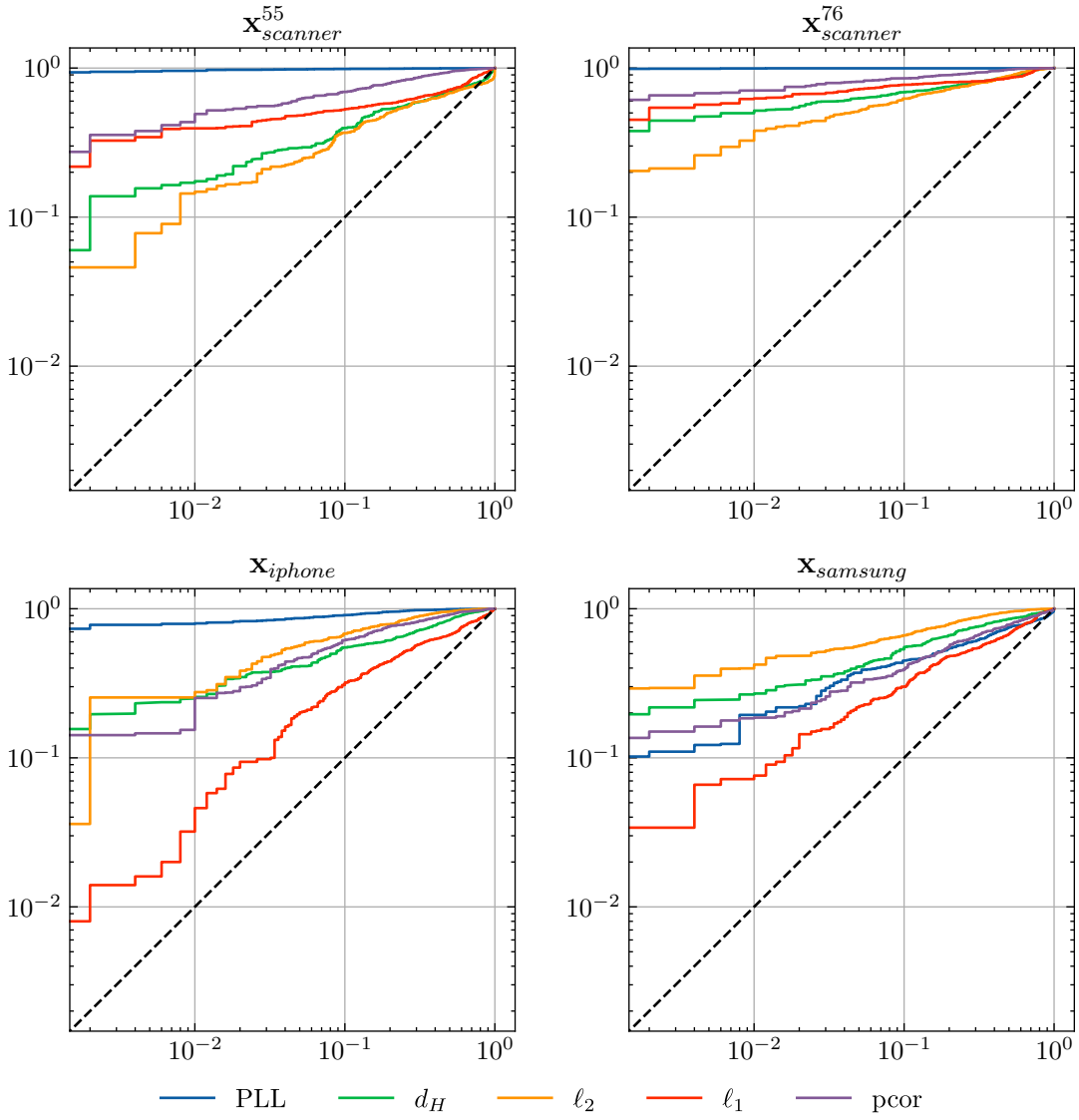


Figure 4.11. ROC curves of the standard metrics used for the authentication of CDP compared with the PLL score (in blue). The four different plots correspond to the four different datasets and the colors correspond to the five different metrics.

4.6 Conclusion

In this Chapter, we introduced the Binary Pattern-based Channel model, a new mathematical model for the description of the Printing-Imaging pipeline of CDP based on an adaptation of the Binary Symmetric Channel model of information theory carefully adapted for the description of the CDP authentication scheme.

We studied the properties of this probabilistic model and justified its construction based on the geometric properties of CDP. We introduced the notions of neighbouring pattern, pattern reliability and Posterior Log-Likelihood score for a given CDP. Finally, we verified the soundness of the model by testing it on a real dataset of printed CDP, for various printers and acquisition devices. This test demonstrated that the BPC model is able to properly describe the printing-imaging process of CDP and offers relevant statistics that actually describe the underlying physical process.

In the next Chapter, we will show how this model and its statistics provide us with tools that can be used to enhance the authentication process of CDP and build powerful schemes able to detect even the most powerful deep learning based attacks against CDP.

Chapter 5

Novel CDP Authentication Scheme based on BPC

In this Chapter, we present new schemes for the authentication of CDP based on the BPC model presented in Chapter 4. The last Chapter was devoted to presenting a mathematical framework that describes the printing-imaging-binarization process of the CDP and introduced several new concepts such as the notion of neighbouring patterns, pattern reliability and Posterior Log-Likelihood measure. The Chapter also presented various experiments that demonstrated their interest and realism when tested on real datasets. In the present Chapter, we will show how these new notions can be used to create new algorithms for the authentication of CDP and we'll compare them with state-of-the-art approaches. The material that we present in this Chapter has been published in two research papers [67, 68] and a patent [66].

5.1 Introduction

In Chapter 4, we presented the BPC model, a probabilistic model inspired by information theory that gives new insights on why standard authentication approaches fail to distinguish original and fake CDP for deep learning based attacks. The main message being that standard metrics only compute an average measure across the whole template, thus mixing up reliable and unreliable channels, leading to poor separability. The BPC model also points out possible solutions as the model shows different statistics for original CDP and fake CDP (see the codebook experiments in Section 4.5 for more detailed discussion).

5.1.1 Our Contributions

In the present Chapter, we explore the possibilities offered by the BPC model to strengthen the standard authentication schemes of CDP. In more details we:

- Propose a new scheme for the authentication of CDP which includes the training of a reference codebook by the Verifier on a small reference dataset of CDP;
- Propose a new authentication measure based on the PLL score introduced in Chapter 4;
- Propose a method for selecting reliable patterns based on the reference codebook and define a restriction mask for the measure of standard metrics only on reliable patterns;
- Test these new approaches on a real dataset of original and fake CDP for the strongest known attacks and demonstrate the superiority of this approach when compared with standard approaches;
- Propose an approach to measure the information loss when mapping a given probe CDP $\mathbf{y}$ to its sample-based codebook $\mathcal{C}_{\mathbf{y}}$ for the task of discriminating between original and fake CDP;
- Analyze the individual contribution of each pattern to the global classification capability of the proposed schemes.

5.2 Research Questions

How can the BPC model be used to strengthen the authentication scheme of CDP ?

By adding a new enrollment step for the Defender in the CDP pipeline, we are able to store a reference Codebook on the secured server. This reference codebook can then be used during authentication to compute the PLL score and the masked metrics. We compare these approaches to the standard metrics and demonstrate their strength and stability.

Is the authentication scheme based on the BPC model stable across different printers and acquisition device ?

We test the authentication scheme based on the BPC model for various modalities of the same dataset. By reporting the ROC curves for the different experiments, we are able to demonstrate that the codebook approach generalizes to many different setups and is able to operate with modalities unseen during the training of the codebook.

Is the mapping from a probe CDP to its sample-based codebook a sufficient statistic for the task of discriminating between original and fake CDP ?

The mapping from a printed probe $\mathbf{y}$ to its sample-based codebook $\mathcal{C}_{\mathbf{y}}$ is lossy from an information theoretic point of view. We perform experiments to try and determinate how much information is loss in view of being able to discriminate between original and fake CDP.

How does each pattern defined in the BPC model individually contribute to the global authentication capability of the authentication schemes based on codebooks ?

Measuring the masked metrics on one fixed type of pattern $\omega \in \Pi$, we show experimentally that the notion of reliable pattern in terms of bit-flipping probability is indeed correlated to the actual authentication capability in terms of average probability of error (2.30).

5.3 CDP Authentication Algorithms Based on Codebooks

We briefly recall the concepts and notations of the PI channel for CDP authentication, seen as a 3-player game and illustrated in Fig. 5.1. The Defender (in green) generates and prints template $\mathbf{t}^n$ on an object package o^n and sends it to the public domain. The Attacker (in red) uses o^n to create a counterfeited version c^n of it. Finally, the Verifier (in blue) scans the object package and authentifies the probe $\mathbf{y}^n$ to decide whether it is an original package or a counterfeit.

5.3.1 Predictor Channel and Reference Codebook

The core idea of building an authentication system based on the BPC model relies on the introduction of a new processing channel for the Defender, called the Predictor channel (highlighted in light green in Fig. 5.1). This extra channel is used for the training of a reference codebook $\mathcal{C}_{ref}$. In more details, during the process of object production and printing and before sending the objects to the public domain, the Defender:

1. selects a small amount N_{train} of printed objects $\{o^n\}_{n=1}^{N_{train}}$;
2. uses a standard capturing device to create a small database $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{N_{train}}$ consisting of pairs of digital templates and captured original printed codes;
3. computes the codebook with estimated probability of bit-flipping $P_b(\omega)$ for the central pixel of each pattern $\omega \in \Pi$.

Algorithm 1, details the steps of computing a reference codebook for a given training set of pairs $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{N_{train}}$. The idea is to create a dictionary $\mathbb{D}_b$ whose keys are the different types

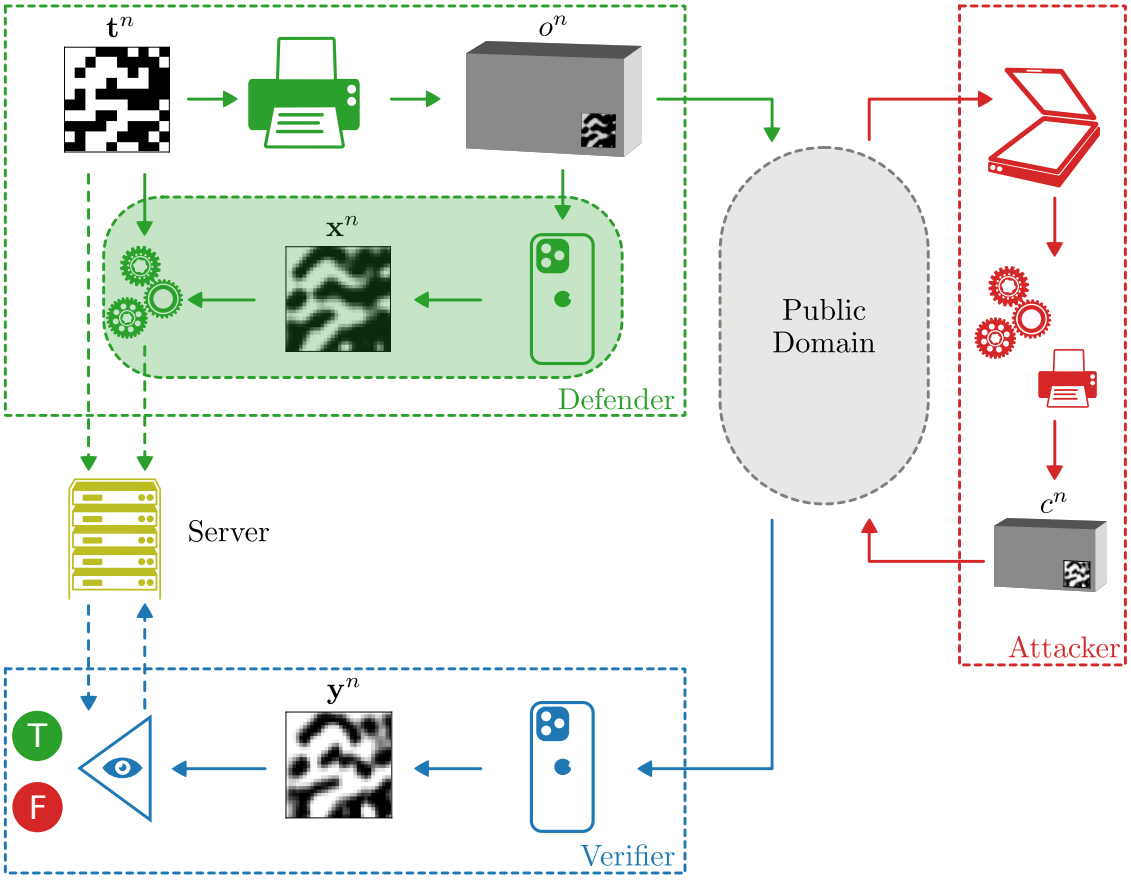


Figure 5.1. Illustration of the CDP framework seen as a 3-player game with the added predictor channel in light green. Note the added predictor channel for the Defender which consists in capturing a small set of objects o^n with a standard capturing device and learning a reference codebook based on the pairs $(\mathbf{t}^n, \mathbf{x}^n)$ that is stored on the Server together with the reference digital templates $\mathbf{t}^n$.

of neighbourhoods. For each $\omega_{ij} \in \Pi$, $\mathbb{D}_b[\omega_{ij}]$ lists the boolean values $(\tilde{t}_{ij} \neq t_{ij})$. Finally, the statistics $P_b(\omega)$ for each type of pattern ω are computed by averaging the corresponding list.

Algorithm 1 Algorithm for codebook estimation

Input: training set $\{(\mathbf{t}^n, \mathbf{x}^n)\}_{n=1}^{N_{train}}$

Output: estimated codebook $\mathcal{C} = (\omega, P_b(\omega))_{\omega \in \Pi}$

Initialisation:

create a dictionary $\mathbb{D}_b$ with the set Π as keys and empty lists as values.
for $n = 1$ to N_{train} **do**
  estimate $\tilde{\mathbf{t}}^n$ from $\mathbf{x}^n$
  for symbol t_{ij}^n in $\mathbf{t}^n$ **do**
    extract pattern ω_{ij}^n in $\mathbf{t}^n$
    extract symbol $\tilde{t}_{ij}^n$ in $\tilde{\mathbf{t}}^n$
    append boolean value $(\tilde{t}_{ij}^n \neq t_{ij}^n)$ in dictionary $\mathbb{D}_b$ at key ω_{ij}^n
  end for
end for
for ω in Π **do**
  compute mean value: $P_b(\omega) = \text{mean}(\mathbb{D}_b[\omega])$
  store the couple $(\omega, P_b(\omega))$
end for
return codebook $\mathcal{C} = (\omega, P_b(\omega))_{\omega \in \Pi}$

The strength of this scheme lies in the fact that the Defender only needs to run this algorithm on a small subset N_{train} of the total number of printed objects (see Section 4.4.2 for a detailed discussion of this matter), and thus does not need to enroll all printed CDP before sending them to the public domain which would be unrealistic in practical applications. We now consider several authentication schemes based on the BPC model.

5.3.2 Authentication Scheme Based on PLL Score

The first authentication scheme is a direct implementation of the PLL measure (4.9). It starts by learning the codebook $\mathcal{C}_{ref}$, running Algorithm 1 on the training set. At the authentication stage, given a probe $\mathbf{y}$, we perform the following steps (illustrated in Fig. 5.2):

1. Estimate $\tilde{\mathbf{t}}$ from the probe $\mathbf{y}$;
2. Given the reference template $\mathbf{t}$, the estimated $\tilde{\mathbf{t}}$ and the codebook $\mathcal{C}_{ref}$, compute the Posterior Log-Likelihood score of $\tilde{\mathbf{t}}$ with (4.9);

3. Compare the score with a chosen threshold γ fixed on a validation set to decide whether $\mathbf{y}$ is original or fake.[1]

It should be pointed out here that symbols t_{ij} located too close to the border of the template do not have a well-defined neighbourhood ω_{ij}. We propose two solutions to address this problem:

- the first solution is simply to ignore these symbols and run the model only on the symbols located on the inside of $\mathbf{t}$;
- another solution is to consider a white padding surrounding template $\mathbf{t}$ as this is the natural padding for $\mathbf{x}$ when printing CDP on white paper.

5.3.3 Authentication Scheme Based on Masked Metrics

The authentication based on reliable patterns is illustrated in Fig. 5.3. The idea behind this scheme is to use the reference codebook of bit-error $\mathcal{C}_{ref}$ to select only the patterns $\omega_{ij} \subset \mathbf{t}$ that have a low probability of bit-error $P_b(\omega_{ij}) \leq \mu$ for some well-chosen fixed threshold $\mu \in [0, 1]$ (see the red box in Fig. 5.4 and the various masks obtained for different choices of μ in Fig. 5.5). In this way, we discard all regions in $\mathbf{y}$ that are known to produce high error for original samples $\mathbf{x}$. The authentication steps are the following:

1. for each neighbourhood ω_{ij} in $\mathbf{t}$, search the probability of bit-flipping $P_b(\omega_{ij})$ in the codebook;
2. define an attention mask $m_{ij} := (P_b(\omega_{ij}) \leq \mu)$ for some fixed threshold $\mu \in [0, 1]$;
3. choose any standard metric $d(\mathbf{t}, \mathbf{y})$ that is computed pixel-wise such as ℓ_1, ℓ_2, Hamming distance d_H or Pearson correlation. Note that some upsampling of $\mathbf{t}$ might be necessary for computation;
4. weight the chosen metric $d(\mathbf{t}, \mathbf{y})$ by using the binary mask, upsampling it if needed:

$$d^m(\mathbf{t}, \mathbf{y}) = \frac{1}{M} \sum_{i,j} m_{ij} \cdot d(t_{ij}, y_{ij}), \quad M = \sum_{i,j} m_{ij}. \tag{5.1}$$

   where $d(t_{ij}, y_{ij}) = |t_{ij} - y_{ij}|$ for ℓ_1, $d(t_{ij}, y_{ij}) = (t_{ij} - y_{ij})^2$ for ℓ_2, ...

[1] If no fake samples are available in the validation set, one can simply fix the threshold at the lowest value of the Posterior Log-Likelihood score.

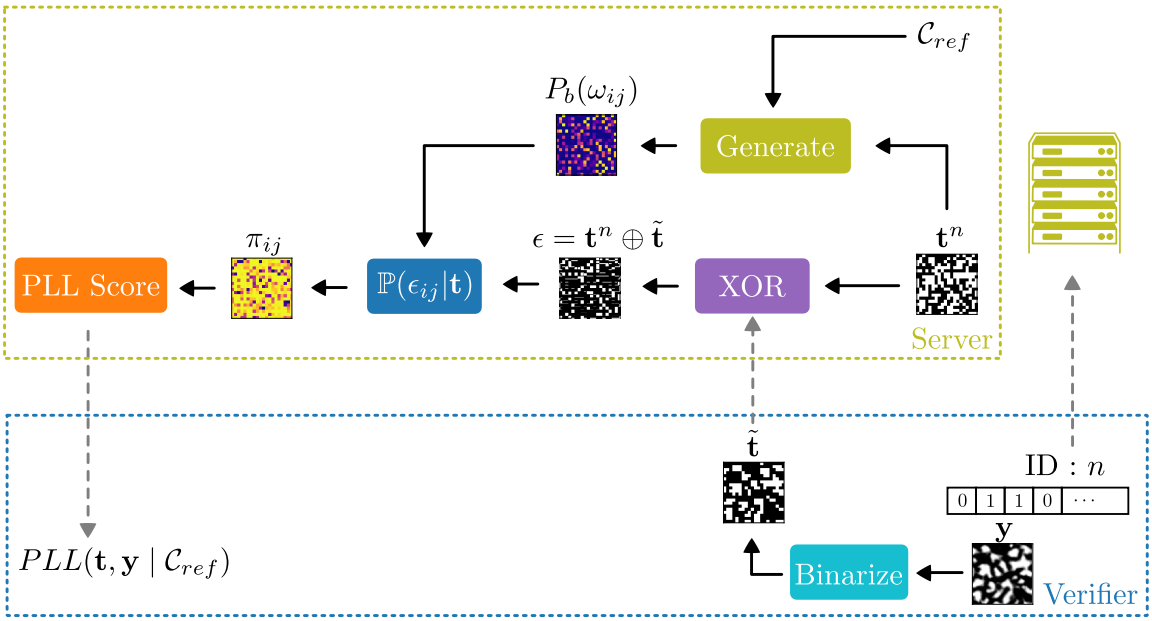


Figure 5.2. A schematic representation of the authentication process based on PLL score. The Verifier (in blue) sends a binarized probe $\tilde{\mathbf{t}}$ together with an identification number n to the server (in yellow). The server then performs the computation of the PLL score based on the reference codebook $\mathcal{C}_{ref}$ and the digital template $\mathbf{t}^n$ and sends back the final result to the Verifier.

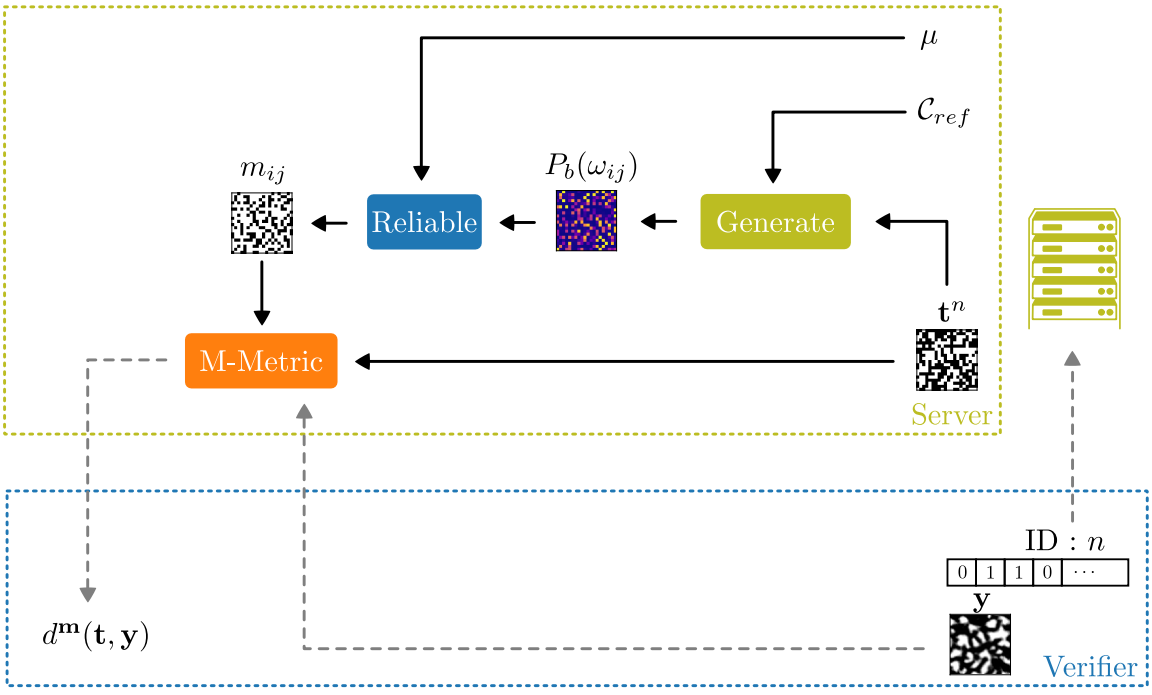


Figure 5.3. An illustration of the authentication process based on masked metrics. The Verifier (in blue) sends a probe CDP $\mathbf{y}$ to the server together with an identification code n to the server (in yellow). The server then computes an attention mask based on the digital template $\mathbf{t}^n$, the reference codebook $\mathcal{C}_{ref}$ and the reliability threshold μ. This mask is finally used to compute the masked metric and the final result is sent back to the Verifier.

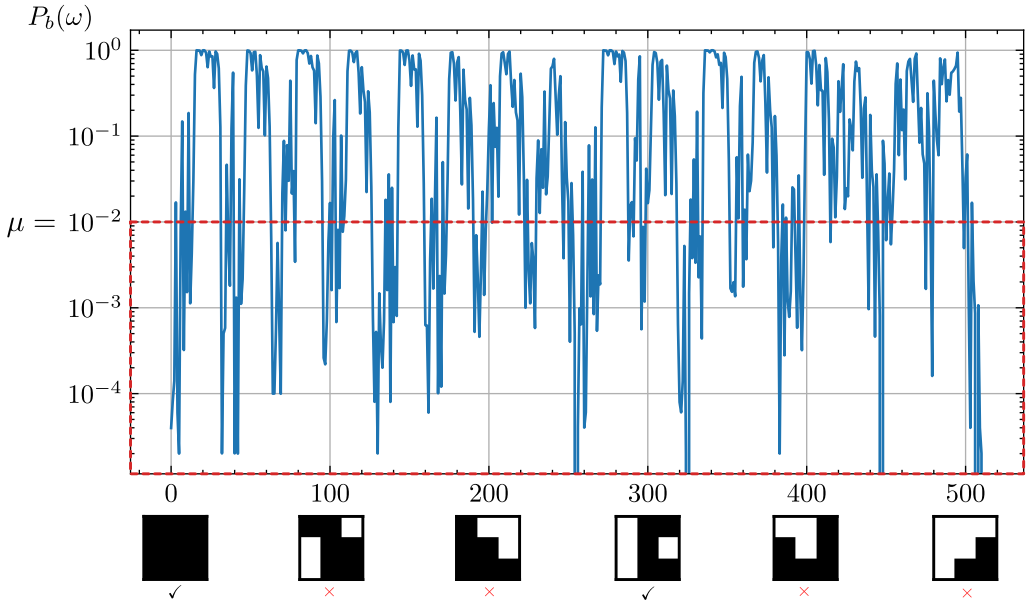


Figure 5.4. This plot illustrates the selection of reliable patterns for a given codebook $\mathcal{C}$ and a choice of threshold $\mu = 10^{-2}$. The patterns on the bottom illustrate some reliable and unreliable patterns.

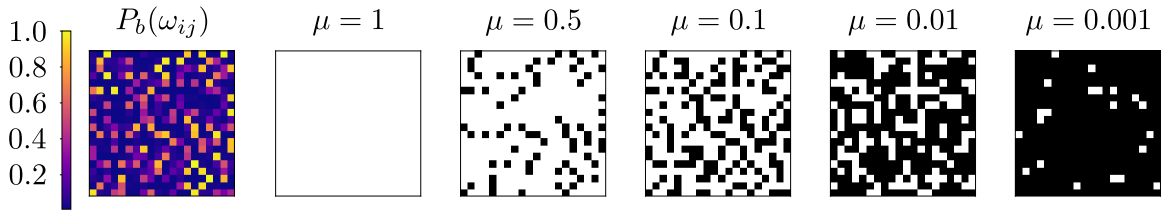


Figure 5.5. An example illustrating the computation of the attention mask. The image on the left represents the probability of bit-flipping for each symbol in a binary template $\mathbf{t}$. The other images show the attention mask for various choices of threshold μ.

5.4 Experimental Results

5.4.1 Experimental Setup

Similarly to Chapter 4, the experiments in this Chapter are based on the 1×1 base dataset. This dataset presents various modalities which are detailed in Section 4.4. The reference codebook is trained on a subset of original CDP constituted of $N_{train} = 50$ randomly selected samples and the authentication tests are performed on 500 randomly selected triples $\{(\mathbf{t}^n, \mathbf{x}^n, \mathbf{f}^n)\}_{n=1}^{500}$. The standard metrics for authentications that we use in the experiments are the same as the one presented in Section 3.3.1. These metrics are being compared to the two new approaches based on the PLL score and the masking technique for various values of $\mu \in [0, 1]$.

Chronologically, we started experimenting with this new approach based on the scanner capturing device and only after did we test the same setup with smartphones. The experiments are thus divided accordingly with results on scanner presented first and results on smartphones presented afterwards.

5.4.2 Authentication Accuracy of BPC Model (Scanner Dataset)

To compare all different approaches in a unified way, we test both originals $\mathbf{x}_{scanner}^{55}$ and $\mathbf{x}_{scanner}^{76}$ separately against all four kind of ML fakes $\mathbf{f}_{scanner}^{55/55}$, $\mathbf{f}_{scanner}^{55/76}$, $\mathbf{f}_{scanner}^{76/55}$ and $\mathbf{f}_{scanner}^{76/76}$. The setup compares the three standard metrics: ℓ_2, d_H, pcor and the PLL score. We also apply the masking strategy described in Section 5.3.3 to all metrics for a fixed reliability threshold $\mu = 0.01$, found empirically. For each metric, we compute the associated ROC curves and report the AUC score. The AUC score is averaged over ten runs with randomization of training/testing set. All results are summarized in Table 5.1, Table 5.2 and in Fig. 5.6.

A first observation at the Tables 5.1 and 5.2 shows that discriminating between originals and fakes is more accurate for $\mathbf{x}_{scanner}^{55}$ than for $\mathbf{x}_{scanner}^{76}$. In general, the results show that the metric PLL outperforms the other metrics. On average, M-PLL, its masked version, appears as the best metric with a very high AUC score on all types of fakes.

The masked metrics show a great improvement in AUC score over all their non-masked counterparts. This is further illustrated in Fig. 5.6, where we compare side-by-side masked and non-masked metrics for PLL and Hamming metrics. Surprisingly, the ℓ_2 metric proves to be the best metric for discriminating $\mathbf{x}_{scanner}^{76}$ and $\mathbf{f}_{scanner}^{55/76}$. This result should however be mitigated by the following observations:

- metric M-PLL performs very close to ℓ_2 and even outperformed it on certain runs;
- the high variability in performance of ℓ_2 on different types of fakes makes it highly unreliable for authentication, as shown by its average score.

Table 5.1. Results (in %) of the Area Under Curve (AUC) for original type $\mathbf{x}^{55}_{scanner}$ and various fakes and metrics. Best results for each type of fakes are highlighted. Average performance are computed for each metric.

	$\mathbf{f}^{55/55}_{scanner}$	$\mathbf{f}^{55/76}_{scanner}$	$\mathbf{f}^{76/55}_{scanner}$	$\mathbf{f}^{76/76}_{scanner}$	Average
metrics					
PLL	99.88	99.89	**100**	**100**	99.94
ℓ_2	59.60	59.05	35.85	27.77	45.57
pcor	87.11	88.41	94.75	92.36	90.66
d_H	63.39	63.82	69.46	61.12	64.45
masked					
M-PLL	**99.98**	99.96	**100**	**100**	**99.99**
M-ℓ_2	99.97	99.95	**100**	**100**	99.98
M-pcor	96.30	92.46	99.35	98.58	96.67
M-d_H	**99.98**	**99.97**	**100**	**100**	99.99

Table 5.2. Results (in %) of the Area Under Curve (AUC) for original type $\mathbf{x}^{76}_{scanner}$ and various fakes and metrics. Best results for each type of fakes are highlighted. Average performance are computed for each metric.

	$\mathbf{f}^{55/55}_{scanner}$	$\mathbf{f}^{55/76}_{scanner}$	$\mathbf{f}^{76/55}_{scanner}$	$\mathbf{f}^{76/76}_{scanner}$	Average
metrics					
PLL	87.24	85.69	**99.97**	**99.98**	93.22
ℓ_2	97.47	**99.03**	85.85	82.12	91.12
pcor	88.97	90.49	95.79	93.88	92.28
d_H	85.36	86.45	89.35	83.14	86.08
masked					
M-PLL	**99.29**	98.97	99.94	99.84	**99.51**
M-ℓ_2	97.04	94.76	99.94	99.85	97.90
M-pcor	97.11	95.45	98.42	97.72	97.17
M-d_H	99.22	98.89	99.94	99.85	99.48

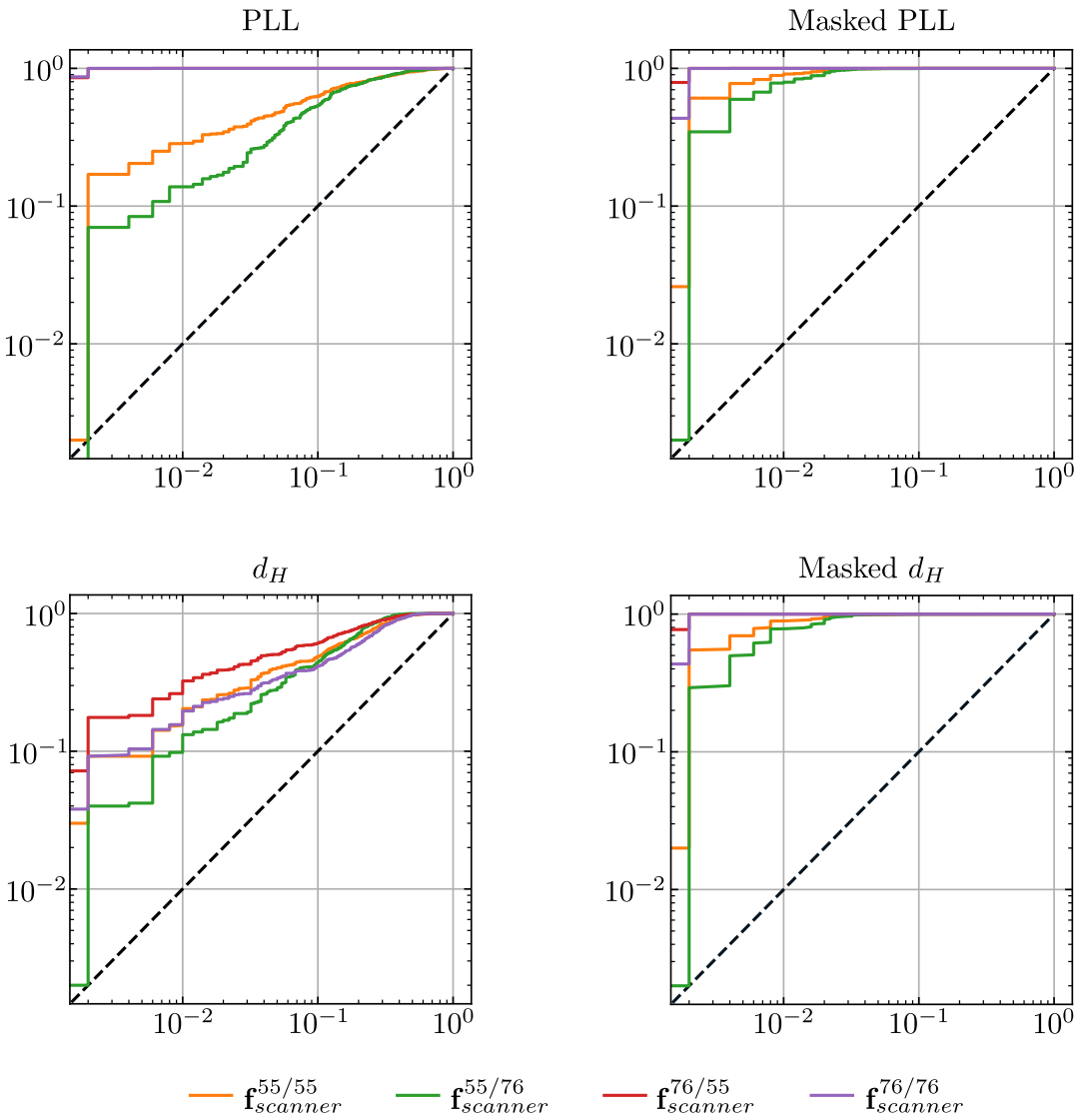


Figure 5.6. ROC Curves for the authentication of CDP based on the scanner dataset $\mathbf{x}^{76}_{scanner}$ against four types of fakes. The left column shows the results for the PLL score and the Hamming distance metric and the right column shows the result for their masked versions with a reliable threshold $\mu = 0.01$.

5.4.3 Authentication Accuracy of BPC Model (Smartphone Dataset)

We test here the two novel schemes of authentication based on the BPC model on the smartphone dataset and compare the metrics presented above both in standard form and restricted to reliable patterns as presented in Section 5.3.3. To this end, we test several different thresholds $P_b(\omega) \leq \mu$ and compute a masked version of the metrics using (5.1). Fig. 5.7 shows the different ROC curves obtained for the various metrics for the iPhone dataset $(\mathbf{x}_{iphone}, \mathbf{f}_{iphone})$ (left column) and the Samsung dataset $(\mathbf{x}_{samsung}, \mathbf{f}_{samsung})$ (right column).

As one can observe, the restriction of the metric to patterns with low probability of bit-flipping drastically enhances the performance of the metrics. For instance, the standard Hamming distance performs poorly on the iPhone dataset (blue curve), but with a correct choice of threshold $\mu = 0.01$ (red curve), it is able to separate originals and fakes almost perfectly. Even the PLL metric, which was already outperforming other metrics is enhanced by the mask. Another point to notice is that if we set the threshold value too low, the performances of the authentication scheme are getting worse, as there is only a few patterns left for authentication. Note that the choice $\mu = 1.0$ corresponds to the standard metrics, where all patterns contribute to the final aggregated score.

To compare all these results in a unified way and to establish the achievable authentication accuracy, we measure the performance of the different authentication schemes by using the minimal average probability of error P_{err}^{min}, which measures the separability between originals and fakes (2.30). This measure, presented in Section 2.2.2, computes an average between the probability of miss and the probability of false acceptance. We find this value to be more indicative of the authentication capability of a given scheme than the Area Under Curve (AUC).

The results are presented in Table 5.3. The reference codebook is trained on either the iPhone or the Samsung dataset and this codebook is then used for the authentication with both smartphones to test cross-device performances. Results are presented both for standard metrics and for masked metrics with a fixed threshold $\mu = 0.01$. The results are averaged over all 6 capture runs with randomization of training/testing set. The first observation we can make is that the proposed authentication schemes significantly outperform the traditional approaches. Indeed, the error in classification drops from an average 20% of error for classical metrics, which is useless in practical applications, to a solid 1% error for the iPhone dataset for the best metrics. The performances on Samsung dataset are not as impressive as the quality of the images are poorer than those acquired with iPhone but, nevertheless, we still see a clear improvement by using our approach. Cross-device performances, i.e., training the codebook on one device and performing authentication on another, are comparable to same-device performances but slightly less accurate. As such, this demonstrates how important it is to carefully select the acquisition device.

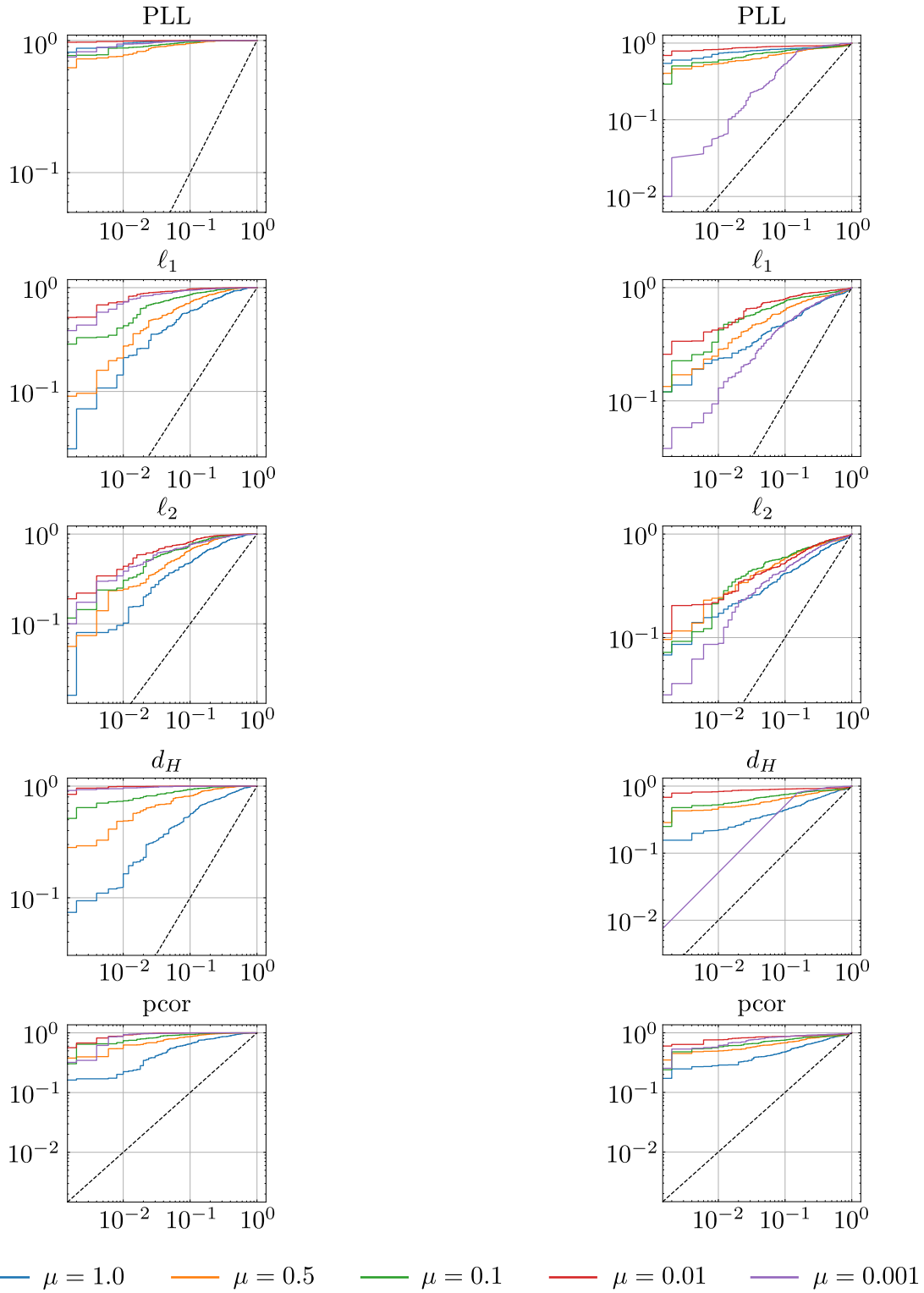


Figure 5.7. ROC Curves for the authentication of CDP based on the iPhone dataset (left column) and the Samsung dataset (right column) for five different metrics. The various colours represent different choices of the reliable threshold μ and thus different masks.

Table 5.3. Results (in %) of the minimal average probability of error in classification P_{error}^{min}. Results are shown both for the iPhone and the Samsung dataset and compares standard metrics with masked metrics for a reliability threshold $\mu = 0.01$. Both same-device and cross-device performances are evaluated.

	iPhone dataset		Samsung dataset	
	$\mathcal{C}_{iphone}$	$\mathcal{C}_{samsung}$	$\mathcal{C}_{iphone}$	$\mathcal{C}_{samsung}$
metrics				
PLL	1.88	2.52	14.63	11.93
ℓ_1	20.07	—	30.20	—
ℓ_2	22.91	—	33.65	—
d_H	21.15	—	32.69	—
pcor	17.12	—	29.87	—
masked				
M-PLL	**0.80**	1.09	**10.91**	**7.89**
M-ℓ_1	6.64	8.44	14.11	14.67
M-ℓ_2	12.52	15.49	22.43	23.09
M-d_H	0.87	**1.07**	12.24	8.78
M-pcor	1.99	1.93	13.29	10.36

5.4.4 Classifier Baseline

In order to show the soundness of the codebook approach jointly with machine learning, we perform the following experiment involving a Support Vector Machine (SVM) classifier [84] trained directly on codebooks. The goal of this machine learning based experiment is to demonstrate that the information contained in the codebooks is statistically sufficient to perform reliable classification. Our approach can be summarized in the following way for both one-class and two-class SVM models: for each pair $(\mathbf{t}^n, \mathbf{x}^n)$ and, if available, $(\mathbf{t}^n, \mathbf{f}^n)$ in the dataset where $n = 1, ..., N$:

1. Run the codebook estimation Algorithm 1 based on the pair $(\mathbf{t}^n, \mathbf{x}^n)$ or $(\mathbf{t}^n, \mathbf{f}^n)$ only. This outputs two lists of sample-based codebooks $\{\mathcal{C}_{\mathbf{x}}^n\}_{n=1}^N$ and $\{\mathcal{C}_{\mathbf{f}}^n\}_{n=1}^N$ which act as statistics of the particular couples $(\mathbf{t}^n, \mathbf{x}^n)$ and $(\mathbf{t}^n, \mathbf{f}^n)$;
2. Divide these lists into a training set $\{(\mathcal{C}_{\mathbf{x}}^n, \mathcal{C}_{\mathbf{f}}^n)\}_{n=1}^{N_{train}}$ and a testing set $\{(\mathcal{C}_{\mathbf{x}}^n, \mathcal{C}_{\mathbf{f}}^n)\}_{n=1}^{N_{test}}$;
3. - *For one-class SVM:* Train the model based on $\{\mathcal{C}_{\mathbf{x}}^n\}_{n=1}^{N_{train}}$;
   - *For two-class SVM:* Train the model based on $\{(\mathcal{C}_{\mathbf{x}}^n, \mathcal{C}_{\mathbf{f}}^n)\}_{n=1}^{N_{train}}$;
4. Use the test set to compute the average probability of error P_{error} score.

This experiment is performed in two setups. A first setup, referred to as "full SVM", where the model is trained on the full codebook with all 512 binary patterns taken as input, that is both reliable and unreliable channels. The second setup, referred to as "masked SVM", where the model is trained only on reliable patterns, with $\mu = 0.01$. For this experiment, we select non-overlapping training set and testing set with $N_{train} = N_{test} = 500$ chosen randomly across the full smartphone dataset composed of $N = 1440$ samples, and we use the radial-basis function (RBF) kernel. The hyperparameters of the SVM are optimized on the testing set using grid-search.

The results are presented in Table 5.4. One-class SVM trained on the full codebook performs similarly to standard non-masked metrics at the task of discriminating original and fake samples. On the other hand, training the model only on reliable patterns greatly improves the capacity of the SVM model to classify them correctly. The results of two-class SVM are also very interesting. Indeed, classification based on codebooks shows perfect separability demonstrating that the codebooks represent a sufficient statistic for the classification task. However, training the two-class SVM requires the knowledge of fakes whereas training one-class SVM is agnostic to this requirement. In general, focusing only on reliable patterns tends to decrease the accuracy of the two-class SVM. Our current hypothesis is that, by putting a hard threshold selection on patterns $P_b(\omega) \leq \mu$, we discard some patterns ω that still contain relevant features.

These observations should however be mitigated by the fact that the two-class SVM does not represent practical situations as having a full access to fake samples is an unrealistic assumption. In this regard, one-class SVM represents a more realistic scenario for which our approach based on reliable patterns shows an improvement.

Table 5.4. 1-class SVM and 2-class SVM trained on codebook features. The table shows the average probability of error (in %) in classification for a trained model. The full SVM column is trained on the full codebook while the masked SVM column is trained only on reliable patterns. The results are averaged across five random selections of training and testing set.

	iPhone dataset	Samsung dataset
full SVM		
1-class	21.86	31.56
2-class	0.00	0.00
masked SVM		
1-class	2.30	17.24
2-class	0.10	1.06

5.4.5 Pattern-Wise Contribution to Authentication

The final experiment we conduct on the smartphone dataset is a more in-depth study of the choice of optimal reliability threshold $P_b(\omega) \leq \mu$. Indeed, the test with different thresholds demonstrated that the optimal choice of μ relies on a tradeoff between keeping too many unreliable patterns that are uninformative in terms of authentication score ($\mu \uparrow 1$) and having too few of them thus decreasing the overall capacity of distinguising originals and fakes ($\mu \downarrow 0$). This tradeoff is well illustrated in Fig. 5.8 where we represent the choice of threshold μ on the x-axis and the minimum probability of error P_{err}^{min} on the y-axis. This plot clearly indicates that the probability of error reaches a minimal value around $\mu = 0.01$ for all different metrics. Results based on iPhone dataset and Samsung dataset have similar behaviour.

Yet another way of measuring the contribution of each individual pattern $\omega \in \Pi$ to the authentication capacity is to measure a masked metric $d^{m_\omega}(\mathbf{t}, \mathbf{y})$ where the mask m_ω is non-zero only on the central pixel of a chosen pattern $\omega \in \Pi$. This is done in Fig. 5.9 for the Hamming distance. The graphics here show that reliable patterns in terms of probability of bit-flipping help discriminate between originals and fakes, while unreliable patterns mix them up.

5.5 Conclusion

In this Chapter, we explored the applications of the Binary Pattern-based Channel (BPC) model for the development of novel authentication schemes for CDP-based physical object protection.

We proposed two novel authentication schemes based on this model which give a viable alternative to the standard metrics, while still maintaining full interpretability of the results. We showed that these new authentication schemes can correctly detect ML-based attacks, in contrast to the standard metrics based on digital templates. Compared to modern deep learning approaches, our model requires very few training data and is very efficient to be run in practice, while still offering great performances against powerful ML attacks.

In the next Chapter, we will delve deeper into the study of how the notion of pattern reliability links to better authentication capabilities. Indeed, the schemes presented in this Chapter can be further enhanced by studying the underlying mathematical model, providing theoretical optimal decision threshold for each pattern separately and exploring various aggregation strategies. This study will also give theoretical guarantees for these authentication schemes which we only demonstrated experimentally here.

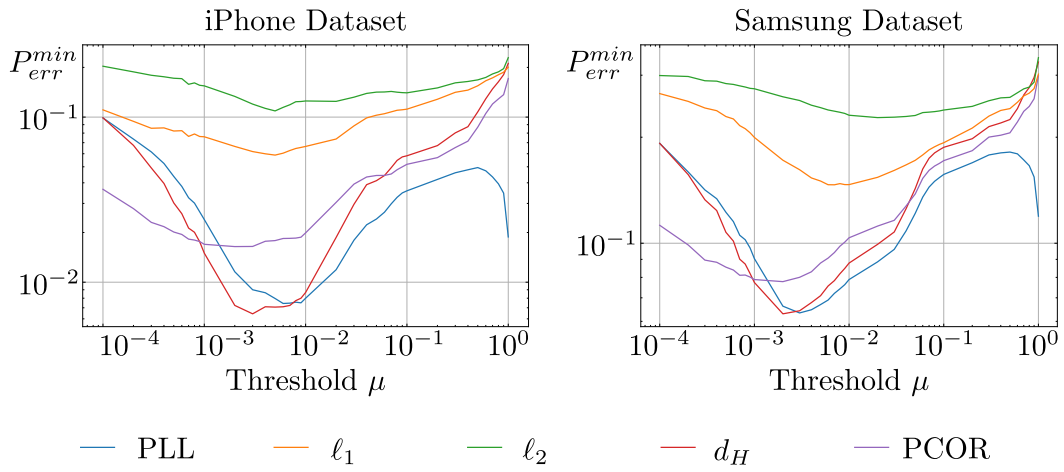


Figure 5.8. Visualisation of the minimal probability of error P_{error}^{min} (2.31) as a function of threshold μ on reliable patterns for iPhone and Samsung dataset. Results are shown for various masked metrics.

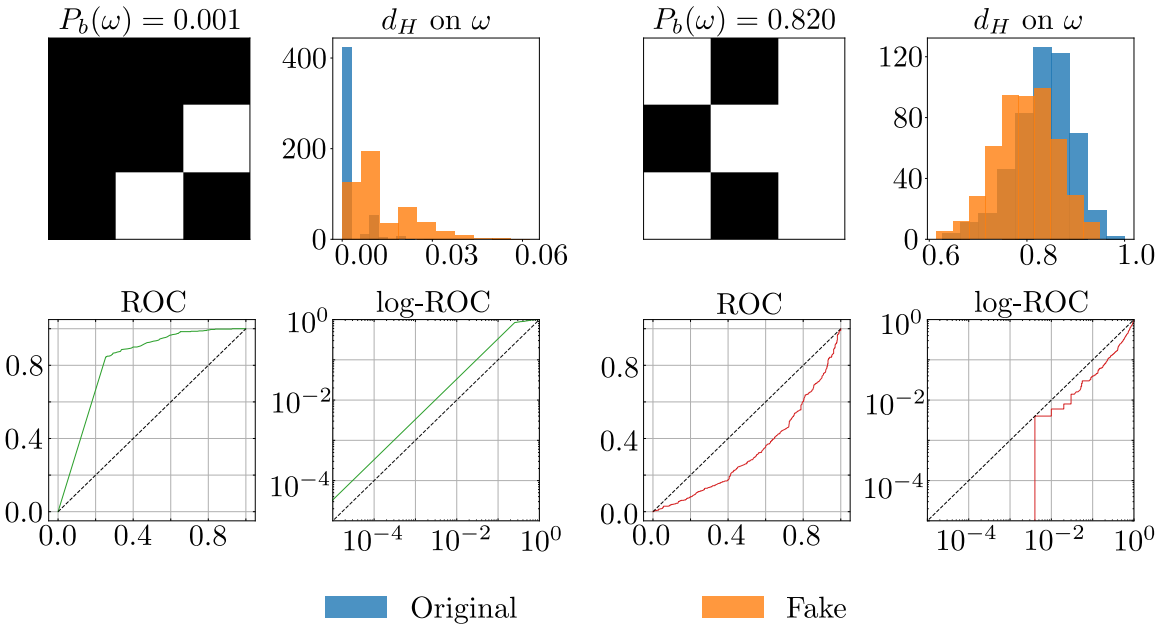


Figure 5.9. Visualisation of the pattern-wise Hamming distance $d^{m_\omega}(\mathbf{t}, \tilde{\mathbf{t}})$. The pattern on the left has a small probability of bit-flip while the pattern on the right has a high probability of bit-flip. The ROC curve measures the separability between originals and fakes in both cases.

Chapter 6

Provable Performance Guarantees for CDP Authentication

Given the critical importance of CDP in preventing counterfeiting, there is an urgent need for provable performance guarantees across various criteria. This Chapter aims to establish a theoretical framework to derive optimal criteria for the analysis, optimization, and future development of CDP authentication technologies, ensuring their reliability and effectiveness in diverse security scenarios. In this regard, we are pushing further the theoretical ground for CDP authentication provided by the BPC model by relating it to well-established statistical studies. In Chapter 5, we explored empirically how the BPC model could be used to enhance the existing authentication schemes and demonstrated experimentally the validity of our approach. In this Chapter we explore how the BPC model, on its own, provides theoretical guarantees and optimal criteria for the task of discriminating original and fake CDP and place these approaches on well-known statistical grounds.

6.1 Introduction

In Chapters 4 and 5, we demonstrated that different CDP patterns, composed of 3×3 elements, have varying probabilities of bit-flipping after printing, scanning or mobile phone image acquisition and binarization. We identified 512 different patterns and characterized their reliability based on the context of their 3×3 neighborhoods. However, those studies employed simple rules, using masking based on empirically optimized thresholds, and focused only on reliable channels characterized by low bit-flipping probabilities. Patterns with high bit-flipping probabilities were disregarded completely. Furthermore, simple metrics such as Hamming distance or Pearson correlation were used for the reliable patterns based CDP authentication.

6.1.1 Our Contributions

In contrast, this new study explores more sophisticated rules that go beyond hard decision-making and the binary characterization of reliable and unreliable patterns. Instead, this Chapter introduces three different criteria for the measure of quality of the acquired CDP based on well-known statistical tests namely: The Hamming distance test; the cross-entropy between the probability of bit-flipping and a reference probability; and Neymann-Pearson likelihood ratio. We investigate different fusion rules and decision strategies for binarized CDP. Specifically, we consider two approaches: making a decision first and then aggregating the results versus aggregating everything first and then making a decision. This paper presents theoretical findings on how to optimize these strategies for enhanced CDP authentication. In more details, this Chapter's main contributions are:

- Providing a theoretical analysis of the authentication capacity of the BPC model in terms of miss and false acceptance;
- Deriving theoretical optimal decision thresholds for the detection of fake CDP based on Hamming distance;
- Reformulating the PLL score as a cross-entropy term between estimated and reference distributions;
- Interpreting CDP authentication in the setting of the Neymann-Pearson test and linking NP optimality criterion with the optimal approach based on Hamming distance;
- Introducing a systematic description of patternwise score linear aggregation and proposing new aggregation strategies;
- Testing the proposed approaches on a real dataset of CDP captured with a smartphone;

6.2 Research Questions

Can the BPC model be used to derive theoretical performance guarantees for CDP authentication?

The BPC model provides a theoretical description for the behaviour of the estimated template $\tilde{\mathbf{T}}$ given a fixed reference template $\mathbf{T} = \mathbf{t}$. By studying the distribution of $\tilde{\mathbf{T}}$ for both inlier and outlier classes, we are able to give theoretical optimal decision bounds for the separability of original and fake CDP for each pattern $\omega \in \Pi$, reformulating the problem as an overlap between two Binomial distributions.

What does the Neymann-Pearson approach to hypothesis testing give for the BPC model for CDP authentication?

The Neymann-Pearson approach gives an optimal decision tool for the testing of two competiting hypotheses $\mathbb{H}_0$ and $\mathbb{H}_1$ if you have access to the likelihood distribution function under both hypotheses. The BPC model provides such a description and thus we can use it to obtain an optimal decision strategy for CDP authentication.

How does the pattern reliability measure relate to the actual performance of CDP authentication?

Having derived optimal decision strategies for each pattern $\omega \in \Pi$, we compare the theoretical results provided by the BPC model with actual performances obtained on a real dataset of CDP captured with a smartphone device and show results that confirm the soundness of the theoretical model.

What is the best strategy to aggregate multiple results into a final decision score?

We study how to aggregate the results obtained independently for each pattern $\omega \in \Pi$ into a final decision score. The approach based on reliability of patterns proposed in Chapter 5 is placed in a larger set of linear aggregation techniques and compared to these different approaches in a systematic way. We provide experiments that show optimal results based on 2C-SVM and demonstrate the importance of multishot capture technique to enhance the overall results of authentication.

6.3 Information Flow in Copy Detection Patterns

In this section, we reformulate the classical authentication process of CDP using an information theoretic approach [85]. When trying to identify a probe sample $\mathbf{y}$, one typically uses binarization and compares the binarized probe $\tilde{\mathbf{t}}$ with the original template $\mathbf{t}$ using Hamming distance $d(\mathbf{t}, \tilde{\mathbf{t}})$. A decision threshold $\gamma \in [0, 1]$ is then fixed to decide whether the probe $\mathbf{y}$ is genuine or counterfeited:

$$d(\mathbf{t}, \tilde{\mathbf{t}}) \gtrless \gamma B, \tag{6.1}$$

where B denotes the total number of symbols in $\mathbf{t}$. From the point of view of information theory and following the analysis done in [68], this process can be described as a Markov chain: $\mathbf{T} \to \mathbf{Y} \to \tilde{\mathbf{T}}$, where $\mathbf{T}$ is a random binary matrix in which each symbol has equal probability of being black or white, $\mathbf{Y}$ is a random matrix with real value entries between 0 and 1 and $\tilde{\mathbf{T}}$ is a binary matrix with fixed probability for each symbol. In this framework, each symbol transition can be described as a noisy binary channel:

$$\tilde{T}_{ij} = T_{ij} \oplus \delta_{ij}, \tag{6.2}$$

where $P_b(i, j) = \mathbb{P}(T_{ij} \neq \tilde{T}_{ij})$ is the probability of bit-flip for symbol T_{ij}, $\delta_{ij} = 1$ if there is a bit-flip and $\delta_{ij} = 0$, otherwise (see the illustration in Fig. 6.1). In this setting, the focus is on being able to describe the posterior distribution $\mathbb{P}(\tilde{\mathbf{T}}|\mathbf{T})$.

In Chapter 4, we introduced the BPC model whose main contribution was to precisely describe the variability of the probability of bit-flipping $P_b(i, j)$ as a function of the pixel position (i, j). This model makes the assumption that the probability of bit-flipping of the symbol T_{ij} only depends on a small subset of symbols directly surrounding symbol T_{ij}, which we denote as $\mathbf{\Omega}_{ij}$. Typically:

$$\mathbf{\Omega}_{ij} = \{T_{i\pm a, j\pm b} \mid 0 \leq a, b < h/2\}, \tag{6.3}$$

for an $h \times h$ square pattern centered around T_{ij}. Following the notations introduced in Chapter 4, we write $\omega \in \Pi$ the set of all possible neighbourhoods and $|\Pi| = 2^{h^2}$ its size. Apart from this local dependency, the model assumes independence of the probability of bit-flipping with respect to the location of the pattern. That is, if $\mathbf{\Omega}_{ij} = \mathbf{\Omega}_{kl}$ are two identical patterns located in different coordinates, then:

$$P_b(i, j) = P_b(k, l). \tag{6.4}$$

Based on these assumptions, the characterization of the binary channels $T_{ij} \to \tilde{T}_{ij}$ reduces to the study of the binary channels $T_\omega \to \tilde{T}_\omega$, where T_ω indicates the central symbol of pattern ω. The advantage of this approach is the reduction of dimension, from the study of B different channels to $|\Pi| = 2^{h^2}$ (in our experiments, $B = 51'984$, $h = 3$ and $|\Pi| = 512$).

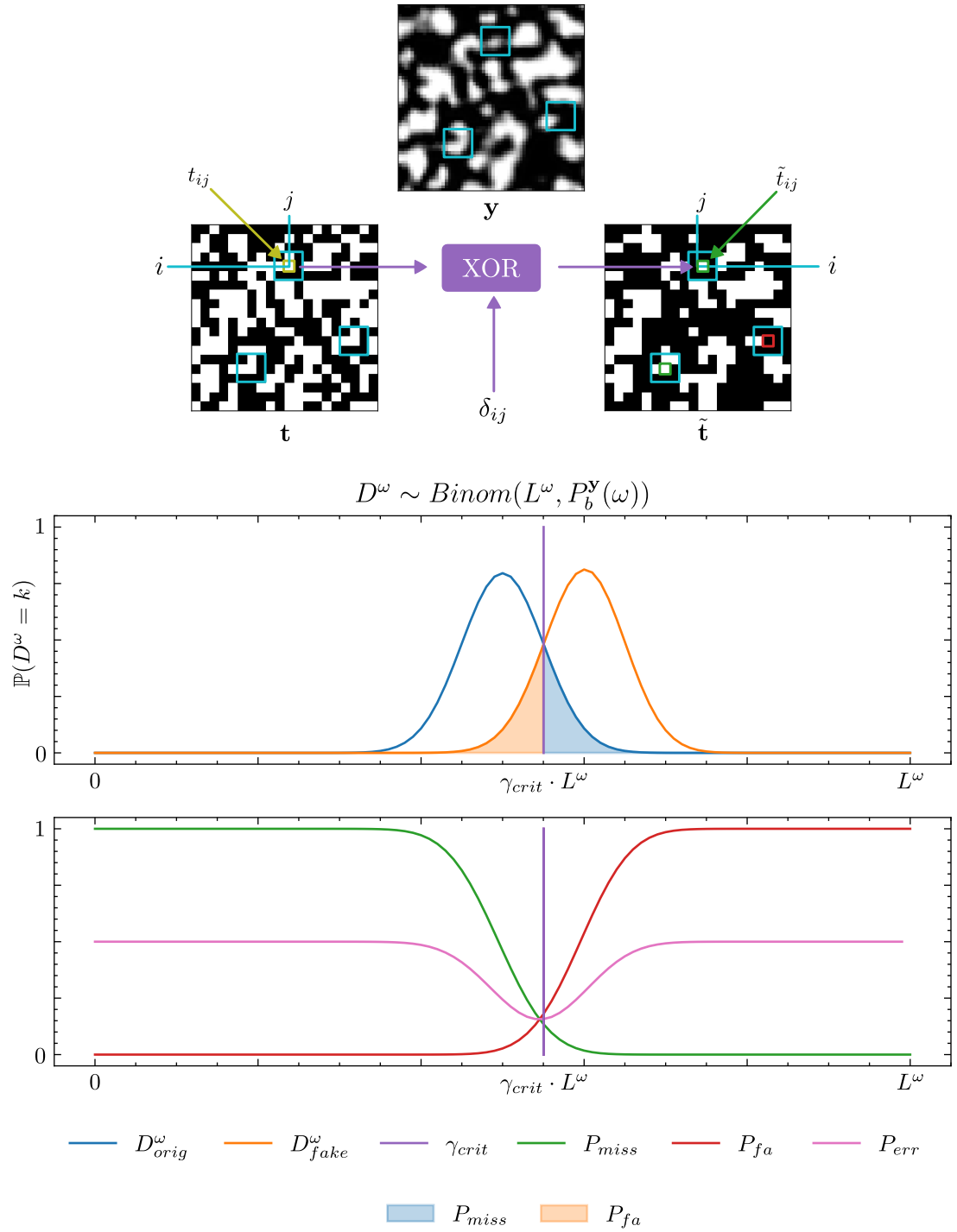


Figure 6.1. An illustration of the theoretical framework under study. Given a fixed pattern $\omega \in \Pi$ and a CDP probe $\mathbf{y}$, one identifies all appearances of the pattern ω in the digital template $\mathbf{t}$. To each location (i, j) of ω in $\mathbf{t}$, one can associate a random bit-flipping δ_{ij}. The Hamming distance D^ω between $\mathbf{t}$ and $\tilde{\mathbf{t}}$ for the pattern ω follows a Binomial distribution which is compared with the reference Binomial distributions of original and fake CDP. Based on the reference distributions, an optimal separation bound γ_{crit} is computed which minimizes the average probability of error P_{err}.

According to this model, one can compute the Hamming distance for each pattern ω:

$$D^{\omega} = d^{\omega}(\mathbf{T}, \tilde{\mathbf{T}}) = \sum_{\substack{i,j: \\ \mathbf{\Omega}_{ij}=\omega}} \delta_{ij}, \tag{6.5}$$

and compute an estimator for the pattern-wise probability of bit-flipping:

$$P_b^{\mathbf{y}}(\omega) \approx \frac{D^{\omega}}{L^{\omega}}, \tag{6.6}$$

where L^{ω} denotes the number of pattern ω appearing in $\mathbf{T}$ and $P_b^{\mathbf{y}}(\omega)$ is the probability of bit-flipping for pattern ω. Note that from a purely theoretical point of view, the quantity L^{ω} is itself a random variable since its very definition relies on $\mathbf{T}$. Although all following derivations can be done in this general setting, it will be simpler to consider the situation where the digital template is fixed $\mathbf{T} = \mathbf{t}$ and so is L^{ω}. As such, the model transforms a probe CDP $\mathbf{y}$ into a sample-based codebook $\mathcal{C}_{\mathbf{y}} = \{P_b^{\mathbf{y}}(\omega)\}_{\omega \in \Pi}$.[1]

6.4 A Statistical Viewpoint

In this chapter, we use the BPC model [68] as a baseline and describe the problem of authentication of CDP as a collection of independent channel-wise statistical tests. Under this model and given a fixed template $\mathbf{t}$, each pattern ω gives rise to a random binomial variable D^{ω} independent of each other. Based on this mathematical description, we focus on different ways to measure channel performances and study how they can be optimally combined together to strengthen authentication. That is, given a probe $\mathbf{y}$, decide whether it is an original $\mathbf{x}$ or a fake $\mathbf{f}$. Following the notations and concepts presented in Section 2.2, we denote $\mathbb{H}_0$ the hypothesis that probe $\mathbf{y}$ is an original and $\mathbb{H}_1$ the hypothesis that $\mathbf{y}$ is a fake. Under $\mathbb{H}_0$, the reference original codebook is denoted as $\mathcal{C}_0 = \{P_b(\omega)\}_{\omega=0}^{M-1}$ while, under $\mathbb{H}_1$, the reference fake codebook is denoted as $\mathcal{C}_1 = \{Q_b(\omega)\}_{\omega=0}^{M-1}$. Given a reference template $\mathbf{t}$, an unidentified probe $\mathbf{y}$, and a fixed pattern ω, we perform the following steps (see Fig. 6.1 for an illustration):

- Identify each appearance of pattern ω in the reference template $\mathbf{t}$ and extract the central pixel of each of these patterns: $t_1, ..., t_{L^{\omega}}$;
- Binarize the probe $\mathbf{y} \mapsto \tilde{\mathbf{t}}$ and extract the corresponding central pixels: $\tilde{t}_1, ..., \tilde{t}_{L^{\omega}}$;
- Compute the bit-flipping values $\delta_k = t_k \oplus \tilde{t}_k$ for each $k = 1, ..., L^{\omega}$, where δ_k denotes the indicator function of bit-flipping with $\delta_k = 1$, if $t_k \neq \tilde{t}_k$, and 0, otherwise.

[1] Profitting from the fact that, on average, $L^{\omega} = B/|\Pi| \approx 100$, one can estimate the probability of bit-flipping empirically for each of the patterns solely based on one pair $(\mathbf{t}, \tilde{\mathbf{t}})$ and compute a sample-based codebook for a given probe $\mathbf{y}$.

The BPC model indicates that the bit-flipping values $\delta_1, ..., \delta_{L^\omega}$ are sampled independently according to a Bernoulli distribution with a fixed probability $P_b^{\mathbf{y}}(\omega)$ for which we have the following statistical estimator:

$$P_b^{\mathbf{y}}(\omega) \approx \hat{P}_b^{\mathbf{y}}(\omega), \tag{6.7}$$

where we define

$$\hat{P}_b^{\mathbf{y}}(\omega) := \frac{1}{L^\omega} \sum_{k=1}^{L^\omega} \delta_k. \tag{6.8}$$

The goal of authentication is now to decide whether this empirical probability is more likely to be $P_b(\omega)$ or $Q_b(\omega)$ and provide bounds on this decision.

6.4.1 Hamming Distance Test

In the following sections, all analyses are conducted on a fixed pattern ω and then extended to all ω. As such, and to ease the notations, we remove the explicit reference to the pattern in the indices, except when formulating final results:[2]

$$D^\omega \to D, \quad L^\omega \to L, \quad P_b(\omega) \to P_b, \quad Q_b(\omega) \to Q_b.$$

Following the analysis in [85], the Hamming distance is described by a Binomial distribution:

$$D = \sum_{k=1}^{L} \delta_k \sim Binom(L; P_b^{\mathbf{y}}). \tag{6.9}$$

If we suppose that we have access to reference codebooks for original and fakes, we can formulate the Hamming distance test and measure the probability of miss as:

$$\begin{aligned} P_{miss}(\gamma) &= \mathbb{P}_{\mathbb{H}_0}(D > \gamma L) \\ &= \sum_{k=\lfloor \gamma L \rfloor + 1}^{L} \binom{L}{k} P_b^k (1 - P_b)^{L-k}, \end{aligned} \tag{6.10}$$

the probability of false acceptance as:

$$P_{fa}(\gamma) = \mathbb{P}_{\mathbb{H}_1}(D \leq \gamma L) = \sum_{k=0}^{\lfloor \gamma L \rfloor} \binom{L}{k} Q_b^k (1 - Q_b)^{L-k}, \tag{6.11}$$

[2]Note that the notation L used in this Chapter is a shorthand for L^ω and should not be mistaken with the $L \times L$ introduced in Chapter 3 where it represented the width and height of the template. To avoid confusion, we do not use the notation $L \times L$ in this Chapter and refer to B for the total number of symbols in the template.

and the average probability of error as:

$$\min_{\gamma\in[0,1]} P_{err}(\gamma) = \min_{\gamma\in[0,1]} \frac{P_{miss}(\gamma) + P_{fa}(\gamma)}{2}, \tag{6.12}$$

assuming both hypotheses $\mathbb{H}_0$ and $\mathbb{H}_1$ are equiprobable (see the plots in Fig. 6.1). One way to choose an optimal decision threshold γ is to minimize the average probability of error:

$$P_{err}^{min} := \min_{\gamma\in[0,1]} P_{err}(\gamma). \tag{6.13}$$

Our goal is to better study these probability of errors and derive an optimal bound for each channel by making an explicit link between this approach and the Neymann-Pearson test.

6.4.2 Posterior Likelihood Approach

The authentication problem can be studied from the point of view of the likelihood function. Given a sequence of realisations $\delta_1, ..., \delta_L \in \{0, 1\}$ and a reference probability of bit-flipping P_b, one can measure the likelihood that these samples were drawn from a Bernoulli distribution with parameter P_b. We write the likelihood function:

$$\begin{aligned}\mathbb{P}(\tilde{t}_1, ..., \tilde{t}_L \mid \mathbf{t}) &= \prod_{k=1}^{L} \mathbb{P}(\tilde{t}_k \mid \mathbf{t}) \\ &= \prod_{k=1}^{L} P_b^{\delta_k} \cdot (1 - P_b)^{1-\delta_k} \\ &= P_b^D \cdot (1 - P_b)^{L-D},\end{aligned} \tag{6.14}$$

where D is the Hamming distance defined in (6.9) and follows a Binomial law. Note that this derivation is based on the assumption of the BPC model that each bit-flipping δ_k is identically distributed and independent. Taking the logarithm on both sides, one arrives to the formula:

$$\log \mathbb{P}(\tilde{t}_1, ..., \tilde{t}_L|\mathbf{t}) = D \cdot \log P_b + (L - D) \cdot \log (1 - P_b)\,. \tag{6.15}$$

These types of derivations are well known [70] and explicit the link between minimization of Hamming distance and maximization of log-likelihood. Taking the expectation with respect to the posterior distribution on both sides yields the following result:

Proposition 6.4.1. *The average log-likelihood is given by:*

$$\mathbb{E}\left[\frac{1}{L^\omega} \log \mathbb{P}(\tilde{t}_1, ..., \tilde{t}_{L^\omega}|\mathbf{t})\right] = -\mathcal{H}(P_b^{\mathbf{y}}(\omega); P_b(\omega)), \tag{6.16}$$

where the right-hand side designates the cross-entropy (2.37) *between the unknown probe distribution* $P_b^{\mathbf{y}}(\omega)$ *and the reference distribution* $P_b(\omega)$.

Proof. The proof is straightforward from (6.15) since $D \sim Binom(L, P_b^{\mathbf{y}})$, and thus,

$$\mathbb{E}\left[\frac{D}{L}\right] = P_b^{\mathbf{y}}.$$

■

Formula (6.16) is very useful to test the adequacy of a given probe $\mathbf{y}$ to a certain reference probability P_b. In practice one can use (6.8) to estimate these quantities empirically. This result also gives new insights on the Posterior Log Likelihood (PLL) score introduced in Chapter 4.

Corollary 6.4.1. *The PLL score* (4.9) *can be computed as an average on all channels:*

$$PLL(\mathbf{t}, \mathbf{y} \mid \mathcal{C}) = -\sum_{\omega=0}^{M-1} L^{\omega} \cdot \mathcal{H}(\hat{P}_b^{\mathbf{y}}(\omega); P_b(\omega)). \tag{6.17}$$

6.4.3 Neymann-Pearson Optimal Bound

Using Neymann-Pearson terminology, we designate the type I error in this problem, as accepting a probe when it is a fake (false acceptance). The type II error would be to reject the probe as a fake when it is an original (true miss). The problem is, by nature, asymmetrical as it is much more harmful in real life scenarios to accept a fake product than to throw away a genuine one (think for instance of fake drugs). Neymann-Pearson lemma [74], then provides an optimal decision criterion for a significance test at level α. That is, if we fix the Type I error to be equal to α, the optimal decision statistic is given by the likelihood ratio test:

$$\mathcal{R} = \frac{\mathbb{P}_{\mathbb{H}_0}(\tilde{t}_1, ..., \tilde{t}_L \mid \mathbf{t})}{\mathbb{P}_{\mathbb{H}_1}(\tilde{t}_1, ..., \tilde{t}_L \mid \mathbf{t})} > \rho. \tag{6.18}$$

Proposition 6.4.2. *The likelihood ratio test* (6.18) *is equivalent to:*

$$\mathcal{H}(\hat{P}_b^{\mathbf{y}}(\omega); Q_b(\omega)) - \mathcal{H}(\hat{P}_b^{\mathbf{y}}(\omega); P_b(\omega)) > \frac{\log \rho}{L^{\omega}}, \tag{6.19}$$

and this test is asymptotically optimal as long as $Q_b(\omega) \neq P_b(\omega)$:

$$\lim_{L^{\omega} \to +\infty} \mathbb{P}_{\mathbb{H}_0}\left(\frac{\log \mathcal{R}}{L^{\omega}} < 0\right) = \lim_{L^{\omega} \to +\infty} \mathbb{P}_{\mathbb{H}_1}\left(\frac{\log \mathcal{R}}{L^{\omega}} > 0\right) = 0. \tag{6.20}$$

Proof. By computing $\frac{1}{L} \log \mathcal{R}$ and following the same derivation as (6.14), one arrives to the following formula:

$$\frac{\log \mathcal{R}}{L} = \frac{D}{L} \log \frac{P_b}{Q_b} + \frac{L - D}{L} \log \frac{1 - P_b}{1 - Q_b}. \tag{6.21}$$

Equation (6.19) then follows by taking the expectation on both sides as in (6.16). Under hypothesis $\mathbb{H}_0$, the random variable $D \sim Binom(L; P_b)$ follows a binomial distribution and

thus:

$$\mathbb{E}\left[\frac{\log \mathcal{R}}{L}|\mathbb{H}_0\right] = P_b \cdot \log \frac{P_b}{Q_b} + (1-P_b) \cdot \log \frac{1-P_b}{1-Q_b}$$

$$= \mathcal{D}_{KL}(P_b \parallel Q_b), \tag{6.22}$$

where $\mathcal{D}_{KL}(P_b \parallel Q_b)$ denotes the Kullback-Leibler divergence (2.36) between the Bernoulli distributions given by P_b and Q_b. With similar computations one can show that:

$$\mathbb{E}\left[\frac{\log \mathcal{R}}{L} \mid \mathbb{H}_1\right] = -\mathcal{D}_{KL}(Q_b \parallel P_b). \tag{6.23}$$

Finally, with similar approach, one shows that the variance $\mathrm{Var}\left(\frac{\log \mathcal{R}}{L}\right) \to 0$, when $L \to +\infty$ and so these quantities concentrate around their means, yielding asymptotic optimality, independant of ρ by Chebyshev's inequality (2.21):

$$\mathbb{P}_{\mathbb{H}_0}\left(\frac{\log \mathcal{R}}{L} < 0\right) = \mathbb{P}_{\mathbb{H}_0}\left(\frac{\log \mathcal{R}}{L} - \mathcal{D}_{KL}(P_b \parallel Q_b) < -\mathcal{D}_{KL}(P_b \parallel Q_b)\right)$$

$$\leq \mathbb{P}_{\mathbb{H}_0}\left(\left|\frac{\log \mathcal{R}}{L} - \mathcal{D}_{KL}(P_b \parallel Q_b)\right| > \mathcal{D}_{KL}(P_b \parallel Q_b)\right)$$

$$\leq \frac{\mathrm{Var}\left(\frac{\log \mathcal{R}}{L}\right)}{\mathcal{D}_{KL}(P_b \parallel Q_b)^2} \longrightarrow 0.$$

The other case is demonstrated similarly.

■

Note that (6.19) can be seen as a direct extension of the approach that we described in Section 5.3.2. Indeed, the PLL metric was solely based on a single cross-entropy term, whereas the Neymann-Pearson test is based on the difference of two cross-entropies. PLL can thus be considered as an analog to a one-class decision based solely on P_b, while NP-test is a two-class decision which requires the knowledge of both P_b and Q_b.

Proposition 6.4.3. *The Neymann-Pearson test and the Hamming distance test $\mathcal{R} > \rho$ and $D < \gamma L$ are equivalent if $P_b \leq Q_b$ and:*

$$\frac{\log \rho}{L} = \mathcal{D}_{KL}(\gamma \parallel Q_b) - \mathcal{D}_{KL}(\gamma \parallel P_b). \tag{6.24}$$

When $\rho = 1$, that is when both likelihoods are equal, we can derive the critical separation boundary independent of L:

$$\gamma_{crit} = \left(1 + \frac{\log P_b - \log Q_b}{\log(1-Q_b) - \log(1-P_b)}\right)^{-1}. \tag{6.25}$$

Proof. To prove (6.24), we start with relation (6.21) and use the facts that

$$D < \gamma L, \quad \log\left(\frac{P_b}{Q_b}\right) \leq 0, \quad \text{and,} \quad \log\left(\frac{1-P_b}{1-Q_b}\right) \geq 0,$$

to derive the lowerbound:

$$\begin{aligned}\frac{\log \mathcal{R}}{L} &= \frac{D}{L}\log\frac{P_b}{Q_b} + \frac{L-D}{L}\log\frac{1-P_b}{1-Q_b} \\ &> \gamma \log\frac{P_b}{Q_b} + (1-\gamma)\log\frac{1-P_b}{1-Q_b} \\ &= \mathcal{D}_{KL}(\gamma \parallel Q_b) - \mathcal{D}_{KL}(\gamma \parallel P_b).\end{aligned}$$

Equation (6.25) is obtained from (6.24) by setting $\rho = 0$ and isolating γ. ■

6.5 Channels Aggregation Strategies

All statistical approaches described in Section 6.4, were computed independently for each channel and different approaches to optimal thresholds selection were given. Applying a decision threshold to each channel transforms the real-valued sample-based codebook $\mathcal{C}_{\mathbf{y}} = \{P_b^{\mathbf{y}}(\omega)\}_{\omega\in\Pi}$ into a binary decision vector: $\{\Delta_\omega\}_{\omega\in\Pi}$, where $\Delta_\omega \in \{0,1\}$. In both situations, we are left with the problem of optimally aggregating these pattern-wise scores into a final decision score S_{final}. We propose here a generic way of describing linear aggregation strategies. A linear aggregation strategy is a weighted sum:

$$S_{final}^{AD} = \sum_{\omega} P_b^{\mathbf{y}}(\omega)\cdot\alpha_\omega \quad \text{or,} \quad S_{final}^{DA} = \sum_{\omega}\Delta_\omega\cdot\alpha_\omega, \tag{6.26}$$

where the coefficients $\alpha_\omega \in \mathbb{R}$ are chosen according to an aggregation strategy. A final decision based on this aggregation is performed by applying a threshold to S_{final}. These two approaches will later be referred to as Aggregate first - Decide next (AD) for the left one and Decide first - Aggregate next (DA) for the one on the right. In Chapter 5, the (AD) aggregation was performed according to two simple criterions:

(S1) *Averaging*: Taking the average of all statistics with equal contribution: $\alpha_\omega = 1$.

(S2) *Reliable Patterns*: Selecting only those patterns for which $P_b(\omega) \leq \mu$ for a fixed threshold μ and taking the average of all statistics on these patterns:

$$\alpha_\omega = \begin{cases} 1, & \text{if } P_b(\omega) \leq \mu, \\ 0, & \text{else.}\end{cases}$$

These approaches, although showing promising results, were chosen empirically and lacked a deeper investigation which could lead to a better understanding of the decision process. In this paper, we now introduce a systematic approach, based on the average probability of error:

(S3) *Minimal Probability of Error*: Fixing a threshold $\nu \in [0,1]$ and only aggregating the patterns which have a probability of classification error smaller than ν:

$$\alpha_\omega = \begin{cases} 1, & \text{if } P_{err}^{min}(\omega) \leq \nu, \\ 0, & \text{else.} \end{cases}$$

(S4) *2C-SVM Optimal*: Training a 2-class linear SVM [84] on the statistics and extract the coefficients learned by the model as "optimal" machine learning coefficients α_ω. We then sort them according to their absolute value and aggregate the k firsts: $|\alpha_{\omega_1}| \geq |\alpha_{\omega_2}| \geq |\alpha_{\omega_3}| \geq \dots$.

Strategies (S2) and (S3) are illustrated in Fig. 6.2 where the choice of threshold is shown on scatterplots. For each $\omega \in \Pi$, we draw a point $(P_b(\omega), P_{err}^{min}(\omega))$. Fixing a threshold $P_b(\omega) \leq \mu$ corresponds to dividing the set vertically (top), whereas fixing a threshold $P_{err}^{min}(\omega) \leq \nu$ corresponds to dividing the set horizontally (bottom).

6.6 Experimental Results

6.6.1 Experimental Setup

The dataset[3] of CDP that we use in the following experiment is the same as the one used in Chapters 4 and 5. It consists of 1440 unique triples of digital template, original printed and fake CDP $(\mathbf{t}, \mathbf{x}, \mathbf{f})$ enrolled with an *iPhone 12 Pro* during 6 capture runs. The CDP acquired in this way are then processed using histogram matching to a reference CDP with a uniform histogram. In the following experiments, the dataset is divided into two randomly chosen train set and test set, which consist of 500 triples. The train set is used to compute reference codebooks of originals $\{P_b(\omega)\}_{\omega\in\Pi}$ and fakes $\{Q_b(\omega)\}_{\omega\in\Pi}$.

[3]The dataset is publicly available and can be found here: https://sipcloud.unige.ch/index.php/s/tYKffnKNRgSwBAN

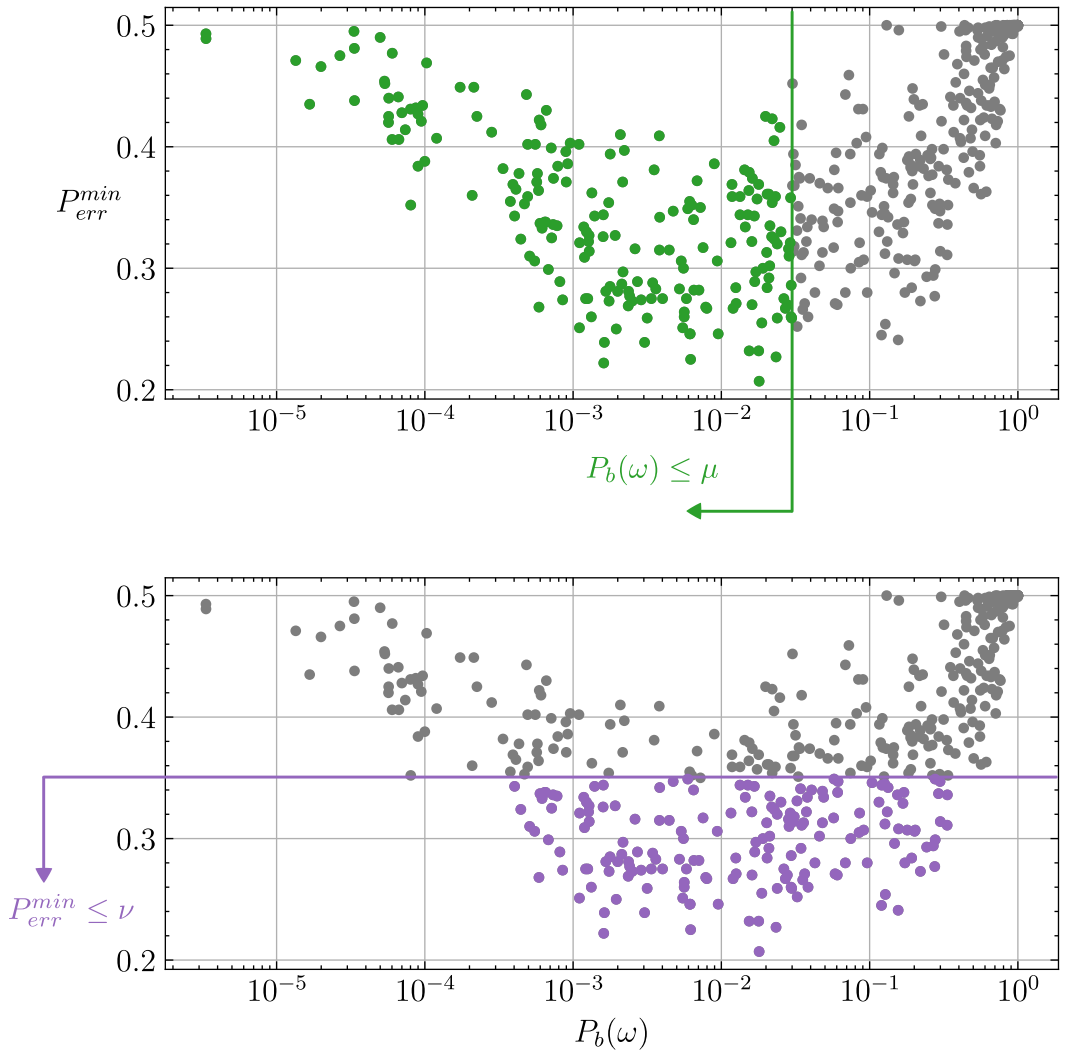


Figure 6.2. A visual illustration of the selection of pattern according to strategy (S2) (top) and strategy (S3) (bottom). Each point in the scatterplot corresponds to a choice of pattern ω and its location is determined by the pair $(P_b(\omega), P_{err}^{min}(\omega))$.

6.6.2 Reliability & Authentication Performance

The results of this experiment are shown in Fig. 6.3. For each pattern ω independently, we measure the minimal average classification error P_{err}^{min} both theoretical (in orange) by referring to the optimal threshold selection γ_{crit} derived in (6.25) for $L^\omega = 100$, and experimental (in blue) by performing an empirical optimal threshold selection on the test set. The results in the top plot show that the theoretical predictions of the average probability of error based on the BPC model closely match the empirical measure based on the testing set. Indeed, the orange and blue dots are very close together especially for small values of P_b.

We complement the study by visualizing the same results for a multi-shot setting where all 6 capture runs are used and fused before computing the Hamming distance test, effectively increasing the number L^ω of symbols in each CDP by a factor of 6 (the behaviour of these symbols is of course strongly correlated but still enhances the prediction capacity). The results are shown in the bottom plot of Fig. 6.3 which illustrates the benefit of using multishot-based predictions (in green) instead of singleshot (in blue). Although not as powerful as increasing the number B of symbols in the CDP (compare with the theoretical multi-shot prediction in violet), we still see a net improvement in classification accuracy, as predicted by Proposition 6.4.2.

We also note that pattern reliability, that is low probability of bit-flipping, does not necessarily imply low classification error. Indeed, the plots clearly indicate that the best performing patterns are those located between $P_b(\omega) = 10^{-3}$ and $P_b(\omega) = 2 \cdot 10^{-2}$. This observation is a motivation for us to investigate strategy (S3) based on a bound ν on P_{err}^{min} for the aggregation strategy and compare it with strategy (S2).

In Fig. 6.4 we see another representation of the comparison between theoretical and empirical separability results. In this representation, the color gradient of the image represents the theoretical average probability of error for two binomial distributions with $L^\omega = 100$ and different P_b and Q_b indexed respectively by the x-axis and y-axis of the image. Typically, we see that the average probability of error is always equal to 0.5 when $Q_b \leq P_b$ and decreases rapidly when $Q_b > P_b$. On top of this image, we superimposed the scatterplot of actual pairs $(P_b(\omega), Q_b(\omega))$ from the training set with the color of each dot representing the empirical average P_{err}^{min} using the same color gradient. Similarly to what we saw in Fig. 6.3, the experimental values match the theoretical ones in most cases. Another interesting observation is the way the experimental $(P_b(\omega), Q_b(\omega))$ pairs populate the plane with a clear indication that the most pairs with $P_b(\omega) > 0.5$ do not contribute to reliable authentication.

6.6.3 Aggregation Strategies Comparison

In this second experiment, we test the various aggregations strategies introduced in Section 6.5. Fig. 6.5 shows a visual comparison of the four different strategies for the selection of the k best optimal pattern. The patterns are ranged from best to worst according to the

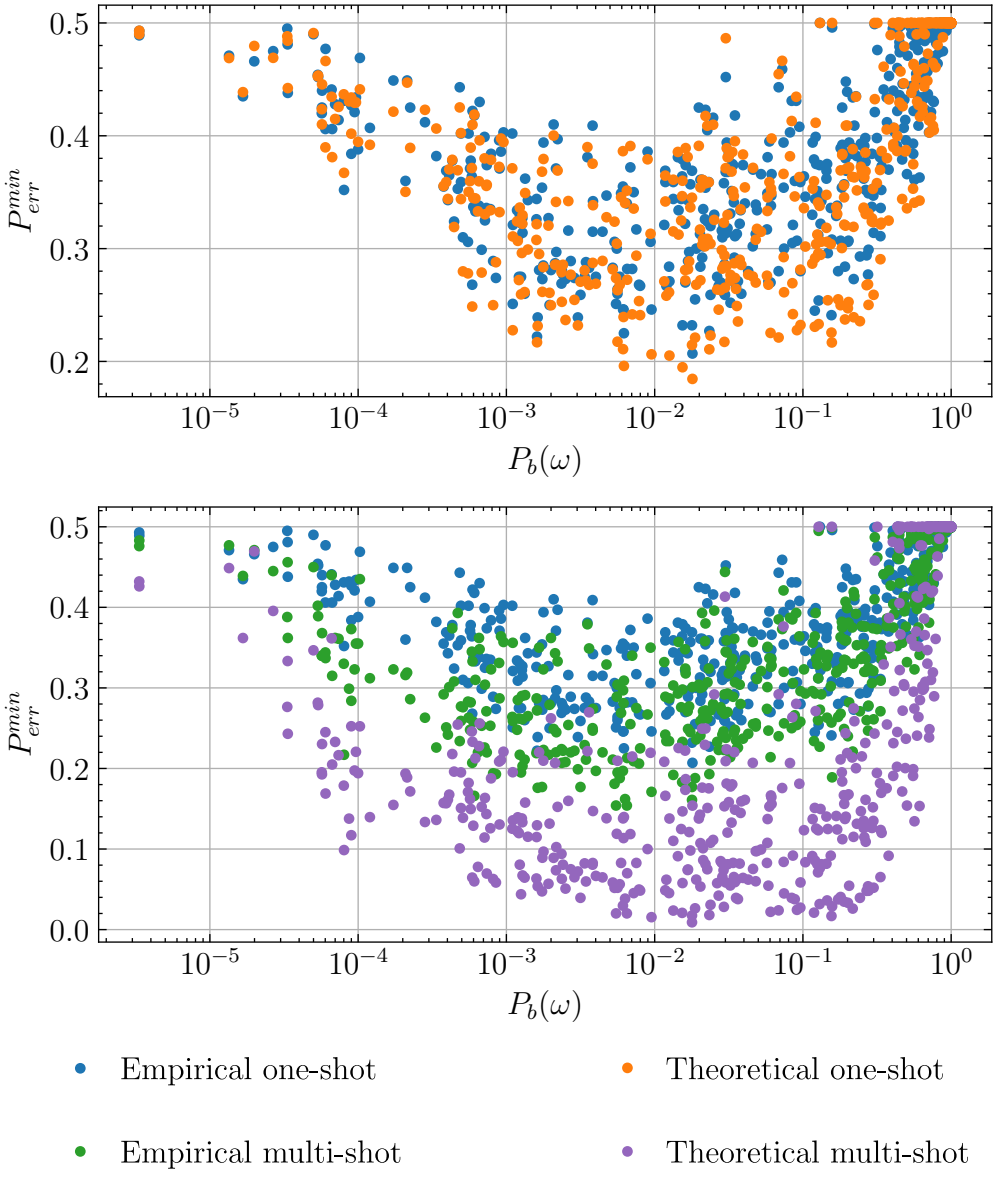


Figure 6.3. Average probability of error between originals and fakes samples as a function of the probability of bit-flipping $P_b(\omega)$ of each channel. The minimal probability of error P_{err}^{min} is computed using the theoretical model (in orange/violet) and empirically on the test set (in blue/green). The same comparison is performed for one-shot setting (top) and multi-shot setting (bottom).

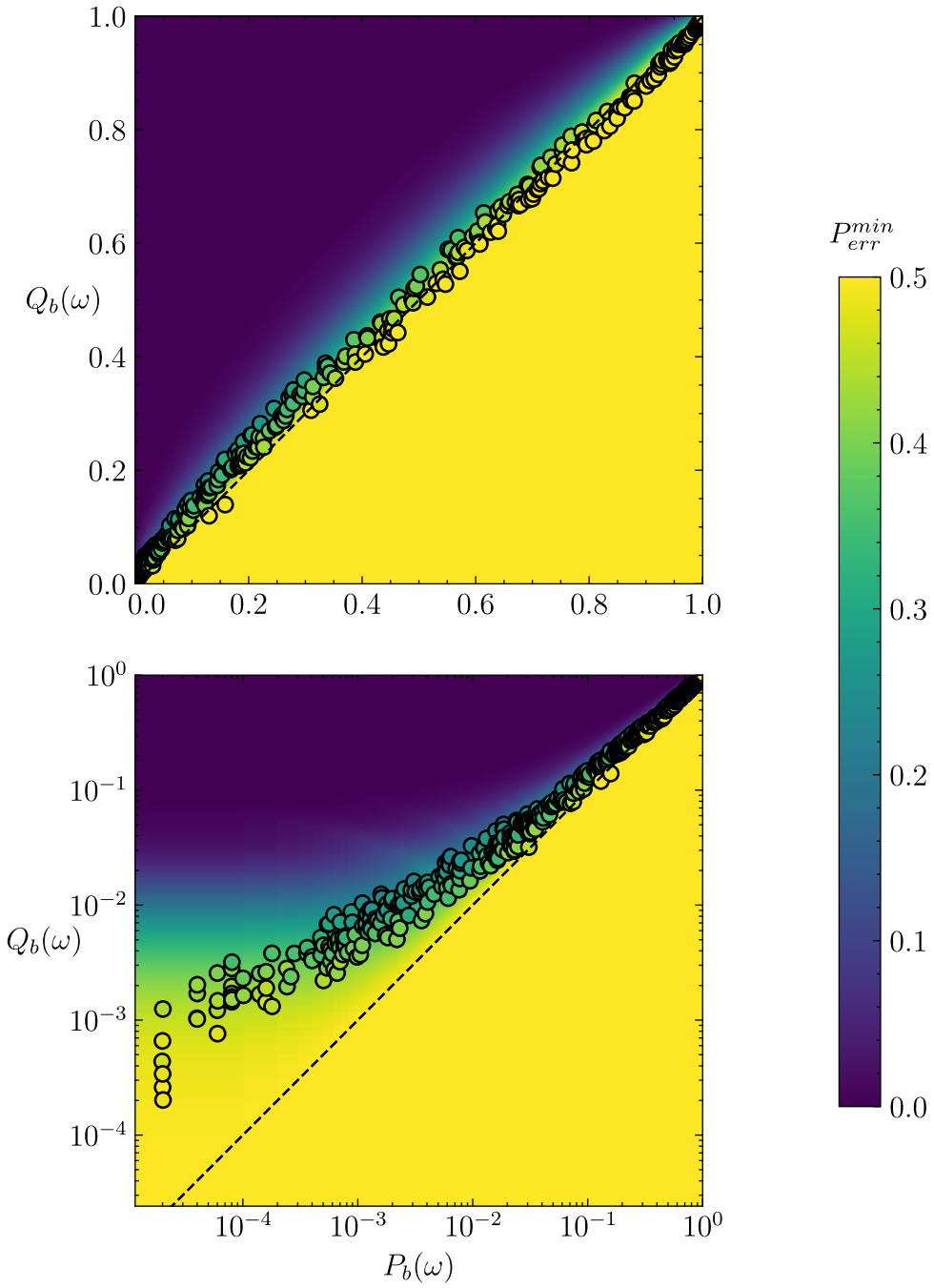


Figure 6.4. 2D color representation of the minimal average probability of error P_{err}^{min} (see the color gradient) for two binomial distribution of fixed length $L^{\omega} = 100$ as a function of $P_b(\omega)$ (x-axis) and $Q_b(\omega)$ (y-axis) for standard scale (on top) and logarithmic scale (on bottom). The scatterplots represent the experimental probability of error computed on a real dataset of CDP using the same gradient scale.

gradient color going from dark blue to yellow and represented on the 2D plane similar to Section 6.5. The strategy (S1) just selects all patterns undistinctively, strategy (S2) orders them from left to right (according to P_b), strategy (S3) from bottom to top (according to P_{err}^{min}) and strategy (S4) according to 2C-SVM coefficients. An interesting observation is the seemingly random selection of patterns performed by the ML-based strategy.

In order to compare these aggregation strategies on the same basis, Fig. 6.6 shows the final average error of classification as a function of the number k of patterns used for authentication for each approach, both for oneshot and multishot technique.

Surprisingly, even though the strategy (S3) based on the minimal P_{err}^{min} is optimal for each pattern taken individually, aggregating according to this strategy does not improve the final decision score on this dataset. Indeed, the aggregation strategy (S2) based on the minimization of $P_b(\omega)$ introduced in Chapter 5 still outperforms this theoretically optimal criterion, while still being solely based on a reference codebook of originals. Finally, the fully supervised aggregation scheme (S4) based on linear 2-class SVM performs optimally and, contrary to the other aggregation schemes, does not worsen when the number of patterns used gets too big. A noticeable difference with the previous strategies is the fact that the coefficients α_ω are real-valued, with both positive and negative values contributing to the final score. This approach, although theoretically interesting, requires the knowledge of both original CDP and fake CDP to be applied and we conjecture that it should be prone to errors against other types of unseen fakes [61].

6.6.4 Global Aggregated Classification Performance

In this third experiment, we compare the final aggregated average classification error for the different aggregating strategies introduced in Section 6.5. A direct comparison is done between single-shot aggregated performance and multi-shot aggregated performance, where all 6 capture runs are used and fused before computing the Hamming distance test.

When looking at Table 6.1, a striking observation is the outstanding enhancement of performance when using multi-shot prediction. Indeed, all aggregation strategies benefit from this approach. We analyze this improvement as an illustration of Prop. 6.4.2. Without much surprise, the fully supervised linear 2C-SVM aggregation strategy (S4) performs the best. It is still remarkable to notice that such an approach is able to perfectly separate originals and fakes when all other approaches still miss by a few. Finally, the (DA) approach also shows an interesting enhancement in performance compared to the (AD) approach which was used in the previous papers.

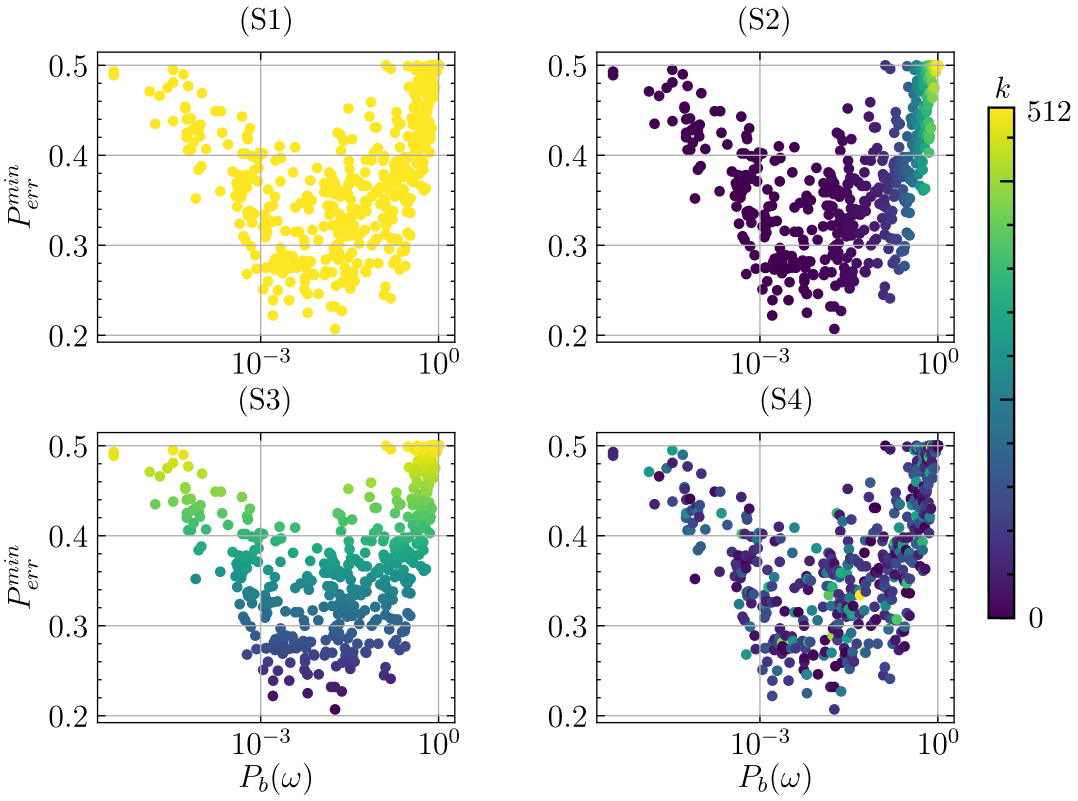


Figure 6.5. A comparison of the four different strategies for the selection of the k best patterns, visualized on the (P_b, P_{err}^{min})-plane. The dark blue patterns are the most reliable while the yellow patterns are the least reliable according to each strategy.

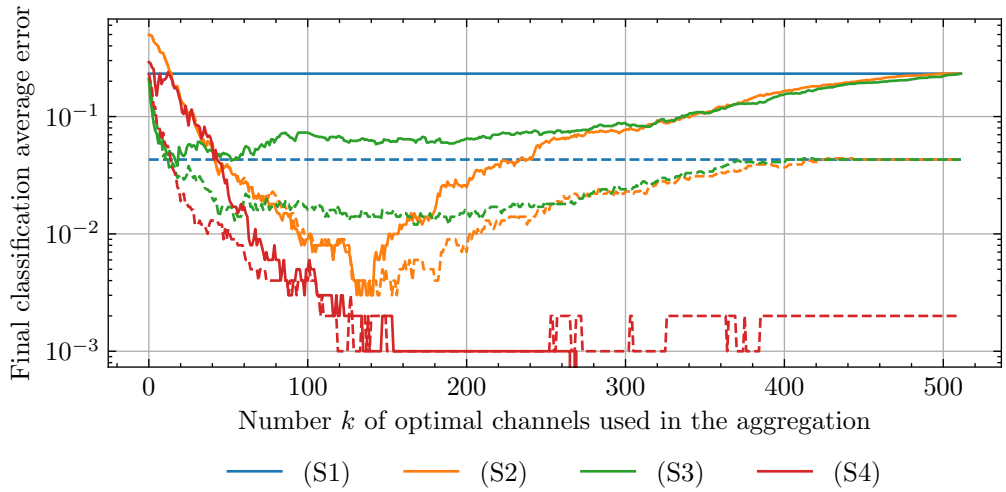


Figure 6.6. A plot of the average classification error probability based on the final score S_{final} for each aggregation strategy as a function of the number of channels aggregated. Continuous curves correspond to (AD) and dashed curves correspond to (DA). The best k channels are different for each strategy and depend on the optimality criterion.

Table 6.1. Results of minimal average P_{err}^{min} (in %) for the aggregation of channels based on four different strategies with Aggregate-Decide (AD) or Decide-Aggregate (DA). Both single-shot and multi-shot settings are compared.

		Aggregation Strategy			
		(S1)	(S2)	(S3)	(S4)
*A*D	single-shot	20.29	0.36	3.07	**0.10**
	multi-shot	14.30	0.04	0.13	**0.00**
*D*A	single-shot	4.12	0.40	0.96	**0.11**
	multi-shot	1.20	0.05	0.06	**0.00**

6.7 Conclusion

In this Chapter, we presented a theoretical analysis of the authentication performance of the BPC Model introduced in Chapter 4. Relying on the theoretical description of pattern-wise Hamming distances, we reformulated the authentication problem in terms of separability of Binomial distributions. We then derived optimal bounds for each channel for the classification task, explicitely linking the separability problem with the likelihood of both inlier and outlier class distributions. In a third part, we introduced a structured description for the aggregation strategies of the different channels. By comparing with a real dataset of smartphone acquired CDP, we demonstrated that the BPC model correctly simulates the authentication process and allows to reason about the design of optimal aggregation strategies.

Chapter 7

Conclusion and Future Work

This thesis takes place in the field of Physical Object Security in view of the development of secure and reliable authentication schemes and is a part of the broader project of *Information-theoretic analysis of deep identification systems* [47], directed by Prof. Slava Voloshynovskiy and supported by the Swiss National Fund. Among the wide variety of technologies available for object protection, each with their own advantages and disadvantages, this thesis concentrated on the Copy Detection Pattern technology, a framework introduced in 2004 which shows a great potential as an anticopy approach based on a secret template element and whose security relies on the random distorsions occuring during printing. The main advantage of this technology lies both in its relative simplicity of implementation and the high level of security it provides against product counterfeiting. However, this technology being relatively new compared to others, there was still a lack of understanding in the proper security guarantees provided by its scheme and as such, the question of performance reliability and overall security remained mostly an unexplored territory. Recent advances in this field showed that this scheme is actually badly challenged by attacks based on deep learning binarization algorithms.

This PhD thesis is thus a proposition for addressing these open questions and tackles the problem of understanding and modelling the printing-imaging scheme of CDP. By proposing theoretical models, this work tries to unveil some of the elements that constitute the strength of this technology and point out its weaknesses. At the same time, this thesis identifies and proposes new and enhanced approaches for the authentication of CDP that constitute a reinforcement of its technology. We will now review in more details the main contributions and conclusions that can be drawn from each chapter of this research and give several direction for future researches that would benefit from further investigations.

The first direction we investigated in this work is the modelling of the printing-imaging channel of CDP using a deep-learning architecture based on a pair of encoder and decoder trained jointly at the task of mimicking the printing-imaging process underwent from a digital template to a printed CDP, followed by its binarized estimation. This work effectively

demonstrates that deep learning architectures are able to act as digital twins of the physical process of printing and acquiring CDP and create digital copies of CDP that are undistinguishable from originals for standard metrics. The chapter then proposes various approaches for the use of this digital twin architecture to enhance the authentication of real CDP and demonstrates the performance gain. Attention is also drawned to the relative instability of the deep architecture training and thus the unreliable variability of the proposed scheme across random training initializations.

As an intermediate conclusion for this first research, we conclude that despite the great performances shown by the digital twin architecture, this model lacks the stability and reliability required of such model in order to build secure framework for critical applications. In view of these limitations, we think that the digital twin architecture would benefit from deeper investigations, especially in trying to understand how the clone CDP deviate from the original CDP and linking it to the performance of the proposed scheme. The same study can be made for deep classifiers trained on clone CDP. For instance, a study of the decision boundary of such classifiers based on a gradient attention map [86] could potentially shed some light on the way these classifiers take decisions.

The second direction we studied in this thesis describes the printing-imaging channel of CDP using a probabilistic model inspired by information theory that we called the BPC model. The critical contribution of the BPC model is the recognition and precise description of the local dependencies that each template symbol undergoes during the printing-imaging-binarization process of CDP authentication. Studying this model in more details, we introduce key notions such as pattern-based probability of bit-flipping, reliability codebook and Posterior Log-Likelihood (PLL) measure. The experiments performed on real datasets of acquired CDP demonstrate that the notions introduced by this model reflect actual intrinsic characteristics of CDP, stable across various printing and capturing modalities.

We conclude this second research by the observation that the BPC model seems to capture essential features of the CDP authentication scheme. By providing an interesting tradeoff between simplicity of the model and description capability, the BPC model seems to land on the right spot to actually provide useful statistics for enhanced authentication while still being fairly easy to implement in practice. We note however that the proposed model makes strong independence assumptions that do not seem to completely hold in practice. For instance, the posterior independence assumption, which simplifies most of the computations, is certainly not fully true in practice for neighbouring symbols. As such, an empirical investigation of a two-point correlation function (similar to the Ising model [87]) would certainly give more insights on this hypothesis, potentially leading to a new model where the posterior independence hypothesis would be relaxed. Another direction of research that we think would be worth exploring is the stationary hypothesis. Indeed, recent researches [44] highlighted the fact that the printing mechanism is a non-uniform process, thus apparently contradicting this hypothesis. In this regard, we think that a study of the variability of the probability of

bit-flipping for a given pattern located on various fixed spots in a template would give new insights on this hypothesis.

The third direction that has been investigated is the design of novel authentication schemes for CDP based on the BPC model. In this chapter, we describe a new authentication scheme, introducing a predictor channel for the Defender, that allows him to store a reference codebook on the server in addition to the reference template. We then explore how the Verifier can benefit from this codebook by deriving two novel authentication strategies based on PLL score and reliability masks. Finally, we demonstrate the gain in performance for authentication by comparing these novel schemes with traditional CDP authentication for various modalities.

The conclusion for this chapter of the thesis is straightforward as we were able to actually demonstrate the usefulness of the BPC model for practical applications. This chapter also adressed the threat posed by deep-learning attacks on the CDP framework as the results show that BPC-based authentication is able to properly detect these powerful attacks. Still, the approaches proposed in this chapter could use some deeper investigations and this is exactly what we have been doing with the fourth direction explored in the thesis, attempting at better understanding the theoretical performance guarantees provided by this model. Hence, we postpone the discussion of possible extensions of this work after we have fully discussed the fourth direction of contribution.

The fourth and final direction explored in the thesis is the study of theoretical guarantees for authentication schemes based on the BPC model. This chapter describes the authentication of CDP as a binary hypothesis test and uses the BPC model to derive theoretical optimal strategies. The chapter also explores a systematic description of linear aggregation techniques in order to arrive to an optimal final decision for authentication. Finally, the theoretical performances predicted by the BPC model are shown to be consistent with empirical results and the various aggregation strategies provide new interesting possibilities for future developments.

As a conclusion to this last chapter of research, we have provided new evidences that CDP authentication schemes can be accompanied with theoretical guarantees of performance and stability. As such, this fourth direction of research further consolidates the confidence one can put into security schemes based on CDP and provides new tools for industries and researchers to measure the degree of confidence they can put in such systems. In order to pursue the researches explored in this chapter, we think that the proposed approaches could be extended to other datasets of CDP, captured with different devices and in more various settings. The question of cross-device performance also remains largely unexplored even though early results show promising performances. On a theoretical level, we think that the linear aggregation theory proposed in the chapter could be further investigated. In particular, we think that the optimal coefficients learned by the 2C-SVM might be connected to deeper information-theoretic quantities yet to be explored. Finally, we think that the performance

guarantees computed for each individual pattern could be exploited for the design of new template that would outperform the standard design of CDP whose sole objective is simply based on entropy maximization.

This thesis is now coming to an end, and it is time for us to reflect on the work presented here as a whole. Our goal for this thesis was to build a better understanding of the CDP security framework so that this new promising technology would rest on firm ground for its future development and implementation in modern industries. With the four directions explored in this thesis, we think that we succeeded in showing that this technology is able to provide guarantees for its performances even against powerful copy attacks and that the Defender is provided with multiple tools to measure and prevent counterfeiting attacks against his system. Fighting counterfeiting is, in essence, an imbalanced problem since an attacker will always have the advantage of proposing threats that are unexpected and unpredictable. Hence, it is of critical importance for the Defender to develop a deep comprehension of his scheme in order to maximize his capacity to detect unknown attacks and we think that the multiple approaches we provided in this work constitute a significant progress in this direction.

The field of physical object security based on CDP proved to be a tough research challenge, connecting many apparently disconnected areas in engineering, computer science and mathematics. From the study of lightning condition and capture stability with smartphones, post-processing algorithms and server maintenance for deep learning, to Shannon's information theory and probabilistic lattice models, my journey exploring these fields took me to many places I would not have been able to foresee when I first started engaging with this problem. And yet, at the end of six years of research, all these puzzle pieces assemble to make a beautiful and yet always incomplete painting of a subject I have learned to appreciate. In the end, what appears as the most important contribution to my personal development as a researcher, is the realization of this equilibrium. That research is by essence incremental and even the smallest contribution to a field of research should be valued and cherished as it represents a step in the direction of unveiling a little bit more of the vast surrounding world of unknown.

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Appendix A

List of publications, patents and awards

Awards

- Best Paper award at IEEE International Workshop on Information Forensics and Security (WIFS), Shanghai, China 2022.
- Best Paper award at IEEE International Workshop on Information Forensics and Security (WIFS), Roma, Italy, 2024.

Patents

- S. Voloshynovskiy, O. Taran, *J. Tutt* and T. Holotyak, "Object Authentication Using Digital Blueprints and Physical Fingerprints", European Patent App. EP 4 102 399 A1, 12.2022
- S. Voloshynovskiy, *J. Tutt*, R. Chaban, O. Taran and T. Holotyak, "Authentication of Physical Objects based on Reliable Parts of Copy Detection Patterns", European Patent App. EP 4 386 586 A1, 06.2024[1]

Journal Papers

- *J. Tutt*, O. Taran, R. Chaban, B. Pulfer, Y. Belousov, T. Holotyak, and S. Voloshynovskiy, "Authentication of Copy Detection Patterns: A Pattern Reliability Based Approach," IEEE Transactions on Information Forensics and Security, vol. 19, pp. 3124-3134, 2024.
- Y. Belousov, G. Quétant, B. Pulfer, R. Chaban, *J. Tutt*, O. Taran, T. Holotyak, and S. Voloshynovskiy, "A Machine Learning-Based Digital Twin for Anti-Counterfeiting

[1]S. Voloshinovskiy and J. Tutt have equal contribution

Applications with Copy Detection Patterns," IEEE Transactions on Information Forensics and Security, vol. 19, pp. 3395-3408, 2024.

International Conferences

- *J. Tutt* and S. Voloshynovskiy, "Provable Performance Guarantees of Copy Detection Patterns," in Proc. 16th IEEE International Workshop on Information Forensics and Security (WIFS), Roma, Italy, 2024.
- *J. Tutt*, O. Taran, R. Chaban, B. Pulfer, Y. Belousov, T. Holotyak, and S. Voloshynovskiy, "Mathematical model of printing-imaging channel for blind detection of fake copy detection patterns," in Proc. IEEE International Workshop on Information Forensics and Security (WIFS), Shanghai, China, 2022.
- Y. Belousov, B. Pulfer, R. Chaban, *J. Tutt*, O. Taran, T. Holotyak, and S. Voloshynovskiy, "Digital twins of physical printing-imaging channel," in Proc. IEEE International Workshop on Information Forensics and Security (WIFS), Shanghai, China, 2022.
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- R. Chaban, O. Taran, *J. Tutt*, Y. Belousov, B. Pulfer, T. Holotyak, and S. Voloshynovskiy, "Printing variability of copy detection patterns," in Proc. IEEE International Workshop on Information Forensics and Security (WIFS), Shanghai, China, 2022.
- O. Taran, *J. Tutt*, T. Holotyak, R. Chaban, S. Bonev, and S. Voloshynovskiy, "Mobile authentication of copy detection patterns: how critical is to know fakes?," in Proc. IEEE International Workshop on Information Forensics and Security (WIFS), Montpellier, France, 2021.
- R. Chaban, O. Taran, *J. Tutt*, T. Holotyak, S. Bonev, and S. Voloshynovskiy, "Machine learning attack on copy detection patterns: are 1x1 patterns cloneable?," in Proc. IEEE International Workshop on Information Forensics and Security (WIFS), Montpellier, France, 2021.